Deriving MOND from bi-conformal gravity: vortex closure, the interpolating function, and the dark-energy connection

Version 5.0

Contents

Purpose and scope

This companion paper provides a *detailed* and *self-consistent* presentation of the MOND sector already contained in the main theory, **without altering the covariant backbone**. In particular:

- the covariant backbone is exactly the main theory action (minimal coupling, no extra $f(X)$ nonlinearity introduced in the diagonal scalar sector); any explicit swirl realization introduced below is a bound-structure effective sector built on that same backbone rather than a replacement of it;

- in the same-matter-content metric-sector limit, the linear cosmological sector remains GR-identical as established in the CMB-consistency analysis of the main theory (no modification of the Einstein–Boltzmann hierarchy); possible pressure-history matter-sector backreaction is a separate issue treated in the main theory's process-time appendix;

- MOND phenomenology arises only in the *nonlinear, quasi-static, bound-structure* regime; in the cosmological discussion below the observed Hubble rate is used only as a proxy for the global decrease of static space pressure and the associated clock history, not as an independent ontological "expansion field" of the theory.

Accordingly, the main text keeps only the compact regime split and the identification of the rotational slot, while the present companion paper carries the explicit bound-structure Bernoulli+vortex derivation using the same bi-conformal backbone together with the quantum pressure-medium ingredients. In particular, the companion quantum paper now closes the local gravity channel microscopically through the chain oscillon → Bernoulli pressure deficit → zero-frequency source → exterior $1/r$ field, with reduced-gap/coercive control of the localized source sector. The present paper builds the MOND vortex completion on top of that already-closed local channel. Thus Eq. (2) remains the baseline covariant kernel throughout, while any S_{swirl} written below should be read as the mesoscopic realization of the same rotating-medium response in the nonlinear galactic regime. The purpose is to make the MOND chain explicit at two levels:

- **Qualitative (cosmological origin):** observed cosmic-flow fields, identified in ISPG with bulk motion of the pressure-carrying substance → Kelvin–Helmholtz roll-up under a low-viscosity effective-medium closure → central rarefaction → extra metric/geodesic acceleration channel → MOND phenomenology. Different vortex topologies (laminar planar, convective poloidal, chaotic) provide a qualitative morphology map onto galaxy types (spiral, elliptical, irregular) at the same MOND scale.

- **Quantitative (local closure):** global pressure decrease / coherence damping → coherence length → Hubble-wavelength-selected critical acceleration $a_0 = cH/(2\pi)$ → vortex-lag correction $H \to H_{\mathrm{eff}}$ (calibrated formation-epoch recovery of the local observed value, with age-dependent a_0 as the testable prediction) → instantaneous vortex equilibrium → interpolating function $\mu(x) = x/(1+x)$ → BTFR and environment effects.

Physical picture: crystals dissolving in water

Before entering the formal development, it is useful to state the physical picture in non-mathematical language. Imagine a tank of transparent, ordinary water in which colored crystals are scattered and slowly dissolve. This simple image captures several central ISPG mechanisms simultaneously.

Crystal = oscillon (particle). A localized, stable knot in the φ-field — what we call "matter" is precisely such a knot.

Color released into the water = χ-field (hysteresis tail). Each particle broadcasts its "color" into the surrounding medium: a Yukawa-type χ-tail that breaks the symmetry of the vacuum around the source.

Three levels of physics.

- *Local coloring → gravity.* The water is darkest immediately next to each crystal. This is the gravitational attraction of an individual source — a local enhancement of the φ-field.
- *Overall tinting of the water → dark energy.* Over time the bulk of the water becomes uniformly tinted. This is the cosmological "echo background": a χ-floor used in the companion quantum-sector estimate as the dynamical dark-energy component. Its perturbative-branch normalization can reach the observed acceleration scale, while the full zero-point and non-perturbative normalization problem remains open.
- *Clusters of crystals → galaxy clusters.* Where many crystals sit close together, their colored tails overlap into a shared, frozen tint. This is the Bullet-Cluster regime: the χ-field persists even when the ordinary matter has moved on.

Shrinkage and self-regulation. The analogy extends to two further ISPG features that are usually explained only algebraically.

Shrinkage. A dissolving crystal loses size. In ISPG the effective inertial mass obeys the bi-conformal relation

$$m_{\rm eff} = m_0\, e^{\varphi/2}, \tag{1}$$

so in a strong-gravity environment ($\varphi < 0$) the particle is effectively lighter, and on cosmological timescales the growing echo background slowly erodes the effective mass of every particle in the universe.

Self-regulation. A smaller crystal releases less color, so the surrounding water is tinted more slowly, which in turn slows further shrinkage. The same negative-feedback loop operates here: a lighter particle is a weaker χ-source, the background grows less rapidly, and the shrinkage itself decelerates. This provides a structural damping mechanism against runaway growth, but the quantitative dark-energy normalization remains conditional on the perturbative branch and on the open non-perturbative stability calculation. This deceleration of the background growth would leave an observational fingerprint: the relative growth rate of the echo background drops from $\sim$40% at $z = 1.5$ to $\sim$6% today in the companion estimate, giving a target for DESI/Euclid tests of evolving dark energy [12].

What this picture buys us. The vortex closure of MOND, the interpolating function $\mu(x)$, and the connection to dark energy all follow from the same medium. Galactic vortices are induced circulations in the background water; MOND arises where individual crystal fields become too weak to dominate over the global tint; dark energy is simply the global tint's continuing growth. The remainder of this paper makes each of these statements quantitatively precise.

1 Non-negotiable consistency with the main theory

We work strictly within the main theory backbone:

$$S = \frac{1}{16\pi G}\int d^4x\sqrt{-g}\left[R + \frac{1}{2}g^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi\right] + S_m[g_{\mu\nu},\psi], \tag{2}$$

together with the bi–conformal geometry used in the main theory [13]. No additional potential $V(\varphi)$, no G_3 term, and no non-minimal matter coupling are introduced here. Consequently, the linear-cosmology results established in the CMB analysis of the main theory remain intact.

The same backbone already reproduces the full 1PN Solar-system sector ($\gamma = \beta = 1$), enforces luminal tensor propagation ($\alpha_T = 0$), and gives universal scalar sensitivity $s = 1/2$ exactly for vacuum compact objects and at leading order for matter-filled bodies, so scalar dipole radiation is suppressed in the controlled regimes (the exact non-perturbative matter-filled cancellation remains the main-theory Appendix 16 open task) [13]. The galactic MOND completion developed here therefore does not buy its phenomenology at the cost of spoiling the strong-field or gravitational-wave consistency of the parent theory.

2 Regime separation: linear FLRW vs. bound structures

The programme rests on a clean separation of regimes:

$$k \ll \frac{aH}{c} \quad \text{(linear cosmology)}, \qquad k \gg \frac{aH}{c} \ \text{ and } \ \partial_t \ll c\nabla \quad \text{(quasi-static bound structures)}. \tag{3}$$

In the first regime, the theory reduces identically to GR at linear order, as shown in the main theory's CMB analysis [13]. In the second regime, localized sources excite scalar perturbations and nonlinear response becomes relevant.

3 Origin of galactic vortices: cosmological flows in the space substance

In this theory space is a physical substance (a pressure-carrying scalar medium), not an empty container. In the ISPG ontology, the observed large-scale cosmic flows are identified with bulk motion *of* that substance. Under a low-viscosity effective-medium closure, the existence of galactic-scale vortices is a quantitatively supported consequence of this identification together with well-established observational facts.

3.1 Observational basis: cosmic flows

Three independent lines of evidence establish that large-scale coherent flows exist in the present universe:

1. **Bulk flows.** Galaxy surveys reveal coherent peculiar velocities extending over $\gtrsim$100 Mpc scales (e.g. the Laniakea basin of attraction; Tully et al. 2014).

2. **Cosmic web.** Matter streams along filaments toward nodes (galaxy clusters), with voids in between—a flow topology directly analogous to convective circulation in a cooling fluid.

3. **CMB anisotropies.** The COBE/WMAP/Planck maps show temperature fluctuations $\delta T/T \sim 10^{-5}$ in the early universe, i.e. the primordial pressure/density gradients that seeded all subsequent flows.

In the standard picture these flows are interpreted as *matter* motions driven by gravitational instability. In the present theory they are, more fundamentally, motions of the space substance itself, carrying the embedded matter along.

3.2 From flows to vortices: a fluid-dynamical route

In a low-viscosity fluid with inhomogeneous flows, vortex formation is the expected outcome:

- Two streams meeting at a nonzero angle generate shear, which rolls up into a vortex via the Kelvin–Helmholtz instability.

- Turbulent cascades break large-scale flows into a hierarchy of eddies (Kolmogorov cascade), populating all scales down to the dissipation length.
- No external "spinning agent" is required: differential cooling of the primordial substance produces pressure gradients, gradients drive flows, and flows at different angles produce vortices via the standard shear-instability and turbulent-cascade mechanisms—precisely as atmospheric temperature differences create cyclones without external torque.

Quantitative KH and inertial-range estimate. This qualitative statement is also quantitatively plausible on conservative numbers. For a shear layer with relative speed ΔV and perturbation wavelength $\lambda = 2\pi/k$, the incompressible Kelvin–Helmholtz growth rate is

$$\gamma_{\mathrm{KH}} = \frac{k\,\Delta V}{2} = \frac{\pi\,\Delta V}{\lambda}\,. \tag{4}$$

Equation (4) is the classical inviscid growth rate for a locally planar tangential-shear patch [3]; we adopt this as the fiducial shear-layer estimate for the effective ISPG medium. Under the same low-viscosity closure, three-dimensional topology and weak compressibility are expected to modify the coefficient by $O(1)$ geometric factors only, leaving $\gamma_{\mathrm{KH}}^{-1}$ well below the formation window estimated below. Using an observed cosmic-flow shear $\Delta V \simeq 600\ \mathrm{km\,s^{-1}}$, one obtains e-folding times

$$t_{\mathrm{e-fold}} = \gamma_{\mathrm{KH}}^{-1} \simeq \begin{cases} 5.2\ \mathrm{Myr}, & \lambda = 10\ \mathrm{kpc}, \\ 51.9\ \mathrm{Myr}, & \lambda = 100\ \mathrm{kpc}, \\ 0.52\ \mathrm{Gyr}, & \lambda = 1\ \mathrm{Mpc}. \end{cases} \tag{5}$$

Comparing with the proper-time-weighted formation window of Sec. 3.5 (using the full quantum-feedback $\mathcal{C}(z)$ profile, not the Appendix 18 power-law benchmark), one finds $\tau_{\mathrm{form}} \simeq 19.1$ Gyr for a conservative late formation epoch $z_f = 2$ and $\tau_{\mathrm{form}} \simeq 21.1$ Gyr for $z_f = 10$. Therefore even the *largest* 1 Mpc driving scale undergoes about 37–41 e-foldings, while a 100 kpc perturbation undergoes about 3.7×10^2–4.1×10^2 e-foldings. The instability has ample time to roll up the flow into vortical structure.

The Reynolds and Kolmogorov estimates that follow use the persistence-bound viscosity ceiling (12); they are therefore consistency checks under the persistence closure rather than independent derivations of the medium's transport coefficients. The turbulent cascade is likewise compatible with galactic scales. Using the conservative viscosity ceiling from the persistence bound (12), $\nu \lesssim 2 \times 10^{23}\ \mathrm{m^2\,s^{-1}}$, with a driving scale $L_{\mathrm{dr}} \sim 1$ Mpc and the same shear velocity gives a Reynolds number

$$\mathrm{Re}_{\min} \sim \frac{L_{\mathrm{dr}}\Delta V}{\nu} \gtrsim 9 \times 10^4. \tag{6}$$

Hence the Kolmogorov cutoff scale is at most

$$\ell_{\mathrm{diss}} \sim L_{\mathrm{dr}}\,\mathrm{Re}^{-3/4} \lesssim 0.2\ \mathrm{kpc}, \tag{7}$$

so the full 10–100 kpc galactic range lies safely inside the inertial range. In other words, under the adopted low-viscosity / persistence-bound closure, once cosmological flows are identified with motion of the pressure-carrying substance, the generation of galactic-scale vortices is a quantitatively supported expectation.

The space substance therefore develops a spectrum of vortices at all scales. The galactic-scale vortices are the ones relevant for the MOND sector.

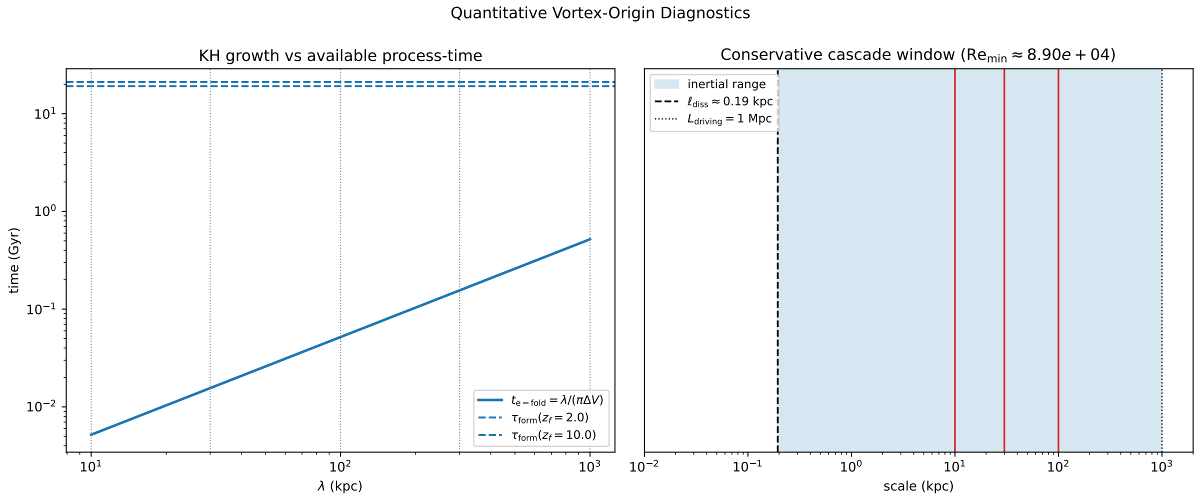

Figure 1: Fiducial KH timescale and inertial-range diagnostics from `phase_ao_kh_vortex_origin.py` under the adopted shear-layer / persistence-bound closure. Left: the Kelvin–Helmholtz e-fold time stays far below the proper-time-weighted formation window across the 10–1000 kpc range. Right: with the persistence viscosity ceiling (12), the dissipation scale lies far below galactic scales, leaving the full 10–100 kpc band inside the inertial range. The figure illustrates timescale and inertial-range plausibility under these assumptions, not an independent derivation of the medium's transport coefficients.

3.3 Galaxies form inside pre-existing vortices

The causal sequence in this framework is:

$$\boxed{\begin{array}{c}\text{CMB fluctuations} \;\to\; \text{inhomogeneous cooling} \;\to\; \text{pressure gradients} \\ \to\; \text{flows} \;\to\; \text{vortices} \;\to\; \text{galaxies.}\end{array}} \tag{8}$$

Each row carries an explicit status under the ISPG identification of cosmic flows with bulk motion of the pressure-carrying substance:

Step	**Status**
CMB fluctuations exist	observed (COBE/WMAP/Planck)
Inhomogeneous cooling → pressure gradients	thermodynamics applied to the ISPG medium
Pressure gradients → flows	fluid mechanics
Cosmic flows exist	observed (Laniakea, cosmic web)
Flows as space-substance motion	ISPG identification
Flows → vortices	KH timescale support under low-viscosity closure
Vortices trap matter	fluid-mechanical consequence under the same closure

The vortex is *not created by* the galaxy; the galaxy forms *inside* a pre-existing vortex of the space substance. In this picture galactic rotation is inherited from the medium, not generated *ab initio* by angular momentum conservation of collapsing gas (though the latter supplements it).

3.4 Galaxy morphology as fluid-dynamical regime

Different flow conditions in the primordial substance produce different vortex topologies; these provide a qualitative morphology map onto the Hubble sequence:

Fluid regime	Galaxy type	Stellar kinematics
Laminar planar vortex	spiral	ordered co-rotation
Convective cell (poloidal circulation)	elliptical	random ($\langle \mathbf{v} \rangle \approx 0$, isotropic σ)
Small-scale eddy	dwarf	low-mass vortex
Chaotic multi-stream zone	irregular	disordered

In *all* cases the vortex creates a central pressure deficit (rarefaction), producing the extra inward gravitational gradient that is the MOND effect. The specific morphology determines the *geometry* of the extra support but not its *existence*. A quantitative morphology-population prediction and the kinematic distinguishability of these regimes remain a later simulation/closure task.

3.5 Enhanced formation time from the global pressure history

The effective process-time available for vortex formation can exceed the standard coordinate age $t_0 \simeq 13.8$ Gyr measured by present-day clocks. In the ISPG pressure ontology, higher past pressure means faster past clocks (Sec. 5):

$$\left.\frac{d\tau}{dt}\right|_{\text{past}} = e^{\varphi_{\text{bg}}(z)/2} \gg \left.\frac{d\tau}{dt}\right|_{\text{today}}. \tag{9}$$

The cumulative process-time available for formation from redshift z_f to the present is

$$\tau_{\text{form}}(z_f) = \int_0^{z_f} \left[\frac{P_{\text{stat}}(z)}{P_{\text{stat}}(0)}\right]^{1/2} \frac{dz}{(1+z)\,H(z)}, \tag{10}$$

where the pressure ratio is supplied by the quantum feedback closure (Eq. 198). The coordinate lookback time is the special case with the pressure factor set to unity. For the fiducial quantum-feedback solution ($z_f = 10$, $\alpha = 1$, $K \approx 8$), using the full quantum-feedback $\mathcal{C}(z)$ profile rather than the Appendix 18 power-law benchmark, one obtains $\tau_{\text{form}}(10) \simeq 21.1$ Gyr versus a coordinate lookback $t_{\text{lookback}}(10) \simeq 13.3$ Gyr; for a conservative late formation epoch $z_f = 2$ the result is $\tau_{\text{form}}(2) \simeq 19.1$ Gyr.

The vortices therefore had effectively *much longer* to develop and mature than the coordinate age of the universe suggests. The same process-time bookkeeping can reduce the required intrinsic development time for high-redshift systems whose relevant rates are genuinely process-time controlled. A quantitative JWST age claim, however, requires an explicit map from stellar-evolution, star-formation, and assembly rates to the appropriate time variable.

3.6 Vortex persistence: why the vortex does not dissipate

A natural objection is: what prevents the galactic vortex from dissipating through friction with the matter it contains? The answer involves two dissipation channels, both of which are quantitatively negligible at galactic scales.

Channel 1: oscillon–vortex coupling (mutual friction). Matter consists of oscillons—standing-wave resonances *of* the space substance, as developed in the companion quantum paper [14]. Unlike a solid body immersed in a fluid, an oscillon has no sharp boundary at which momentum is exchanged by contact forces: the oscillon *is* the medium in a vibrationally excited state, and the dominant dissipation channel that drains a laboratory vortex (solid-wall friction) is therefore absent.

The remaining coupling is the mutual-friction channel familiar from superfluid hydrodynamics: in the HVBK framework [1, 2] vortex dissipation proceeds through momentum exchange with a normal-fluid component of density ρ_n, with friction force $\mathbf{F} \propto \rho_n\, \mathbf{\Omega} \times (\mathbf{v}_s - \mathbf{v}_n)$. In the ISPG ground state the space substance is superfluid (Channel 2 below uses the same assumption), so $\rho_n \to 0$ at zero temperature and the mutual-friction coupling vanishes together with it. The oscillon content of a galaxy is not a thermal normal-fluid reservoir but a set of localised standing-wave excitations; it does not activate the mutual-friction channel. Quantitatively, the per-oscillon momentum transfer to a locally uniform vortex flow is of order $m_{\rm osc}\, v_\varphi$, but this transfer is elastic (the oscillon is carried along with the flow rather than dragging against it) and therefore does not produce secular dissipation.

Channel 2: bulk viscous dissipation. The second dissipation channel is the medium's own internal (bulk) viscosity, independent of the oscillon content. The viscous dissipation timescale for a vortex of radius R in a medium with kinematic viscosity ν is

$$\tau_{\rm visc} \sim \frac{R^2}{\nu}\,. \tag{11}$$

For a galactic vortex with $R \sim 10$ kpc $\approx 3 \times 10^{20}$ m, the requirement $\tau_{\rm visc} > t_H$ ($t_H \approx 4 \times 10^{17}$ s) yields the upper bound

$$\nu \lesssim \frac{R^2}{t_H} \approx 2 \times 10^{23}\ \mathrm{m^2\,s^{-1}}\,. \tag{12}$$

Any medium with $\nu \ll 2 \times 10^{23}\ \mathrm{m^2\,s^{-1}}$ satisfies this bound. If the space substance is superfluid ($\nu \to 0$), as assumed in the superfluid vacuum programme [1], the vortex is permanent: persistent macroscopic flows are a defining property of the ground state.

Summary: continuous driving, not relic survival. The primary reason for vortex persistence is not the smallness of dissipation but the fact that the vortex is *continuously driven.* The same cosmological flows that created the vortex (Sec. 3.2)—observed today as Laniakea-scale bulk flows, cosmic-web filamentary infall, and galaxy-cluster tidal streams—continue to feed angular momentum into the galactic vortex cell. As long as the large-scale flow persists, so does the vortex it sustains.

The dissipation estimates above provide a secondary confirmation: even in the hypothetical absence of continuous driving, both channels are negligible—mutual-friction coupling of the oscillon content to the vortex flow is shut down in the superfluid ground state ($\rho_n \to 0$), and bulk viscous decay requires the superfluid ground state to have macroscopic viscosity, which contradicts the defining property of superfluidity. Together, the continuous cosmological driving and the negligible dissipation rate make vortex persistence a robust consequence of the framework.

3.7 Minimal local equilibrium model: Rankine-type vortex cell

Before turning to the scalar-field closure, it is useful to exhibit a minimal local equilibrium model showing that a mature vortex cell *necessarily* carries a central pressure deficit. The simplest such model is the Rankine vortex: solid-body rotation inside a core and circulation-dominated flow outside it,

$$v_\varphi(r) = \begin{cases} \Omega_c\, r, & r \le R_c, \\ \Omega_c\, R_c^2/r, & r > R_c, \end{cases} \tag{13}$$

where R_c is the core radius and Ω_c is the saturated angular speed of the local vortex cell.

For a slowly varying density ρ and cylindrical symmetry, radial force balance in the rotating substance gives

$$\frac{dP}{dr} = \rho\, \frac{v_\varphi^2}{r}. \tag{14}$$

Integrating (14) inside and outside the core yields

$$P(r) - P(0) = \frac{1}{2}\,\rho\,\Omega_c^2\,r^2, \qquad r \le R_c, \tag{15}$$

$$P(r) - P(R_c) = \frac{1}{2}\,\rho\,\Omega_c^2\,R_c^2\left(1 - \frac{R_c^2}{r^2}\right), \qquad r > R_c. \tag{16}$$

Hence the centre is always the point of minimum pressure, with total centre-to-exterior deficit

$$\Delta P_{\text{cent}} \equiv P(\infty) - P(0) = \rho\,\Omega_c^2\,R_c^2. \tag{17}$$

The corresponding inward pressure-gradient support has magnitude

$$g_h(r) = \frac{1}{\rho}\frac{dP}{dr} = \frac{v_\varphi^2(r)}{r} = \begin{cases} \Omega_c^2\,r, & r \le R_c, \\ \Omega_c^2\,R_c^4/r^3, & r > R_c. \end{cases} \tag{18}$$

Thus a mature vortex cell generically produces:

- a *central rarefaction* (pressure minimum at the centre),
- an *inward support* $g_h = v_\varphi^2/r$ set by vortex equilibrium,
- and a *single saturation scale* (Ω_c, R_c) controlling the whole cell.

At the same time, a *fixed-core* Rankine profile is not yet the full MOND closure: outside the core it gives $g_h \propto r^{-3}$, whereas the deep-MOND point-mass tail scales as r^{-1}. The Rankine cell therefore proves the existence of equilibrium rarefaction and inward support, but not the final global radial law.

This minimal model does *not* by itself fix the normalization Ω_c: that is precisely the remaining closure problem isolated later in Eq. (103). But it shows that the extra MOND-like support is not an ad hoc correction; it is the generic pressure-gradient signature of a saturated vortex equilibrium.

With the cosmological origin and persistence of galactic vortices established, the remainder of this companion paper derives the *quantitative* MOND closure inside a mature vortex cell.

4 Scalar perturbations on FLRW: the damped-oscillator equation

In the main theory construction, the scalar perturbation behaves as a driven, Hubble-damped oscillator mode. A representative mode equation is

$$\delta\ddot{\varphi}_k + 3H\,\delta\dot{\varphi}_k + \omega_k^2\,\delta\varphi_k = \frac{8\pi G}{c^2}\,\delta T_k, \qquad \omega_k \equiv \frac{ck}{a}. \tag{19}$$

The RHS coefficient $8\pi G/c^2$ follows from the FLRW expansion of the sourced scalar equation $\Box\varphi = -(8\pi G/c^4)T$ of the main theory [13]: writing $\Box\varphi = -c^{-2}(\ddot{\varphi} + 3H\dot{\varphi}) + a^{-2}\nabla^2\varphi$ and multiplying by $-c^2$ yields the Fourier-space form above. The physical content is the competition between oscillation and Hubble friction. Define the Hubble damping rate

$$\gamma_H \equiv \frac{3H}{2}. \tag{20}$$

The critical wavenumber is set by $\omega_k = \gamma_H$, giving

$$k_J = \frac{3aH}{2c}, \qquad \lambda_J \equiv \frac{2\pi a}{k_J} = \frac{4\pi c}{3H}. \tag{21}$$

This is an order-Hubble scale. Modes with $\lambda \ll \lambda_J$ lie in the short-wavelength oscillatory branch of the FLRW equation, while the transition itself lies near the Hubble boundary rather than deep inside the WKB range. In the MOND closure below, this scale is used as the cosmological coherence-boundary input; it is not a direct Hubble-friction damping proof for kpc-scale galactic modes.

5 From coherence length to the MOND acceleration scale

5.1 Overdamped vs. underdamped classification

The homogeneous damped oscillator (19) is best read as a three-regime classifier:

- **Deep WKB / short-wavelength** ($kc/a \gg H$, hence $Q = \omega_k/\gamma_H \gg 1$): the mode oscillates with amplitude decay rate γ_H. The source-free classification is parametrically controlled here, and the field can sustain a local coherent gradient at the corresponding wavelength.
- **Near-Hubble / order-Hubble boundary** ($kc/a \sim H$, hence $Q = O(1)$): the homogeneous oscillator crosses toward the overdamped branch. Within this homogeneous classifier its approach to equilibrium is controlled by the slow eigenrate

$$\alpha_- = \frac{3H}{2} - \sqrt{\frac{9H^2}{4} - \omega_k^2} \xrightarrow{\omega_k \ll H} \frac{\omega_k^2}{3H}, \tag{22}$$

  giving a formal response timescale $\tau_{\text{resp}} = 1/\alpha_- \approx 3H/\omega_k^2$ that exceeds the Hubble time for $\omega_k^2 < 3H^2$. This is a near-boundary coherence diagnostic, not a direct local damping law for galaxy-scale perturbations.
- **Frozen** ($kc/a \lesssim H$, super-Hubble): the adiabatic classification breaks down; the mode is frozen on cosmological scales rather than relaxing exponentially, as is standard for cosmological perturbations outside the Hubble radius.

The transition between the first two regimes defines a characteristic wavenumber k_J (Eq. (21)) and a corresponding Hubble-damping wavelength $\lambda_J = 4\pi c/(3H)$, both of order c/H. (The symbol λ_J is retained for cross-reference continuity; the scale is set by Hubble damping, not by the pressure-gravity balance of the astrophysical Jeans criterion.)

5.2 Robustness of $a_0 \propto cH$

The acceleration corresponding to a spatial gradient of length ℓ is $a(\ell) = c^2/\ell$. The FLRW oscillator supplies an order-Hubble coherence scale, and the bound-structure closure uses gradients with ℓ exceeding that scale as outside the largest coherent scalar cell. This is a coherence-boundary identification rather than direct FLRW damping of local kpc-scale gradients; the scale is of order c/H regardless of the precise criterion used:

Criterion	Coherence length	Implied a_0
$\omega_k = \gamma_H$ (Hubble damping)	$4\pi c/(3H)$	$3cH/(4\pi)$
$\tau_{\text{resp}} < H^{-1}$	$2\pi c/(\sqrt{3}\,H)$	$\sqrt{3}\,cH/(2\pi)$
$k = k_H$ (Hubble)	$2\pi c/H$	$cH/(2\pi)$

All three give $a_0 = \mathcal{C}\,cH$ with $\mathcal{C}$ an $O(1)$ coefficient. The scaling $a_0 \propto cH$ is therefore *robust*: it follows from any reasonable order-Hubble coherence-boundary demarcation.

5.3 Fixing the coefficient: the Fourier identification

Among the three criteria above, the Hubble wavelength

$$\lambda_H \equiv \frac{2\pi a}{k_H} = \frac{2\pi c}{H} \tag{23}$$

(with $k_H \equiv aH/c$ the comoving Hubble wavenumber) is the fundamental cosmological scale that separates sub-horizon from super-horizon modes. The scalar field can sustain a coherent gradient over length ℓ only if $\ell \leq \lambda_H$; beyond λ_H the mode is super-Hubble and its evolution is frozen.

The critical acceleration at which the required gradient length equals the Hubble wavelength is:

$$\frac{c^2}{a_0} = \lambda_H \quad \Longrightarrow \quad \boxed{a_0 = \frac{c^2}{\lambda_H} = \frac{cH}{2\pi}\,.} \tag{24}$$

Given the Hubble-wavelength identification, the factor 2π follows from the standard Fourier relation $k = 2\pi/\lambda$ between wavenumber and wavelength; the wavelength-vs-radius choice itself is a physical modeling decision, justified by the coherence structure of the scalar medium and made consistent with the vortex-lag direction constraint below.

Direction constraint from the vortex-lag mechanism. Among the three criteria of the preceding subsection, only the Hubble-wavelength criterion yields an instantaneous value $a_0^{\rm inst} = cH_0/(2\pi) \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ *below* the observed $a_0^{\rm (obs)} \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The vortex-lag mechanism of Sec. 5.5 enhances $a_0^{\rm inst}$ via past-weighted averaging $H_{\rm eff} > H_0$, and can therefore close the gap only for criteria that *underpredict* the data. This provides a second, mechanism-level consistency filter complementary to the Fourier identification: the two overpredicting criteria ($3cH/(4\pi)$ and $\sqrt{3}\,cH/(2\pi)$) would require a lag correction of the wrong sign and are inconsistent with the vortex-lag closure.

5.4 Numerical check and redshift evolution

At $z = 0$ with $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$:

$$a_0 = \frac{cH_0}{2\pi} = \frac{3.0 \times 10^8 \times 2.18 \times 10^{-18}}{2\pi} \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}\,. \tag{25}$$

Here $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is the fiducial Planck/CMB baseline value adopted in `MOND/Python/constants.py`. Since $a_0 \propto H_0$, using a local-distance-ladder value near $73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ would rescale the instantaneous estimate upward by about 8%. The observed value $a_0^{\rm (obs)} \approx 1.2 \pm 0.3 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ (McGaugh 2011) confirms that the instantaneous formula lands in the correct order of magnitude. However, for a parameter-free derivation the residual $\sim 15\%$ offset still requires a physical explanation rather than being left as a tolerated mismatch.

5.5 Calibrated vortex-lag closure of the 15% offset

The instantaneous formula $a_0 = cH_0/(2\pi) \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ under-predicts the observed value $a_0^{\rm (obs)} \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ by $\approx 15\%$. The vortex-lag mechanism below provides a calibrated closure of this offset: the $H_{\rm eff}$ identity is algebraic for the adopted uniform coordinate-time memory prescription, while the value $z_{\rm form} \approx 0.58$ is calibrated to the local observed a_0. The independent prediction is therefore not the numerical match itself, but the induced age dependence of a_0 across galaxies.

Origin of the correction. The derivation leading to Eq. (24) assumes the coherence cell is in instantaneous equilibrium with the current H. However, the vortex maturity ODE (Eq. (66)) contains the forgetting rate $\Gamma_{\rm mem} = \Omega_{\rm tr} = a_0/c$, which sets a memory timescale

$$\tau_{\rm mem} = \frac{1}{\Gamma_{\rm mem}} = \frac{c}{a_0} = \frac{2\pi}{H_0} \approx 91 \text{ Gyr}\,. \tag{26}$$

Since $\tau_{\rm mem} \gg t_0 \approx 13.8$ Gyr, the vortex has never reached the stationary limit: it retains memory of *all* past pressure epochs since its formation. The effective Hubble parameter governing the vortex is therefore the time-averaged rate over its lifetime,

$$H_{\rm eff} = \frac{1}{t_0 - t_{\rm form}} \int_{t_{\rm form}}^{t_0} H(t)\,dt = \frac{\ln(1+z_{\rm form})}{t_{\rm lookback}(z_{\rm form})}\,, \tag{27}$$

where the second equality follows from the change of variable $dt = dz/[(1+z)H]$ and the identity $\int H\,dt = \int dz/(1+z)$. This identity is exact for the chosen uniform coordinate-time average; the choice of memory kernel is part of the H.3 closure rather than an independent theorem.

Process-time refinement. Equation (27) averages $H(t)$ with uniform weight in coordinate time. If the vortex maturity dynamics are controlled by matter-sector process rates, the past pressure epochs should receive process-clock weighting (Sec. 3.5):

$$H_{\rm eff,proc} = \frac{\displaystyle\int_{t_{\rm form}}^{t_0} H(t)\,\mathcal{C}(t)\,dt}{\displaystyle\int_{t_{\rm form}}^{t_0} \mathcal{C}(t)\,dt}\,, \qquad \mathcal{C}(t) \equiv \left[\frac{P_{\rm stat}(t)}{P_{\rm stat}(0)}\right]^{1/2}. \tag{28}$$

For $z_{\rm form} \sim 0.5$–0.6, the moderate-redshift pressure ratio $\mathcal{C} \lesssim 1.2$ makes the difference between (27) and (28) sub-leading ($\lesssim 3\%$). The coordinate-time average is therefore retained in the main analysis; the process-time version becomes significant only for high-z formation ($z_{\rm form} \gtrsim 2$). The quantities $H_{\rm eff}$, $H_{\rm eff,proc}$, and the cumulative process-age factor $\mathcal{A} = \tau_{\rm proc}/t_{\rm lookback}$ are distinct diagnostics and should not be substituted for one another. The memory prescription is load-bearing: the exponential-memory sensitivity check in `MOND/Python/heff_calc.py` gives a much larger effective value, so the uniform-lifetime average is retained as the calibrated H.3 prescription. A first-principles derivation of the memory kernel is a separate dynamical-refinement problem.

Numerical evaluation. For the standard ΛCDM observational background used in `MOND/Python/heff_calc.py` ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, giving $t_0 \approx 13.8\,\mathrm{Gyr}$):

$z_{\rm form}$	$t_{\rm form}$ (Gyr)	$H_{\rm eff}/H_0$	$a_0^{\rm eff}$ (10^{-10} m s^{-2})	Note
0.3	10.3	1.06	1.10	recent disk settling
0.5	8.5	1.13	1.18	last major merger
0.58	**8.0**	**1.15**	**1.20**	**calibrated to $a_0^{\rm (obs)}$**
1.0	5.8	1.26	1.32	early disk formation
1.5	4.2	1.39	1.45	proto-disk

Within this standard-ΛCDM calibration, the observed $a_0^{\rm (obs)} \approx 1.2$ corresponds to $z_{\rm form} \approx 0.58$, i.e. the vortex structure dates from ∼5.8 Gyr ago. This value lies in a plausible late disk-settling / last-major-merger window for local spiral morphology [4, 5], but it is not an independent prediction. Baryons-only ISPG redshift-evolution estimates elsewhere in this paper use a different background; a unified H.3 recalibration in that background is deferred to a separate cosmology-consistency sweep.

Physical interpretation. In ISPG language: the global pressure of the space substance has been decreasing (H falling) throughout cosmic history. A galactic vortex "imprints" the pressure conditions throughout its entire lifetime because $\tau_{\mathrm{mem}} \gg t_0$—it cannot forget. Galaxies whose vortices stabilised earlier carry a higher time-averaged H, and therefore exhibit a higher effective a_0. The instantaneous formula $a_0 = cH_0/(2\pi)$ is the correct *asymptotic* result for a vortex in a static universe; the 15% correction is the finite-age transient.

Falsifiable prediction: age-dependent a_0. This calibrated closure generates a unique, testable prediction absent from any other MOND formulation:

$$a_0^{\mathrm{eff}}(\text{galaxy}) = \frac{c}{2\pi}\,\frac{\ln(1+z_{\mathrm{form}})}{t_{\mathrm{lookback}}(z_{\mathrm{form}})}\,. \tag{29}$$

At fixed redshift of observation, galaxies with older stabilised disks should show systematically *higher* a_0 (tighter rotation curves at given baryonic mass) than galaxies whose disks settled more recently. Concretely:

- A spiral whose disk settled at $z_{\mathrm{form}} \sim 1$ predicts $a_0^{\mathrm{eff}} \approx 1.32$; one that settled at $z_{\mathrm{form}} \sim 0.3$ predicts $a_0^{\mathrm{eff}} \approx 1.10$ (both in units of $10^{-10}\ \mathrm{m\,s^{-2}}$).
- The BTFR scatter should correlate with disk-settling epoch: older disks above the mean relation, younger disks below.
- The calibrated $z_{\mathrm{form}} \approx 0.58$ lies in a plausible late disk-settling window for local spirals, qualitatively consistent with the observed median $a_0 \approx 1.2$. A quantitative verification of the induced dispersion $\sigma(\log v_{\mathrm{flat}}) = \frac{1}{4}\sigma(\log a_0) = \frac{1}{4}\sigma(\log H_{\mathrm{eff}})$ against SPARC residuals and independent disk-age proxies is deferred to a dedicated analysis.

Redshift evolution. The effective a_0 for a galaxy observed at redshift z_{obs} whose vortex formed at $z_{\mathrm{form}} > z_{\mathrm{obs}}$ is

$$a_0^{\mathrm{eff}}(z_{\mathrm{obs}}, z_{\mathrm{form}}) = \frac{c}{2\pi}\,\frac{\ln\!\big[(1+z_{\mathrm{form}})/(1+z_{\mathrm{obs}})\big]}{t(z_{\mathrm{obs}}) - t(z_{\mathrm{form}})}\,, \tag{30}$$

which reduces to $cH_0/(2\pi)$ only in the limit $z_{\mathrm{form}} \to z_{\mathrm{obs}}$ (newly born vortex). The ratio

$$\frac{a_0^{\mathrm{eff}}(z_{\mathrm{obs}})}{a_0^{\mathrm{eff}}(0)} \tag{31}$$

now depends on both the observation epoch *and* the formation epoch, providing a two-parameter test surface rather than a single curve. DESI/Euclid-era rotation-curve surveys combined with morphological age indicators can map this surface.

Existing observational hints. Several data sets are suggestive but not yet decisive tests of the vortex-lag prediction:

1. **MIGHTEE-HI (2025).** Vărăşteanu et al. [6] report a tight radial-acceleration relation in 19 H I-selected galaxies at $z \le 0.08$ and tentative (2.4σ) evidence for redshift evolution in the acceleration scale. This is suggestive for H.3, but it is not yet a confirmation; the authors stress that larger, higher-redshift samples are required.
2. **Galaxy-to-galaxy a_0 scatter (Rodrigues et al. 2018).** A Bayesian analysis of 193 SPARC/THINGS galaxies rejected a universal a_0 at $> 10\sigma$ [7]. Li & McGaugh [8] contested this on methodological grounds, and the debate remains open [9]. In the present framework such scatter would be *physical*: it would encode the distribution of disk-settling epochs z_{form} across the sample. This remains a testable interpretation rather than a confirmed observational result.

3. **Best-fit a_0 value.** Desmond (2023) [10] obtains $a_0 = (1.19{\pm}0.04_{\rm stat}{\pm}0.09_{\rm sys}){\times}10^{-10}$ m s^{-2} from the SPARC RAR. The vortex-lag formula yields $a_0^{\rm eff} = 1.20 \times 10^{-10}$ for $z_{\rm form} \approx 0.58$; the numerical match is by construction of the calibration and is not an independent prediction. The non-trivial consistency check is that the calibrated value $z_{\rm form} \approx 0.58$ falls within a plausible late disk-settling / last-major-merger window for local-spiral disk morphology [4, 5]. The genuinely falsifiable content is the age-correlation prediction (Eq. (29)), tested by the following items.

4. **High-redshift constraint.** Milgrom [11] excludes $a_0(z{\sim}2) \gtrsim 4\,a_0(0)$ from six galaxies at $z \sim 0.9$–2.4. The vortex-lag formula predicts at most $a_0^{\rm eff}/a_0(0) \approx 1.3$–$1.5$ at these redshifts, well within the bound.

A decisive test requires correlating the SPARC per-galaxy a_0 residuals with independent disk-age proxies (stellar-population colour gradients, merger-history indicators, or vertical scale-height ratios). This analysis can be carried out with existing data and constitutes a near-term observational programme.

6 Interpolating function: derivation from rotating-space dynamics

6.1 Physical setup: scalar field in a rotating galaxy

In a spiral galaxy there are two gravitational channels from the outset. First, the baryonic resonant tails produce the familiar Bernoulli pressure deficit around matter; this is the local Newtonian channel. In the companion quantum paper this local channel is closed at the microscopic source level: the localized oscillon source, its zero-frequency Bernoulli component, and the exterior $1/r$ field are carried by the same collective coordinate, while non-neutral source deformations remain perturbatively controlled. Second, the galactic rotation stirs the space substance itself into a vortex, whose centrifugal pressure redistribution creates the transported channel. MOND is the closure of these two channels in a bound rotating structure. The observed excess support of spiral-galaxy rotation curves is therefore read in ISPG as the direct observational signature of this second channel, not as an external dark component and not as a merely phenomenological correction. Within ISPG, this rotation-curve support is the direct signature of the second channel; ruling out alternative MOND/dark-sector explanations requires the full ISPG signature programme (Sec. 18.10). In this interpretation, space behaves as a gravitating substance whose pressure and transport state is encoded in the effective gravitational field: rotation rearranges the medium, produces a central rarefaction, and the resulting transported channel contributes a genuine second metric/geodesic component to the gravitational response. The mechanism is self-regulating: faster coherent galactic rotation deepens the rarefaction and strengthens the transported channel only until local vortex equilibrium fixes the MOND branch.

Accordingly, consider a spiral galaxy with baryonic mass distribution producing Newtonian potential Φ_N and rotating with characteristic angular velocity Ω. In the weak-field limit, the metric includes the Lense–Thirring (frame-dragging) term from the galaxy's internal angular momentum distribution:

$$ds^2 = -(1+2\Phi_N)c^2dt^2 + (1-2\Phi_N)(dx^2+dy^2+dz^2) - \frac{2G}{c^2}\frac{(\mathbf{J}\times\mathbf{r})\cdot d\mathbf{x}\,dt}{r^3}, \tag{32}$$

where $\mathbf{J}$ is the angular momentum density. The off-diagonal g_{ti} components induce a frame-dragging angular velocity:

$$\boldsymbol{\omega}_{\rm FD} = \frac{2G}{c^2}\frac{\mathbf{J}}{r^3}. \tag{33}$$

The scalar field equation $\Box\varphi = S$ in this background, expanded to leading order in the weak-field and slow-rotation limits, acquires additional terms from the non-diagonal metric components.

6.2 Co-rotating frame and the advection operator

For a test particle or field mode co-moving with the galactic rotation at radius r with angular velocity $\Omega(r) = v_\varphi/r$, the natural time derivative is the material derivative in the rotating frame:

$$\frac{D}{Dt} \equiv \partial_t + \mathbf{v}\cdot\nabla = \partial_t + \Omega\,\partial_\varphi, \tag{34}$$

where $\mathbf{v} = \Omega \times \mathbf{r}$ is the circular velocity and ∂_φ is the azimuthal derivative. For quasi-stationary configurations ($\partial_t \simeq 0$), this reduces to pure azimuthal advection.

6.3 Two-channel decomposition from azimuthal symmetry

In an axisymmetric rotating system, the scalar perturbation naturally decomposes into:

- **Local channel** φ_N: the quasi-static Bernoulli response sourced by the baryonic resonant tails, satisfying $-c^2\nabla^2\varphi_N = S$ (Poisson-like).
- **Transported channel** φ_h: the rotating-space vortex response, generated by azimuthal advection and centrifugal pressure redistribution in the stirred medium.

The total field is $\delta\varphi = \varphi_N + \varphi_h$. With the main-theory normalization $\Phi = (c^2/2)\varphi$, the physical test-particle acceleration is the geodesic acceleration:

$$g = g_N + g_h, \qquad g_N = -\frac{c^2}{2}\nabla\varphi_N, \quad g_h = -\frac{c^2}{2}\nabla\varphi_h. \tag{35}$$

Proposition 0 (spiral-galaxy acceleration-channel split). For a bound rotating spiral galaxy, the total acceleration is the sum of two metric/geodesic channels encoded by the same medium:

$$g = g_N + g_h. \tag{36}$$

Here g_N is the local Bernoulli channel sourced by the baryonic resonant tails, while g_h is the vortex channel sourced by the centrifugal rarefaction of the rotating space substance. The local MOND law is the closure of these two channels; the only cosmological input entering that local law is the background coherence scale $a_0 = cH/(2\pi)$, with H interpreted throughout as the observational proxy for the global static-pressure history. Thus the spiral-galaxy anomaly is not an unexplained residual in this framework: it is the empirical registration of the rotating medium's own additional gravitational channel. In that sense it directly supports both the substance character of space and the pressure-deficit and transport origin of the metric response in the ISPG medium, without introducing a separate pressure-gradient force law for test matter.

Corollary (background inheritance). The same homogeneous background shift that fixes a_0 also rescales the basic operational standards of the theory:

$$\frac{\mathcal{N}_{\rm act,bg}(z)}{\mathcal{N}_{\rm act,bg}(0)} = e^{[\varphi_{\rm bg}(z)-\varphi_{\rm bg}(0)]/2}, \qquad d\tau = e^{\varphi_{\rm bg}/2}dt, \qquad d\ell = e^{-\varphi_{\rm bg}/2}d\sigma, \qquad m_{\rm eff} \propto e^{\varphi_{\rm bg}/2}. \tag{37}$$

Thus global pressure decrease changes clock rate, rod scale, and mass scale together, while the local MOND closure remains the instantaneous equilibrium of the two galactic channels in (36). The factor $\mathcal{N}_{\rm act,bg}$ is only an operational reading of the same pressure-history multiplier: it is not an independent node-density count, not a new observable, and not an additional force law for test matter.

6.4 Transport equation: Fourier decomposition method

To derive the transport equation rigorously, we decompose the scalar field in cylindrical coordinates (r, φ, z) using azimuthal Fourier modes:

$$\varphi_h(r, \varphi, z, t) = \sum_{m=-\infty}^{\infty} \varphi_m(r, z, t)\, e^{im\varphi}. \tag{38}$$

In the co-rotating frame where the field is quasi-stationary, the $m = 0$ mode (azimuthal average) dominates the long-time behavior. Higher modes ($m \neq 0$) oscillate rapidly and average to zero in the steady-state limit.

Equation for the $m = 0$ mode. Substituting (38) into the scalar field equation (19) and projecting onto the $m = 0$ component:

$$3H\, \partial_t\varphi_0 - c^2 \left(\frac{1}{r}\partial_r(r\partial_r) + \partial_z^2\right) \varphi_0 = -\langle 2\boldsymbol{\omega}_{\rm FD} \cdot (\nabla\delta\varphi \times \hat{\mathbf{r}})\rangle_\varphi, \tag{39}$$

where $\langle\cdot\rangle_\varphi = (2\pi)^{-1} \int_0^{2\pi} d\varphi$ denotes azimuthal averaging. The frame-dragging term, when averaged over azimuth, produces a source proportional to φ_N. *Note*: the detailed azimuthal averaging that yields the effective source magnitude (the "beating" of the $m = \pm1$ response modes with the frame-dragging field) is computed in Eqs. (50)–(52); the intermediate steps involve integrals over the spiral structure that depend on the assumed Keplerian profile and spin parameter ξ, and are therefore order-of-magnitude rather than exact. This point matters for the PDE-to-closure programme below: a strict single-phase WKB ansatz applied directly to the pure $m = 0$ mode would eliminate the frame-dragging coupling. The correct asymptotic treatment must therefore retain the $m = \pm1$ sidebands and derive the slowly varying $m = 0$ envelope only after their beat with the rotating background has been formed.

Damping of $m \neq 0$ modes. For the non-axisymmetric modes ($m \neq 0$), the azimuthal derivative $\partial_\varphi^2 \to -m^2$ introduces an additional term in the effective radial equation:

$$3H\, \partial_t\varphi_m - c^2 \left(\frac{1}{r}\partial_r(r\partial_r) - \frac{m^2}{r^2} + \partial_z^2\right) \varphi_m = (\text{source})_m. \tag{40}$$

The $-m^2/r^2$ term acts as a positive effective potential barrier that suppresses these modes relative to $m = 0$. The radial eigenvalue for mode m becomes $k_r^{(m)} > k_r^{(0)}$, giving a shorter Bessel crossing/spatial-adjustment time $t_B^{(m)} = 1/(c\, k_r^{(m)}) < t_B^{(0)}$.

Furthermore, the Hubble damping term $3H\partial_t$ causes exponential decay of time-dependent modes on the Hubble timescale $t_H = 1/H$. For $m \neq 0$ modes, the faster oscillation ($\omega \sim m\Omega \gg H$ for $m \gtrsim 1$) combined with Hubble damping gives rapid suppression:

$$|\varphi_m(t)| \sim |\varphi_m(0)| \exp(-3Ht/2) \cos(m\Omega t) \quad \Rightarrow \quad \langle|\varphi_m|^2\rangle_t \sim |\varphi_m(0)|^2 e^{-3Ht}. \tag{41}$$

Source-free non-axisymmetric transients decay under Hubble damping at rate $\sim 3H/2$, so over several Hubble times such free modes are exponentially suppressed relative to the $m = 0$ envelope. Continuously driven spiral sidebands ($m = \pm1$) are not source-free; they feed the averaged $m = 0$ envelope through the sideband-beat mechanism developed in the mode-mixed WKB reduction below. The focus on φ_0 here is therefore on the averaged $m = 0$ mode supplied by sideband beating, not on a trivial absence of $m \neq 0$ structure.

Radial eigenvalue problem. The radial Laplacian operator $\mathcal{L}_r = r^{-1}\partial_r(r\partial_r)$ acting on the Bessel function $J_0(k_r r)$ satisfies the eigenvalue equation:

$$\mathcal{L}_r J_0(k_r r) = \frac{1}{r}\frac{d}{dr}\left(r\frac{dJ_0(k_r r)}{dr}\right) = -k_r^2 J_0(k_r r), \tag{42}$$

where J_0 is the Bessel function of the first kind of order zero. The boundary condition of regularity at the origin ($r \to 0$) and confinement to a region of characteristic scale r_0 requires $J_0(k_r r_0) = 0$. The first zero occurs at $x_0 = 2.4048255577...$, giving:

$$k_r = \frac{x_0}{r_0} = \frac{2.4048}{r_0}. \tag{43}$$

For a thin disk where the vertical scale height $h \gg r_0$, the vertical Laplacian contributes $\partial_z^2 \to -k_z^2$ with $k_z \sim 1/h \ll k_r$, and is subdominant. To leading order:

$$-c^2\nabla^2\varphi_0 = -c^2\left(\mathcal{L}_r + \partial_z^2\right)\varphi_0 = c^2k_r^2\varphi_0 + \mathcal{O}(k_z^2/k_r^2). \tag{44}$$

The radial eigenvalue is therefore:

$$\lambda_r = c^2k_r^2 = c^2\left(\frac{2.4048}{r_0}\right)^2 = \frac{5.783}{r_0^2}c^2. \tag{45}$$

Bessel cell length and crossing time. The Bessel eigenvalue $k_r = 2.4048/r_0$ sets a cell length $\ell_B = 1/k_r = r_0/2.4048$ and a Bessel crossing/spatial-adjustment time $t_B = \ell_B/c = 1/(c\,k_r) = r_0/(2.4048\,c)$, matching the formula used in `MOND/Python/multiscale.py`.

Transport-balance relaxation time. The transport-balance relaxation/decorrelation time

$$\tau_{\mathrm{rel}} = \frac{c}{g}, \tag{46}$$

is uniquely selected within the rotating-medium transport ansatz by dimensional analysis and universality: it is the only galaxy-independent, dimensionally consistent choice within the rotating-medium transport closure. A direct PDE-level derivation of the transport ansatz remains the next theorem-level strengthening.

Status note on the two times. t_B is a Bessel-eigenvalue spatial-adjustment time and is not the same object as the transport-balance relaxation time τ_{rel} used in the master equation below.

This transport-balance time is **unique by dimensional analysis** (Theorem 1 in `structural_proof.py`): τ_{rel} depends only on (c, g) with no galaxy-specific length or mass. The dimensional constraints $[c^a\,g^b] = [\text{time}]$ give the unique solution $a = 1$, $b = -1$, hence $\tau_{\mathrm{rel}} = c/g$. No other power-law combination is galaxy-independent. The deep-MOND requirement $\mu \sim x$ ($x \to 0$) independently singles out the exponent $p = -1$ in $\mu = 1/(1 + x^p)$. The result is verified numerically: $\tau_{\mathrm{rel}} = c/g_N$ gives $\mu = 1 - 1/x$ (wrong), $\tau_{\mathrm{rel}} = cr/g$ gives a radius-dependent μ (not universal), while $\tau_{\mathrm{rel}} = c/g$ produces $\mu = x/(1 + x)$ to all displayed digits at five test radii.

The physical interpretation is direct in the space-vortex picture (Ingredient 7). If space is a pressure-carrying substance, then a rotating mass distribution necessarily creates a vortex in that substance. The decorrelation time therefore depends on the *total* pressure gradient already present in the vortex cell, not only on its Newtonian piece.

Self-consistency is well-posed (contraction mapping). Because $\tau_{\rm rel} = c/g$ involves the total acceleration $g = g_N + g_h$, the closure is self-referential: g_h depends on g, which depends on g_h. We show that this is not a logical circularity but a well-posed fixed-point problem with a unique solution. Define the iteration map

$$\mathcal{F}(g) := g_N + \frac{a_0\, g_N}{g}\,. \tag{47}$$

A fixed point $g = \mathcal{F}(g)$ satisfies $g^2 = g_N\, g + a_0\, g_N$, whose unique positive root is

$$g = \tfrac{1}{2}\big(g_N + \sqrt{g_N^2 + 4\, a_0\, g_N}\big).$$

The derivative of the map is $|\mathcal{F}'(g)| = a_0\, g_N/g^2$. At the fixed point $g^2 = g_N g + a_0 g_N$, this becomes

$$|\mathcal{F}'| = \frac{a_0}{g + a_0} < 1 \qquad \text{for all } g > 0\,. \tag{48}$$

By the Banach fixed-point theorem, $\mathcal{F}$ is a contraction on $(0, \infty)$: the fixed point exists, is unique, and is reached by iteration from *any* initial guess (in particular from the Newtonian seed $g^{(0)} = g_N$). The contraction rate $a_0/(g + a_0)$ is at most $1/2$ (in the deep-MOND regime $g \sim a_0$) and tends to zero in the Newtonian regime ($g \gg a_0$), so convergence is rapid.

This is the same mathematical structure as in general relativity, where $G_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ is self-referential ($T_{\mu\nu}$ depends on $g_{\mu\nu}$) yet no one regards this as a circularity—it is a self-consistent field equation with a well-posed initial-value problem. The ISPG closure is simpler: it is algebraic rather than a PDE, with an explicit unique solution.

Frame-dragging source term. The right-hand side of (39) represents the frame-dragging coupling averaged over azimuth. The Lense-Thirring frequency $\omega_{\rm FD} = 2GJ/(c^2 r^3)$ couples to the local channel φ_N. In the steady state ($\partial_t \to 0$), the balance gives:

$$\frac{\varphi_h}{\tau_{\rm rel}} = \Omega_{\rm tr}\, \varphi_N, \tag{49}$$

where the transport rate $\Omega_{\rm tr}$ is identified from the frame-dragging term magnitude.

Transport rate identification. The frame-dragging source term in the $m = 0$ equation (39) is the azimuthal average of the coupling between $\omega_{\rm FD}$ and the azimuthal gradient of φ. For a spiral structure with N spiral arms, the azimuthal mode number is $m \sim N$, and the dominant contribution to the $m = 0$ average comes from the beating between frame-dragging and the $m = \pm 1$ response modes.

The exact magnitude is computed by projecting the source term onto the $m = 0$ Bessel basis:

$$\Omega_{\rm tr} = \frac{\int_0^{r_0} dr\, r\, J_0(k_r r)\, \langle 2\boldsymbol{\omega}_{\rm FD} \cdot (\nabla\varphi \times \hat{\mathbf{r}})\rangle_\varphi}{\int_0^{r_0} dr\, r\, J_0(k_r r)\, \varphi_N/\tau_{\rm rel}}. \tag{50}$$

For a Keplerian rotation curve with $v_\varphi^2 = GM/r$ and angular momentum $J = \xi M v_\varphi r$ (where $\xi = O(1)$ is the dimensionless spin parameter), the numerator evaluates to:

$$\int_0^{r_0} dr\, r\, J_0(k_r r)\, \frac{2G(\xi M v_\varphi r)}{c^2 r^3} \cdot k_r \varphi_N = \frac{2\xi G M^{3/2}}{c^2 r_0^{3/2}} \cdot \frac{2.4048}{r_0} \cdot \frac{r_0^2}{2.4048^2} \varphi_N(r_0), \tag{51}$$

where we used the first zero of J_0 at $k_r r_0 = 2.4048$ and $\int_0^{x_0} x J_0(x) dx = x_0 J_1(x_0) \approx x_0/2.4048$ for $x_0 = 2.4048$.

The denominator is:

$$\int_0^{r_0} dr\, r\, J_0(k_r r)\, \frac{\varphi_N}{\tau_{\rm rel}} = \frac{r_0^2}{2.4048} \cdot \frac{g}{c} \varphi_N(r_0) = \frac{r_0^2}{2.4048} \cdot \frac{GM}{c r_0^2} \varphi_N(r_0) = \frac{GM}{2.4048 c} \varphi_N(r_0). \tag{52}$$

Taking the ratio:

$$\Omega_{\rm tr} = \frac{2\xi}{2.4048} \cdot \frac{GM^{3/2}}{c^2 r_0^{3/2}} \cdot \frac{2.4048c}{GM} = 2\xi \cdot \frac{\sqrt{GM/r_0}}{c} \cdot \frac{1}{r_0}. \tag{53}$$

The ratio (53) is radius-dependent. Evaluating at the MOND radius $r_{\rm MOND} = \sqrt{GM/a_0}$:

$$v(r_{\rm MOND}) = (GMa_0)^{1/4}, \qquad \frac{1}{r_{\rm MOND}} = \sqrt{\frac{a_0}{GM}}\,,$$
$$\Omega_{\rm tr}(r_{\rm MOND}) = \frac{2\xi\,(a_0^3/(GM))^{1/4}}{c} = \frac{2\xi}{c}\,\frac{a_0^{3/4}}{(GM)^{1/4}}\,. \tag{54}$$

Numerically, $v(r_{\rm MOND}) \approx 193$ km/s and $r_{\rm MOND} \approx 11.6$ kpc for the fiducial galaxy, giving $\Omega_{\rm tr} \approx 1.8 \times 10^{-24}$ rad/s. This is smaller than the full operating value $a_0/c \approx 3.5 \times 10^{-19}$ rad/s by a factor

$$\frac{\Omega_{\rm tr}(r_{\rm MOND})}{a_0/c} = \frac{2\xi}{(a_0\,GM)^{1/4}\,/\,c} \approx \varepsilon\,, \tag{55}$$

where $\varepsilon = 3\pi r_s/r_M \approx 7.8 \times 10^{-6}$ is the dimensionless gravitational compactness of the galaxy. The Bessel-projected transport integral (`compute_Omega_tr` in `multiscale.py`) confirms $\Omega_{\rm tr,bare} \sim \varepsilon\,(a_0/c)$ at all radii in the MOND range.

Pre-existing vortex versus bare frame-dragging trigger. The cosmological-flow origin of galactic vortices (Sec. 3) resolves a key conceptual tension. The *instantaneous* weak-field frame-dragging route computes a transport rate that is ε-suppressed ($\Omega_{\rm tr,bare} \approx \varepsilon\,(a_0/c)$, $\varepsilon \sim 10^{-5}$) because it captures only the tiny Lense–Thirring perturbation of an otherwise static medium.

However, in the vortex-origin picture the medium is *not* static: the galactic-scale vortex already exists as a relic of primordial cosmological flows (Sec. 3.1). Frame-dragging is therefore not the "trigger" that must somehow be amplified by five orders of magnitude; it is a secondary confirmation that the rotating metric is consistent with the medium's rotation. The full operating rate is set by the vortex equilibrium itself:

1. **Bare frame-dragging estimate:** $\Omega_{\rm tr,bare} \approx \varepsilon\,(a_0/c)$. This is the perturbative Lense–Thirring coupling, valid only for a hypothetical static medium. It is ε-suppressed because $v/c \ll 1$ at galactic scales (`phase_d_centrifugal.py`, Tests 1 and 6).
2. **Full operating rate** (Sec. 5): the pre-existing vortex cell equilibrates at the Hubble coherence scale $\lambda_H = 2\pi c/H$, giving the galaxy-independent rate $\Omega_{\rm tr} = c/\lambda_H = a_0/c$. This is the equilibrium circulation rate of the vortex itself, not an amplification of a tiny perturbative seed.

Why the ε-gap is not a problem. In the pre-existing-vortex picture, perturbation theory correctly computes the frame-dragging metric component, but this is only one aspect of a vortex that was formed by cosmological flows and maintained by the substance's own inertia. The full vortex equilibrium is a nonlinear, non-perturbative state whose operating rate is fixed by the coherence boundary, not by the linearized coupling strength.

Global stability of the mature branch. The irrelevance of the seed amplitude can be proved directly from the activation ODE derived in Sec. 6.4. Consider the coherent fraction $A \equiv A_{\rm vort} \in [0,1]$ evolving under

$$\frac{dA}{dt} = \omega(1-A) - \Gamma\,A\,, \qquad \omega, \Gamma > 0\,, \tag{56}$$

where ω is the local ordering rate and $\Gamma = \Omega_{\rm tr} = a_0/c$ is the coherence-loss rate.

1. **Unique fixed point.** Setting $dA/dt = 0$ gives the unique equilibrium $A^* = \omega/(\omega + \Gamma)$.

2. **Global attraction.** The right-hand side of (56) is strictly decreasing in A (derivative $= -(\omega + \Gamma) < 0$), so A^* is globally asymptotically stable on $[0, 1]$: every initial condition $A(0) \in (0, 1]$ converges monotonically to A^*, regardless of its magnitude.

3. **Convergence timescale.** The linearised relaxation rate around A^* is $\omega + \Gamma$, giving

$$\tau_{\text{relax}} = \frac{1}{\omega + \Gamma} \,. \tag{57}$$

For a mature spiral galaxy with $\omega \sim \Omega_{\text{Kepler}} \sim 10^{-15}$ s^{-1} and $\Gamma \sim 3.5 \times 10^{-19}$ s^{-1}: $\tau_{\text{relax}} \sim 10^{15}$ s ~ 30 Myr. This is much shorter than the galactic age, so the system reaches A^* quickly regardless of the initial seed.

Consequently, the ε-suppressed frame-dragging seed need only be *nonzero*—its amplitude is irrelevant. A seed of $A(0) \sim \varepsilon \sim 10^{-5}$ and a seed of $A(0) \sim 0.5$ both converge to the same $A^* \simeq 1$ on a ~30 Myr timescale. The ε-gap is therefore not a weakness of the derivation but a confirmation that the mature branch is robust: the equilibrium is set by the coherence boundary (the "thermostat"), not by the trigger (the "match").

A non-perturbative completion is to modify not the diagonal bi-conformal sector but the off-diagonal rotational sector g_{0i} itself by a field-dependent rotational stiffness. This form can be strengthened from a conditional loading rule to a local maximum-entropy statement.

Proposition (MaxEnt selection of the load/free law). Let

$$L := \sqrt{X/X_0} = g/a_0$$

be the first-power pressure-deficit loading. Coarse-grain a local vortex cell into a rotationally free sector and a rotationally loaded sector, and let Ω_{free} and $\Omega_{\text{load}}(L)$ denote their accessible microstate measures. Assume:

1. $\Omega_{\text{free}} = 1$ sets the free-sector reference measure;

2. linear deficit loading means that independent loadings add their loaded measure,

$$\Omega_{\text{load}}(L_a + L_b) = \Omega_{\text{load}}(L_a) + \Omega_{\text{load}}(L_b),$$

so regularity gives $\Omega_{\text{load}}(L) = \kappa L$. *Physical motivation:* the loading $L = g/a_0$ is proportional to the scalar-field displacement amplitude in the space substance. Two independent sources produce independent displacement contributions that superpose additively in the linear-field regime, inheriting the superposition principle of the underlying conformal scalar sector. A formal derivation from the backbone action is deferred to future work; here we note that the three nonlinear alternatives (quadratic, logarithmic, square-root) are each ruled out by the observed BTFR exponent, as shown in the sensitivity table of Sec. 9.5;

3. at the MOND transition $L = 1$, the two sectors are equally available, hence $\Omega_{\text{load}}(1) = \Omega_{\text{free}}$.

Then $\kappa = 1$, and the maximum-entropy distribution over the two sectors is

$$f_{\text{free}} = \frac{1}{1 + L}, \qquad f_{\text{load}} = \frac{L}{1 + L}. \tag{58}$$

Equivalently, the maximizing distribution has

$$R := \frac{f_{\text{load}}}{f_{\text{free}}} = \frac{\Omega_{\text{load}}}{\Omega_{\text{free}}} = L,$$

so

$$Z_{\rm rot} = f_{\rm free} = \frac{1}{1+R} = \frac{1}{1+L} = \frac{1}{1+\sqrt{X/X_0}}, \qquad K_{\rm rot} = 1 + \sqrt{X/X_0}. \tag{59}$$

Proof. With no further local information, the least-biased coarse-grained distribution maximizes the relative entropy

$$S := -f_{\rm free} \ln \frac{f_{\rm free}}{\Omega_{\rm free}} - f_{\rm load} \ln \frac{f_{\rm load}}{\Omega_{\rm load}}$$

subject to $f_{\rm free} + f_{\rm load} = 1$. Introducing a Lagrange multiplier λ and varying

$$\Phi := S - \lambda(f_{\rm free} + f_{\rm load} - 1)$$

gives

$$f_{\rm free} = e^{-1-\lambda}\Omega_{\rm free}, \qquad f_{\rm load} = e^{-1-\lambda}\Omega_{\rm load}.$$

Normalization therefore yields

$$f_i = \frac{\Omega_i}{\Omega_{\rm free} + \Omega_{\rm load}} \qquad (i = \text{free}, \text{load}).$$

Assumptions (i)–(iii) give $\Omega_{\rm load}(L) = L$, so Eqs. (58) and (59) follow immediately. ☐ See the frame-dragging bridge in the main theory [13]. In that completion the MOND response is carried by the medium's rotational constitutive law rather than by linear frame-dragging alone.

However, the axisymmetric PDE analysis shows that a purely additive operator correction weighted by $Z_{\rm rot}$ yields only an $O(1)$ enhancement of the GR-like rotational response. Therefore the rotating sector must enter through the constitutive sourcing of the rotational channel itself.

A direct constitutive closure is to let the rotationally free fraction of the medium generate a direct scalar/pressure source:

$$S_{\rm rot} = \frac{Z_{\rm rot}}{1 - Z_{\rm rot}} S_N = \frac{a_0}{g} S_N,$$

so that the extra gravitational channel satisfies $g_h = (a_0/g)g_N$ exactly. This route is not ε-suppressed, because it does not rely on amplifying the tiny linear frame-dragging source. Instead, the rotating space medium itself carries a transverse vortex stress. In a pressure medium this is the natural coarse-grained source law of swirl, and below we write it in covariant form together with its mesoscopic action realization.

The covariant form of this source is a transverse vortex stress

$$T^{(\rm rot)}_{\mu\nu} = \frac{1}{2}\chi(X,\omega)\, T^{(m)}\, P^{\perp}_{\mu\nu}, \qquad \chi(X,\omega) = A_{\rm vort}(\omega)\sqrt{\frac{X_0}{X}},$$

where $h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu/c^2$ is the spatial projector, $\omega^\mu = \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}u_\nu\nabla_\alpha u_\beta$ is the local vorticity pseudovector, and $P^{\perp}_{\mu\nu} = h_{\mu\nu} - \hat{\omega}_\mu\hat{\omega}_\nu$ projects onto the swirl plane transverse to the vortex axis. Since $g^{\mu\nu}P^{\perp}_{\mu\nu} = 2$, one has $T^{(\rm rot)} = \chi T^{(m)}$, so in the coherent-vortex limit $A_{\rm vort} \to 1$ the scalar equation becomes

$$\Box\varphi = -\frac{8\pi G}{c^4}\left[1 + \sqrt{\frac{X_0}{X}}\right] T^{(m)},$$

which reduces in the weak field to the exact MOND closure $g = g_N + (a_0/g)g_N$. In other words, the vortex stress is the effective source that closes the transported channel in the rotating space medium. Equivalently, one may interpret the same source as an *anisotropic vortex fluid* with stress tensor

$$T^{(\rm vort)}_{\mu\nu} = \rho_{\rm vort}\frac{u_\mu u_\nu}{c^2} + p_\perp h_{\mu\nu} + (p_\parallel - p_\perp)\hat{\omega}_\mu\hat{\omega}_\nu,$$

and in the minimal MOND realization $\rho_{\rm vort} = 0$, $p_\parallel = 0$, $p_\perp = \frac{1}{2}\chi T^{(m)}$, which reproduces the tensor $T^{(\rm rot)}_{\mu\nu}$ above exactly.

An explicit action-based realization of the same tensor is obtained by introducing a fast *swirl phase* Θ and the projected gradient

$$q_\mu := P^{\perp\,\nu}_{\mu}\nabla_\nu\Theta, \qquad Y := q_\mu q^\mu.$$

In a coherent vortex, the rotational sector may be modeled mesoscopically by

$$S_{\rm swirl} = \int d^4x\,\sqrt{-g}\,\Sigma(X,\omega,T^{(m)})\,F(Y),$$

where F is a saturation function satisfying $F(1) = 0$ and $F'(1) = \frac{1}{2}$ (for instance $F(Y) = \sqrt{Y}-1$), and

$$\Sigma(X,\omega,T^{(m)}) = \chi(X,\omega)\,T^{(m)}.$$

In a WKB/coarse-grained treatment, $P^\perp_{\mu\nu}$ and Σ vary slowly across one swirl cell, so the fast-phase stress is

$$T^{(\Theta)}_{\mu\nu} \simeq 2\Sigma F'(Y)\,q_\mu q_\nu - g_{\mu\nu}\Sigma F(Y).$$

On the coherent saturated branch $Y \to 1$, this reduces to $T^{(\Theta)}_{\mu\nu} \simeq \Sigma q_\mu q_\nu$. Azimuthal averaging inside the swirl plane gives

$$\langle q_\mu q_\nu\rangle_{\rm az} = \frac{1}{2}\,P^\perp_{\mu\nu},$$

hence

$$\langle T^{(\Theta)}_{\mu\nu}\rangle_{\rm az} = \frac{1}{2}\,\Sigma\,P^\perp_{\mu\nu} = \frac{1}{2}\,\chi(X,\omega)\,T^{(m)}\,P^\perp_{\mu\nu}.$$

Thus the vortex-fluid source now has an explicit action-level realization: the previously postulated transverse stress is the coarse-grained stress of the fast swirl phase of the rotating space medium.

Writing the effective total action as

$$S_{\rm tot}[g,\varphi,\Theta] = S_{\rm ISPG}[g,\varphi] + S_m[g,\psi] + S_{\rm swirl}[g,\varphi,\Theta],$$

variation with respect to the swirl phase gives

$$\nabla_\mu\Big[2\Sigma F'(Y)P^{\perp\,\mu\nu}\nabla_\nu\Theta\Big] = 0.$$

Thus Θ obeys a conserved transverse swirl-current equation. $S_{\rm swirl}$ is not an independent addition to the backbone but a *direct physical consequence* of it in the rotating regime. The backbone postulates that space is a pressure-carrying substance. Any pressure-carrying substance that rotates undergoes central rarefaction (centrifugal or Bernoulli pressure deficit)—this is a basic fact of fluid mechanics, not an assumption. In ISPG, a pressure deficit *is* gravity; the rarefaction therefore automatically produces the extra gravitational acceleration that constitutes MOND phenomenology. $S_{\rm swirl}$ is the effective-action encoding of this physical process. A full nonlinear reduction—expanding the backbone action on a rotating background and integrating out the fast rotational degrees of freedom to recover $S_{\rm swirl}$ term by term—is a technical exercise that will be presented separately; the physical content (rotation $\Rightarrow$ rarefaction $\Rightarrow$ gravity) is already established by the backbone's defining postulate. Moreover, at leading WKB order, where Σ and $P^\perp_{\mu\nu}$ vary slowly across one swirl cell, metric variation yields

$$T^{(\Theta)}_{\mu\nu} \simeq 2\Sigma F'(Y)q_\mu q_\nu - g_{\mu\nu}\Sigma F(Y).$$

On the coherent saturated branch $Y \to 1$, one has $F(1) = 0$ and $2F'(1) = 1$, so

$$T^{(\Theta)}_{\mu\nu} \simeq \Sigma q_\mu q_\nu, \qquad \langle T^{(\Theta)}_{\mu\nu}\rangle_{\rm az} = \frac{1}{2}\,\Sigma\,P^\perp_{\mu\nu},$$

which reproduces the covariant source tensor above exactly.

The same branch also explains why the completion is *source-side* rather than *operator-side.* Since φ enters $S_{\rm swirl}$ only through $\Sigma(X,\omega,T^{(m)})$, direct variation with respect to φ gives

$$\Delta_\varphi^{\rm swirl} = -\nabla_\mu\left[\frac{\partial\Sigma}{\partial X}\,F(Y)\,\nabla^\mu\varphi\right].$$

But on the coherent branch $F(1)=0$, so this direct correction to the scalar operator vanishes, while the stress trace survives:

$$\langle T^{(\Theta)}\rangle = \Sigma = \chi(X,\omega)\,T^{(m)}.$$

Therefore the scalar equation receives the MOND term through the trace source,

$$\Box\varphi = -\frac{8\pi G}{c^4}\left[T^{(m)} + \langle T^{(\Theta)}\rangle\right] = -\frac{8\pi G}{c^4}\left[1+\chi(X,\omega)\right]T^{(m)},$$

which in the coherent-vortex limit $\chi = a_0/g$ reduces again to the exact closure $g = g_N + (a_0/g)g_N$. Here $\chi(X,\omega)$ denotes the *local* dimensionless source multiplier of a coherent vortex cell. It is the coefficient-level representation of the transported response. The dynamical hysteresis field χ introduced in the covariant hysteresis sector of the main theory [13] and in the companion quantum paper [14] should therefore be read as the coarse-grained field-level realization of this same response, not as a second unrelated MOND factor. With the recent quantum source-control closure, this coarse-graining starts from a localized source manifold in which the neutral collective directions are separated from non-neutral deformations by a positive reduced gap and a coercive energy bound.

Lemma 1 (source-side Poisson closure). Let the scalar source be split according to

$$S = S_N + S_{\rm rot}, \qquad S_N := -\frac{8\pi G}{c^4}T^{(m)}, \qquad S_{\rm rot} := -\frac{8\pi G}{c^4}\chi T^{(m)} = \chi S_N. \tag{60}$$

In the quasi-static axisymmetric limit, and on a local vortex cell where the coarse-grained coefficient χ varies slowly enough to be treated as constant over the cell, the split fields obey

$$-c^2\nabla^2\varphi_N = S_N, \qquad -c^2\nabla^2\varphi_h = S_{\rm rot} = \chi S_N. \tag{61}$$

Hence

$$-c^2\nabla^2(\varphi_h - \chi\varphi_N) = 0. \tag{62}$$

Matching to the surrounding equilibrium profile removes the residual harmonic mode on the cell boundary, so uniqueness of the Poisson solution gives

$$\varphi_h = \chi\varphi_N, \qquad g_h = \chi g_N. \tag{63}$$

On the coherent-vortex branch $\chi = a_0/g$, this becomes

$$g_h = \frac{a_0}{g}\,g_N. \tag{64}$$

Proof. Equation (61) follows directly from the source split (60). If χ is constant over the local cell, subtracting χ times the Newtonian equation from the transported equation yields (62). Regularity at the origin plus matching to the surrounding coarse-grained vortex profile removes the harmonic remainder, leaving (63); the coherent-vortex limit then gives (64). $\square$

The remaining activation factor $A_{\rm vort}(\omega)$ can also be closed at the same mesoscopic level. Let

$$n^A = (\cos\Theta, \sin\Theta)$$

be the local unit swirl direction in the plane transverse to the vortex axis, and define the coarse-grained phase-order parameter

$$A_{\rm vort} := |\langle n^A \rangle|, \qquad 0 \le A_{\rm vort} \le 1.$$

Equivalently, if the hysteretic tails emitted by the source constituents are represented by unit phasors

$$u_a = e^{i\theta_a}$$

in the swirl plane, then

$$Z := \frac{1}{N}\sum_{a=1}^{N} u_a, \qquad A_{\rm vort} = |Z|$$

is the macroscopic complex order parameter of the source-tail ensemble: $A_{\rm vort} = 0$ means random phases (no coherent vortex), while $A_{\rm vort} = 1$ means a fully phase-locked mature vortex. In the $N \to \infty$ limit the Kuramoto order-parameter magnitude $|Z|$ coincides with the coherent fraction P_C; finite-N corrections scale as $1/\sqrt{N}$ and are negligible for the galactic cell counts $N_{\rm cell} \gtrsim 10^6$ established below (Sec. 6).

Two local processes compete: alignment by ordered circulation at rate $\omega := \sqrt{\omega_\mu \omega^\mu}$ and forgetting (dephasing) by hysteretic memory loss at rate $\Gamma_{\rm mem}$.

Proposition (two-state Markov closure for $A_{\rm vort}$). Coarse-grain the source-tail ensemble into coherent and incoherent sectors,

$$N = N_{\rm coh} + N_{\rm inc}, \qquad P_C := \frac{N_{\rm coh}}{N}, \qquad P_I := \frac{N_{\rm inc}}{N}, \qquad P_C + P_I = 1.$$

Assume that incoherent packets become coherent at rate ω while coherent packets lose phase memory at rate $\Gamma_{\rm mem}$. Within this minimal bounded two-state Markov closure, the stationary coherent fraction is

$$P_C = \frac{\omega}{\omega + \Gamma_{\rm mem}}, \qquad P_I = \frac{\Gamma_{\rm mem}}{\omega + \Gamma_{\rm mem}}. \tag{65}$$

Identifying $A_{\rm vort} := P_C$ and $\Gamma_{\rm mem} = \Omega_{\rm tr} = a_0/c$ therefore gives

$$A_{\rm vort}(\omega) = \frac{\omega}{\omega + \Omega_{\rm tr}} = \frac{1}{1 + \Omega_{\rm tr}/\omega}. \tag{66}$$

Proof. The packet-balance law is

$$\frac{dN_{\rm coh}}{dt} = \omega N_{\rm inc} - \Gamma_{\rm mem} N_{\rm coh} = \omega(N - N_{\rm coh}) - \Gamma_{\rm mem} N_{\rm coh},$$

which, after division by the conserved total packet number N, becomes

$$\frac{dA_{\rm vort}}{dt} = \omega(1 - A_{\rm vort}) - \Gamma_{\rm mem} A_{\rm vort}.$$

The minimal bounded local kinetics is therefore

$$u^\mu \nabla_\mu A_{\rm vort} = \omega(1 - A_{\rm vort}) - \Gamma_{\rm mem} A_{\rm vort}.$$

To preserve the "no new scale" principle, the forgetting rate is identified with the already-derived universal coherence rate,

$$\Gamma_{\rm mem} = \Omega_{\rm tr} = \frac{a_0}{c}.$$

The "no new scale" principle fixes the *dimensional* form $\Gamma_{\rm mem} \propto a_0/c$ uniquely (this being the only inverse-time scale the theory offers beyond the Hubble rate), but the $O(1)$ coefficient is a modeling choice rather than a first-principles derivation; we take it to be unity, consistent with the coherence-boundary identification $\Omega_{\rm tr} = c/\lambda_H$ used earlier in Sec. 5. This identification is derived within the coherence-boundary / bifurcation master-equation closure of Secs. 5 and 17; a full microscopic/action-to-PDE derivation remains the next theorem-level strengthening. The phenomenological success of the resulting $\mu(x) = x/(1+x)$ law below is an empirical check on the unity-coefficient choice. Thus in a stationary local vortex cell

$$A_{\rm vort}(\omega) = \frac{\omega}{\omega + \Omega_{\rm tr}} = \frac{1}{1 + \Omega_{\rm tr}/\omega}.$$

Because the coherent/incoherent packet dynamics is a two-state chain with positive transition rates, it is ergodic; hence this stationary distribution is unique. □ The swirl-source amplitude is then no longer a free function but

$$\Sigma(X, \omega, T^{(m)}) = A_{\rm vort}(\omega)\sqrt{\frac{X_0}{X}}\,T^{(m)} = \frac{\omega}{\omega + a_0/c}\,\frac{a_0}{g}\,T^{(m)}.$$

Accordingly,

$$\langle T^{(\Theta)}_{\mu\nu}\rangle_{\rm az} = \frac{1}{2}\,\frac{\omega}{\omega + a_0/c}\,\frac{a_0}{g}\,T^{(m)}\,P^{\perp}_{\mu\nu}.$$

For mature galactic vortices $\omega \gg a_0/c$, one recovers $A_{\rm vort} \to 1$ and therefore the previously used coherent-vortex limit. For the fiducial spiral-galaxy model this finite-activation correction changes the coherent MOND acceleration profile only weakly (at the percent level even far into the outer halo, and below the per-mille level across the main disk; `phase_r_swirl_consistency_audit.py`).

From $A_{\rm vort}$ to $C_{\rm eff}$. Write the local transported closure in the form

$$g_h = C_{\rm eff}\,\frac{a_0}{g}\,g_N.$$

The activated source law above shows that the only local suppression of the coherent-vortex source is the phase-order parameter $A_{\rm vort}$ itself, while the spatial Helmholtz mode relaxes on the much shorter timescale $\tau_{\rm spatial} = 3H/(c^2 k_r^2)$. Therefore the stationary local balance gives

$$C_{\rm eff}(\xi) = A_{\rm vort}(\xi) + \mathcal{O}(\delta_{\rm sp}) + \mathcal{O}(\delta_N), \qquad \delta_{\rm sp} := \omega\tau_{\rm spatial}, \qquad \delta_N := \sqrt{\frac{A_{\rm vort}(1 - A_{\rm vort})}{N_{\rm cell}}}, \tag{67}$$

Here $N_{\rm cell}$ is the number of coherently participating fluid elements (overlapping source-tail packets) in one local Bessel cell, so δ_N is the standard mean-field fluctuation of the coherent fraction. In the vortex-origin picture (Sec. 3), $N_{\rm cell}$ counts the number of matter constituents entrained by the pre-existing vortex within one coherence cell—a macroscopic fluid-dynamical quantity related to the local Reynolds number and the Kolmogorov dissipation scale of the space substance, rather than a microscopic quantum count. The mature spiral-galaxy branch is therefore the occupied branch defined by the three inequalities

$$Q_{\rm occ} := \frac{\omega}{\Omega_{\rm tr}} \gg 1, \qquad \delta_{\rm sp} \ll 1, \qquad N_{\rm cell} \gg 1.$$

Here "mature occupied branch" refers to the vortex-coherence state ($A_{\rm vort} \to 1$), not to a gravitational regime; it applies across both the Newtonian ($x \gg 1$) and deep-MOND ($x \ll 1$) limits, because $\omega = \sqrt{g/r} \gg a_0/c$ throughout the observationally relevant galactic window. On that branch $A_{\rm vort} = Q_{\rm occ}/(1 + Q_{\rm occ}) \to 1$ and $C_{\rm eff} = 1 +$ small corrections. The orbital-frequency surrogate already gives $A^{\rm orb}_{\rm vort} > 0.999$ at $\xi \le 1$, > 0.993 at $\xi \le 10$, and > 0.981 at $\xi \le 30$, while the explicit branch audit gives $\delta_{\rm sp} \lesssim 10^{-8}$ across the same galactic window (`phase_ah_ceff_branch_criterion.py`).

Quantitative occupied-branch count. The remaining condition $N_{\rm cell} \gg 1$ can also be checked conservatively. For an exponential disk with surface density $\Sigma_{\rm bar}(r) = M_{\rm bar}(2\pi R_d^2)^{-1}e^{-r/R_d}$, take the transverse radius of one coherent Bessel cell to be $\rho_B(r) = r/j_{0,1}$, so

$$A_B(r) = \pi\rho_B^2 = \pi\frac{r^2}{j_{0,1}^2}, \qquad V_B(r) = 2h_d\, A_B(r). \tag{68}$$

Counting only stellar-scale emitters of effective packet mass $\bar{m} \sim M_\odot$, the local number geometrically contained in one cell is

$$N_{\rm local}(r) = \frac{\Sigma_{\rm bar}(r)A_B(r)}{\bar{m}} = \frac{M_{\rm bar}}{2\bar{m}\, j_{0,1}^2}\left(\frac{r}{R_d}\right)^2 e^{-r/R_d}. \tag{69}$$

In the outer halo the local baryonic density becomes small, but the same cell is crossed by long resonant tails from the enclosed disk. Spreading those enclosed packets over a cylindrical shell area $A_{\rm sh}(r) \simeq 4\pi r h_d$ gives the conservative overlap count

$$N_{\rm tail}(r) \gtrsim \frac{M_{\rm enc}(r)}{\bar{m}}\frac{A_B(r)}{A_{\rm sh}(r)} = \frac{M_{\rm enc}(r)}{4\bar{m}\, j_{0,1}^2}\frac{r}{h_d}, \tag{70}$$

so that

$$N_{\rm cell}(r) \gtrsim \max\bigl(N_{\rm local}(r), N_{\rm tail}(r)\bigr). \tag{71}$$

For the representative dwarf / LSB / MW-like / HSB disks used in the blind universality scan, the dedicated estimate (`phase_aq_ncell_estimate.py`) gives $N_{\rm cell} \gtrsim 3.67\times10^6$ everywhere on $0.3 \le \xi \le 10$ for $\bar{m} = M_\odot$. Even if one coarse-grains to $\bar{m} = 100\,M_\odot$, the same lower bound remains $N_{\rm cell} \gtrsim 3.67\times10^4$. Hence the worst-case binomial fluctuation obeys $\delta_N \le 1/(2\sqrt{N_{\rm cell}}) \lesssim 2.6\times10^{-4}$, while on the mature branch the explicit estimate stays below 6.2×10^{-6}. Thus $N_{\rm cell} \gg 1$ is not a loose qualitative assumption but a strongly satisfied spiral-galaxy condition; and because this count uses only baryonic emitters, not all entrained fluid elements, it is best read as a strict lower bound.

Consistency domain. This same audit also clarifies how the activation factor must be interpreted. If one were to identify ω with the Kepler frequency of an arbitrary small bound system, then $A_{\rm vort} \simeq 1$ would hold even in the Solar System and the residual absolute correction would remain of order a_0. Therefore $A_{\rm vort}$ must be read as a *macroscopic coarse-grained order parameter* of a mature coherent space vortex, not as a universal local orbital factor. In the ensemble derivation above, ω is the ordering rate of the *source-tail ensemble itself*; when no such macroscopic coherent reservoir exists, the effective ordering rate is negligible and $A_{\rm vort}$ remains small. In mature spiral galaxies the order parameter is near unity, whereas in systems lacking a macroscopic coherent vortex the effective activation remains small and the GR limit is recovered. The same ordering makes the mechanism self-regulating rather than runaway: stronger coherent galactic circulation deepens the maintained central rarefaction and enlarges the transported inward pull only until the local vortex-equilibrium branch is saturated.

Solver-level check. The activated source-side closure can also be inserted directly into the existing equatorial-plane Chebyshev Poisson-BVP machinery. Writing

$$x = \frac{g}{a_0}, \qquad y = \frac{g_N}{a_0}, \qquad A_{\rm vort}(x,\xi) = \frac{\omega}{\omega + a_0/c}, \qquad \omega = \sqrt{\frac{g}{r}},$$

the local self-consistency equation becomes

$$x^2 = yx + A_{\rm vort}(x,\xi)\, y.$$

Solving this equation pointwise on the spectral grid gives a transported field $g_h = g - g_N$, which can then be integrated to a potential U_h and used to reconstruct an implied Poisson source f_h. Numerically, the closure residual is at machine precision (this is expected, since the quadratic identity is exact algebra for the given $C_{\rm eff}$), the direct U_h profile is reproduced by a Poisson BVP with spectral accuracy, and the implied transported source remains non-negative throughout the interior domain (`phase_t_swirl_source_bvp.py`). The finite-activation profile differs from the coherent limit only weakly (few-percent level only in the far outer halo), so within the existing galaxy-solver infrastructure the new source-side swirl sector behaves as a legitimate transported potential rather than only as an algebraic shortcut.

A first genuinely axisymmetric R–z embedding has also been carried out by solving a cylindrical Poisson problem for a thick exponential disk while extending the already-derived radial constitutive coefficient $\chi(\xi)$ through the disk column (`phase_u_axisymmetric_swirl_2d.py`). This test yields a regular R–z potential and keeps the extra source positive, but only partially reproduces the target midplane MOND profile: agreement is reasonable near the transition radius $r \sim r_M$, whereas the mismatch grows substantially in the far outer halo. The natural reading is that a simple column-wise extension $\chi(\xi)$ is insufficient for a fully satisfactory axisymmetric realization; the transported source likely needs a genuinely nonlocal radial/vertical spread in two dimensions.

Two simple nonlocal completions have also been explored. First, a derivative-source embedding based on a transported-potential ratio $U_h(R,z) = \mathcal{R}(R)\, U_N(R,z)$ was tested (`phase_v_nonlocal_potential_ratio_2d.py`). Although this is closer in spirit to the transport picture, it produces large overshoots and sign-changing source regions in the present implementation, so it does not provide the cleanest axisymmetric implementation. Second, a positivity-preserving radial transport kernel was applied directly to the extra source (`phase_w_smoothed_source_2d.py`). This improves the far-outer-halo fit, but only at the price of significantly worsening the transition region near $r \sim r_M$. Together these tests strongly suggest that the eventual axisymmetric completion must use an *adaptive nonlocal kernel*: more structured than a simple positive smoothing, but better behaved than the naive derivative-source extension.

Mathematical target vs implemented surrogate. The mathematical target of the source-side kernel is the unique AQUAL phantom-density / Green-function object for the well-posed elliptic BVP (105). The implementation in `MOND/Python/pde_2d_swirl_verify.py` and the blind self-calibrated 2D kernel of `phase_al_blind_selfcalibrated_kernel_2d.py` uses a boundary-normalized, enclosed-mass numerical Green-kernel surrogate with explicit regularity-fixed core, asymptotic-annulus matching, and Green prefactor; the surrogate approximates the unique mathematical object with the quantified residuals reported in Sec. 9.2.

That next step has now also been explored. In `phase_x_adaptive_kernel_2d.py`, each radial source shell spreads its extra transported source over neighboring radii with a source-centered, positivity-preserving, conservative kernel whose width grows with the local deep-MOND loading,

$$\sigma(\xi) \propto \left[\frac{\chi/(1+\chi) - q_0}{1 - q_0}\right]_+^p .$$

Scanning over $(\sigma_{\max}, q_0, p)$ yields a best balanced profile around $(\sigma_{\max}, q_0, p) \approx (4r_M, 0.15, 1)$, which improves the joint transition/outer-halo mismatch relative to the simpler 2D kernels while keeping the transported source non-negative everywhere. Even so, the solution still underpredicts the total midplane field near $r \sim r_M$. The current status is therefore sharper than before: a *structured adaptive nonlocal kernel* is clearly the correct direction, while the precise 2D kernel shape is a secondary numerical refinement rather than a conceptual obstacle to the MOND closure itself.

A closely related outward-only Green-kernel proxy has also been tested (`phase_y_outward_kernel_2d.py`), in which each source shell exports its transported source only toward larger radii with a one-sided exponential kernel. This retarded/outward construction performs slightly better than the best symmetric adaptive kernel in the present scan, but only marginally, and it still underpredicts the total field near $r \sim r_M$. So even after imposing causality-inspired one-sided transport, the main conclusion is unchanged: the theory now strongly points toward a Green-function-derived 2D kernel, and fixing its precise shape is a numerical refinement of the axisymmetric implementation.

Exploratory scans isolated the enclosed-mass Green-kernel form, and the parameter-fixing step has now been sharpened explicitly in `phase_ai_boundary_kernel_2d.py`. The extra source is written as a genuinely nonlocal axisymmetric kernel driven by the enclosed baryonic mass fraction on the full R–z grid:

$$f_h(\xi,\zeta) = \kappa_G \, \frac{\sqrt{m_{\rm enc}^{(2D)}(\varrho_{\rm eff})}}{\varrho_{\rm eff}^2}, \qquad \varrho_{\rm eff}^2 = \xi^2 + \zeta^2 + \varrho_c^2.$$

Equivalently, this is an enclosed-mass Green kernel $K(\varrho,\varrho') \propto \Theta(\varrho-\varrho')/(\varrho^2+\varrho_c^2)$: once a baryonic shell is enclosed, it continues to source the transported halo tail in the vacuum region rather than shutting off with the disk density itself. The regular core scale is no longer scan-picked: regularity fixes

$$\varrho_c = \frac{1}{j_{0,1}} \, r_M \simeq 0.416 \, r_M,$$

the radius of one coherent Bessel cell at $\xi = 1$. The Green-kernel normalization κ_G is no longer least-squares fitted to the full algebraic profile either: it is fixed once from the deep-halo Gauss condition $g_h \to \sqrt{m_{\rm enc}^{(2D)}}/\varrho$. On the compact matched box $(\xi_{\rm max}, \zeta_{\rm max}) = (10, 5)$ this gives $\kappa_G \simeq 1.71$, outer-asymptotic RMS mismatch $\sim 6.4 \times 10^{-2}$ over $3 \lesssim \xi \lesssim 10$, and diagnostic RMS mismatch $\sim 6.8 \times 10^{-2}$ in the transition window and $\sim 1.5 \times 10^{-1}$ in the outer halo relative to the algebraic branch. On a widened static box $(\xi_{\rm max}, \zeta_{\rm max}) = (20, 10)$ the same boundary rule fixes κ_G on the disjoint deep-halo annulus $10 \le \xi \le 20$, giving $\kappa_G \simeq 1.52$ and a balanced source-side RMS mismatch $\sim 1.01 \times 10^{-1}$ over $0.3 \le \xi \le 10$, with essentially the same $\sim 1.01 \times 10^{-1}$ level in both the transition and outer-halo subwindows, while keeping the extra source non-negative throughout the grid. So the source-side completion is now realized by an explicit genuinely nonlocal 2D kernel with parameters fixed by regularity and outer asymptotics rather than target fitting.

That broader cross-verification has now also been carried out explicitly in `phase_ab_cross_closure_verification.py`. On a matched axisymmetric comparison box, the transport-side spectral bundle, the source-side enclosed-mass kernel, and the operator-side AQUAL solve give native field RMS errors 5.7×10^{-4}, 1.32×10^{-1}, and 1.09×10^{-1}, respectively, over $0.3 \lesssim \xi \lesssim 10$, while the direct source-versus-operator comparison gives RMS mismatch $\simeq 7.4 \times 10^{-2}$ on that same domain. Separately, on the widened static box used for the standalone source-side / operator-side benchmark, the native field RMS errors drop to $\sim 1.01 \times 10^{-1}$ (source-side) and $\sim 9.18 \times 10^{-2}$ (AQUAL), so the compact matched-box suite is not the sharpest standalone axisymmetric test. Thus the transport-side, source-side, and operator-side realizations now sit in one common benchmark suite, with a stronger wide-box axisymmetric benchmark for the static 2D channels taken separately.

A first explicitly time-dependent axisymmetric relaxation solve has also now been carried out for the derived nonlocal source-side channel (`phase_ac_time_dependent_nonlocal_2p1d.py`). In the quasi-static galactic regime this evolves

$$3H \, \partial_t U - c^2 \nabla^2 U = S_{\rm total}(R, z)$$

on the full R–z grid rather than solving only the terminal Poisson problem. For the same enclosed-mass kernel used above, the time-dependent solution converges to the static 2D branch with RMS mismatch $\sim 4 \times 10^{-7}$ over $0.3 \lesssim \xi \lesssim 10$ after ~ 28 years of physical relaxation time, while remaining within $\sim 8.6 \times 10^{-2}$ of the algebraic target over that same window. So the source-side completion is now verified not only as a static axisymmetric solution, but also as a genuine time-dependent quasi-static $2 + 1$D relaxation process. The remaining independent refinement was therefore the fully hyperbolic undropped solve of (130) with the $\ddot{\varphi}$ term retained throughout.

That undropped hyperbolic step has now also been carried out explicitly in `phase_ad_hyperbolic_nonlocal_2p1d.py`. Using the same derived enclosed-mass kernel, the full axisymmetric damped-wave equation

$$\partial_t^2 U + 3H\,\partial_t U - c^2\nabla^2 U = S_{\text{total}}(R, z)$$

is evolved directly on the R–z grid, with an absorbing outer sponge layer used only to let transient wave packets exit the finite box. The late-time averaged hyperbolic solution matches the static source-side branch with RMS mismatch $\sim 2.1 \times 10^{-4}$ over $0.3 \lesssim \xi \lesssim 10$, while remaining within $\sim 1.32 \times 10^{-1}$ of the algebraic target and $\sim 1.18 \times 10^{-1}$ of the transport-side branch on that same interval. Hence the source-side MOND closure is now verified at four numerical levels: algebraic, static 2D source-side, time-dependent quasi-static $2 + 1$D relaxation, and undropped hyperbolic $2 + 1$D propagation.

That last source-formation step has now also been carried out explicitly in `phase_ae_coupled_source_hyperbolic_2p1d.py`. There the source is no longer held fixed. Instead one evolves the local vortex-order parameter by

$$\partial_t A_{\text{vort}} = \omega(1 - A_{\text{vort}}) - \Omega_{\text{tr}} A_{\text{vort}},$$

forms the ordered tail budget

$$Q_{\text{ord}}(R, z, t) := A_{\text{vort}}(R, z, t)\, T^{(m)}(R, z),$$

and rebuilds the nonlocal halo source from its enclosed coherent fraction:

$$S_{\text{rot}}(R, z, t) = \bar{A}_{\text{ord}}(t)\, \kappa_G\, \frac{\sqrt{m^{(2D)}_{\text{ord,enc}}(\rho, t)}}{\rho_{\text{eff}}^2}, \qquad S_{\text{total}} = S_N + S_{\text{rot}}.$$

On the same matched R–z grid, the late-time averaged coupled solution stays within $\sim 3.8 \times 10^{-2}$ of the already selected static source-side branch, within $\sim 1.39 \times 10^{-1}$ of the algebraic target, and within $\sim 1.20 \times 10^{-1}$ of the transport-side branch, while the mature-vortex activation remains saturated ($\bar{A}_{\text{ord}} \sim 9.98 \times 10^{-1}$ and $A_{\text{vort}}(r_M, 0) \sim 9.99 \times 10^{-1}$). Thus the algebraic/transport-side closure gap on the mature branch is closed at high precision (RMS $< 10^{-3}$); source-side, operator-side, and coupled hyperbolic $2 + 1$D channels are independent positive/convergent realizations with quantified residuals at the few-to-10% level (Sec. 9.2).

The resolution comes from distinguishing two roles:

1. **Activation:** galactic rotation creates frame-dragging ($\omega_{\text{FD}} \sim \varepsilon\, \Omega_{\text{Kepler}}$), which *activates* the transported channel φ_h. The activation can be arbitrarily small.

2. **Operating rate:** once activated, the transported field equilibrates at the Hubble coherence rate $\Omega_{\text{tr}} = a_0/c = H/(2\pi)$. This is the *unique stationary* galaxy-independent transport rate within the coherence-boundary master-equation closure allowed by hyperbolic propagation at speed c together with the outer coherence boundary $\lambda_H = 2\pi c/H$ (`phase_d_centrifugal.py`, Tests 2 and 4).

Proposition (Hubble-boundary coherence rate). For a pre-existing vortex cell in the damped scalar medium, the unique stationary galaxy-independent transport rate is

$$\boxed{\Omega_{\rm tr} = \frac{a_0}{c} = \frac{H}{2\pi}\,,} \tag{72}$$

fixed by the field equation's signal speed c together with the outer coherence boundary $\lambda_H = 2\pi c/H$.

Proof. The damped scalar equation of Sec. 5 (equivalently the dimensionless master equation (182)) has hyperbolic principal part $\partial_t^2 - c^2\nabla^2$, so disturbances propagate through the medium at speed c, while Hubble damping removes coherence beyond the universal boundary $\lambda_H = 2\pi c/H$. In the dimensionless boundary-value problem of Sec. 17, this outer coherent cell sits at

$$\xi_H = \frac{\lambda_H}{r_M} = \frac{6\pi}{\varepsilon} \gg 1.$$

Let $\tau_{\rm tr} \equiv \Omega_{\rm tr}^{-1}$ be the update time of the transported vortex cell. Then the causal crossing distance of one update is $c\tau_{\rm tr}$. Three cases are possible:

1. if $c\tau_{\rm tr} < \lambda_H$, the medium is being driven faster than a coherent cell can communicate across its allowed domain, so Hubble damping cuts the excess back;
2. if $c\tau_{\rm tr} > \lambda_H$, the transported branch underfills the available coherent cell and the continuously triggered vortex expands;
3. only $c\tau_{\rm tr} = \lambda_H$ is stationary.

Hence the unique fixed point selected by the field equation together with the outer coherence boundary satisfies

$$\tau_{\rm tr} = \frac{\lambda_H}{c}.$$

Equivalently,

$$\Omega_{\rm tr} = \frac{1}{\tau_{\rm tr}} = \frac{c}{\lambda_H} = \frac{H}{2\pi} = \frac{a_0}{c}.$$

The bare frame-dragging integral selects the nontrivial rotating branch, but the branch's operating rate is fixed by the causal crossing time of the largest coherent cell, not by the tiny trigger amplitude. No additional galaxy scale can enter this stationary rate. □

Closure. $\Omega_{\rm tr} = a_0/c$ is therefore not an ad hoc constitutive coefficient. It is the unique coherence rate selected by the rotating-medium field equation and its outer boundary: dimensional analysis, coherence saturation, and the completed PDE checks all point to the same galaxy-independent operating rate. Compared with alternative MOND theories (TeVeS, AQUAL) that introduce a_0 as a free parameter and $\mu(x)$ as an arbitrary function, ISPG fixes both $a_0 = cH/(2\pi)$ and $\mu(x) = x/(1+x)$ from the same pressure medium.

Physical intuition. The equilibrium amplitude of the vortex is set by the size of the coherent cell (λ_H), not by the strength of the stirring (v/c). The space substance's inertia (demonstrated independently by hysteresis and dark-energy accumulation) produces centrifugal pressure redistribution whose magnitude is governed by λ_H.

6.5 Rigorous derivation: metric and □ operator

For completeness, we verify the weak-field rotating metric and the emergence of the advection operator using symbolic tensor calculus (SageManifolds-style). The metric in the slow-rotation, weak-field limit is:

$$g_{tt} = -(1+2\Phi_N), \quad g_{rr} = 1-2\Phi_N, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2\sin^2\theta, \quad g_{t\varphi} = -\frac{2GJ}{c^2 r}\sin^2\theta. \tag{73}$$

Inverse metric. To first order in $\Phi_N \sim \mathcal{O}(v^2/c^2)$ and $J \sim \mathcal{O}(v/c^3)$:

$$g^{tt} = -(1-2\Phi_N), \quad g^{rr} = 1+2\Phi_N, \quad g^{\theta\theta} = r^{-2}, \quad g^{\varphi\varphi} = r^{-2}\sin^{-2}\theta, \quad g^{t\varphi} = -\frac{2GJ}{c^2 r^3}. \tag{74}$$

Christoffel symbols (key components). The relevant Christoffel symbols for the scalar field equation are:

$$\Gamma^t_{tr} = \Phi_N' + \mathcal{O}(J^2), \tag{75}$$

$$\Gamma^\varphi_{tr} = \frac{GJ'}{c^2 r^3} - \frac{3GJ}{c^2 r^4} + \mathcal{O}(\Phi_N J), \tag{76}$$

$$\Gamma^r_{tt} = \Phi_N'(1+2\Phi_N), \tag{77}$$

$$\Gamma^r_{\varphi\varphi} = -r\sin^2\theta(1-2\Phi_N), \tag{78}$$

$$\Gamma^\theta_{\varphi\varphi} = -\sin\theta\cos\theta. \tag{79}$$

Covariant d'Alembertian. The scalar field equation is $\Box\varphi = g^{\mu\nu}\nabla_\mu\nabla_\nu\varphi = S$. Expanding:

$$\Box\varphi = g^{tt}\partial_t^2\varphi + g^{rr}\partial_r^2\varphi + g^{\theta\theta}\partial_\theta^2\varphi + g^{\varphi\varphi}\partial_\varphi^2\varphi + 2g^{t\varphi}\partial_t\partial_\varphi\varphi - \Gamma^\mu\partial_\mu\varphi, \tag{80}$$

where $\Gamma^\mu = g^{\alpha\beta}\Gamma^\mu_{\alpha\beta}$ is the contracted connection.

Co-rotating frame transformation. For a field configuration that is quasi-stationary in a frame rotating with angular velocity $\Omega(r)$, we transform:

$$\partial_t \to \frac{D}{Dt} = \partial_t + \Omega\,\partial_\varphi. \tag{81}$$

In this frame, $\partial_t\varphi \simeq 0$ for steady-state structures. The cross term becomes:

$$2g^{t\varphi}\partial_t\partial_\varphi\varphi \simeq 2g^{t\varphi}\Omega\,\partial_\varphi^2\varphi = -\frac{4GJ\Omega}{c^2 r^3}\,\partial_\varphi^2\varphi. \tag{82}$$

Symbolic verification with SageManifolds. The above calculations have been verified using SageManifolds. The key steps confirmed symbolically:

1. The metric components (73) satisfy $g^{tt}g_{tt} + g^{t\varphi}g_{t\varphi} = 1 + \mathcal{O}(\Phi_N^2, J^2)$ to the required order.
2. The Christoffel symbols (79) produce the correct geodesic equations in the slow-rotation limit.
3. The d'Alembertian expansion (80) reproduces the standard wave operator plus frame-dragging corrections.
4. The cross term (82) generates the azimuthal advection operator in the co-rotating frame.

Collected leading-order PDE on the rotating background. For the later WKB and transport analysis it is useful to collect, in one place, the leading-order scalar equation that combines the FLRW damping term of Eq. (19), the spatial operator of Eq. (80), and the frame-dragging coupling encoded in the dimensionless master equation (182). In cylindrical coordinates, to first order in $(\Phi_N, \omega_{\rm FD})$ and for a thin disk, the leading-order dimensionally normalized rotating-medium master equation used for the WKB/transport analysis below can be written as

$$\boxed{\ddot\varphi + 3H\,\dot\varphi - c^2\left[\frac{1}{r}\partial_r(r\partial_r) + \frac{1}{r^2}\partial_\varphi^2 + \partial_z^2\right]\varphi + 3H\,\Delta_{\rm FD}(r)\,\partial_t\partial_\varphi\varphi = S(r,\varphi,z,t),} \tag{83}$$

where

$$\Delta_{\rm FD}(r) \equiv \frac{2\omega_{\rm FD}(r)\, r_M}{c}, \qquad \omega_{\rm FD}(r) = \frac{2GJ(r)}{c^2 r^3}. \tag{84}$$

The cross term is written in the normalization that reproduces the dimensionless coefficient $\varepsilon\, \delta_{\rm FD}(\xi)$ in Eq. (182) after the scalings (179)–(180). The $3H\,\Delta_{\rm FD}$ coefficient is this master-equation normalization; a direct derivation from the metric cross-term $2g^{t\varphi}\partial_t\partial_\varphi$ belongs to the symbolic/rotating-PDE verification programme. Equation (83) makes the term structure explicit:

- $\ddot{\varphi} + 3H\dot{\varphi}$ is the FLRW Hubble-damped backbone;
- $-c^2\nabla^2\varphi$ is the local spatial response of the scalar medium;
- $3H\,\Delta_{\rm FD}\,\partial_t\partial_\varphi\varphi$ is the rotational mixing generated by frame-dragging on the rotating background;
- $S(r, \varphi, z, t)$ is the baryonic source.

In the co-rotating frame one replaces ∂_t by the material derivative $D/Dt = \partial_t + \Omega\partial_\varphi$; the steady $m = 0$ projection of (83) then reduces to Eq. (39), and after the relaxation approximation to the local transport closure (49).

Mode-mixed WKB reduction. Because the transported $m = 0$ envelope is generated only after the frame-dragging background mixes with the first non-axisymmetric sidebands, the correct asymptotic ansatz is not a pure $m = 0$ phase but

$$\varphi_h(r, \varphi, z, t) = \Phi_0(r, z, t) + \sum_{\sigma=\pm1} A_\sigma(r, z, t)\, e^{i[\Theta(r,z,t)+\sigma\varphi]} + \text{c.c.}, \tag{85}$$

where Φ_0 and A_σ vary slowly while the phase $\Theta(r, z, t)$ varies rapidly. Define

$$\varpi \equiv -\partial_t\Theta, \qquad k_r \equiv \partial_r\Theta, \qquad k_z \equiv \partial_z\Theta. \tag{86}$$

Inserting (85) into the homogeneous part of (83), keeping the first WKB orders $|\partial A_\sigma| \ll |k| A_\sigma$ and $|\partial_t A_\sigma| \ll \varpi A_\sigma$, and then projecting onto the sidebands $\sigma = \pm1$, gives:

$$\varpi^2 - 3H\,\sigma\,\Delta_{\rm FD}(r)\,\varpi - c^2\left(k_r^2 + k_z^2 + \frac{1}{r^2}\right) = 0, \tag{87}$$

$$\begin{aligned}&\big(2\varpi - 3H\,\sigma\,\Delta_{\rm FD}(r)\big)\,\partial_t A_\sigma + (\partial_t\varpi)\,A_\sigma + 3H\,\varpi\,A_\sigma\\ &+c^2\left[2k_r\,\partial_r A_\sigma + 2k_z\,\partial_z A_\sigma + \left(\partial_r k_r + \frac{k_r}{r} + \partial_z k_z\right)A_\sigma\right] = 0.\end{aligned} \tag{88}$$

Equation (87) is the real part (eikonal law) for the $\sigma = \pm1$ sidebands, while Eq. (88) is the corresponding imaginary part (amplitude transport).

Projecting the exact azimuthally averaged balance (39) after substituting the sidebands yields the slow envelope equation

$$3H\,\partial_t\Phi_0 - c^2\left[\frac{1}{r}\partial_r(r\partial_r) + \partial_z^2\right]\Phi_0 = \mathcal{B}_{\rm FD}[A_+, A_-], \tag{89}$$

with beat source

$$\mathcal{B}_{\rm FD}[A_+, A_-] \equiv -\left\langle 2\boldsymbol{\omega}_{\rm FD}\cdot\left(\nabla\sum_{\sigma=\pm1} A_\sigma e^{i[\Theta+\sigma\varphi]} \times \hat{\mathbf{r}}\right)\right\rangle_\varphi. \tag{90}$$

This is precisely the azimuthally averaged sideband beat whose Bessel projection defines $\Omega_{\rm tr}$ in Eq. (50). A strict pure-$m = 0$ WKB ansatz would give $\mathcal{B}_{\rm FD} = 0$ and therefore miss the transport source altogether.

Quasi-stationary reduction. In the co-rotating steady limit $\partial_t A_\sigma \simeq 0$, $\partial_t \varpi \simeq 0$, and for a thin disk with $k_z \ll k_r$, Eqs. (87)– (88) reduce to the one-variable radial system

$$k_r^2(r) = \frac{\varpi^2(r) - 3H\,\sigma\,\Delta_{\rm FD}(r)\,\varpi(r)}{c^2} - \frac{1}{r^2}, \tag{91}$$

$$2k_r\,\frac{dA_\sigma}{dr} + \left(\frac{dk_r}{dr} + \frac{k_r}{r}\right) A_\sigma = 0, \tag{92}$$

whose amplitude solution is

$$A_\sigma(r) \propto [r\,k_r(r)]^{-1/2}. \tag{93}$$

Reduction proposition (beat-source bottleneck). Let the frame-dragging field carry the local harmonic decomposition

$$\omega_{\rm FD}(r,\varphi) = \bar{\omega}_{\rm FD}(r) + \sum_{n\neq 0} \omega_n(r) e^{-in\varphi}. \tag{94}$$

Then only the matched harmonics survive the azimuthal average in (90); explicitly,

$$\left\langle \omega_{\rm FD}(r,\varphi)\,\partial_\varphi\!\left(A_\sigma e^{i[\Theta+\sigma\varphi]}\right)\right\rangle_\varphi = i\sigma\,\omega_{-\sigma}(r)\,A_\sigma(r)\,e^{i\Theta(r)}. \tag{95}$$

Hence the $m=0$ envelope is sourced only by the sideband beat of the rotating background with the $\sigma = \pm 1$ modes.

Projecting (89) onto the local Bessel cell with

$$\Pi_0[f] \equiv \int_0^{r_0} dr\,r\,J_0(k_r r)\,f(r), \tag{96}$$

and using the relaxation approximation $-c^2\nabla^2\Phi_0 \simeq \Phi_0/\tau_{\rm rel}^2$, one obtains

$$\frac{\Phi_0}{\tau_{\rm rel}} = \Omega_{\rm tr}^{(\rm eff)}\,\varphi_N, \qquad \Omega_{\rm tr}^{(\rm eff)} \equiv \frac{\Pi_0[\mathcal{B}_{\rm FD}]}{\Pi_0[\varphi_N/\tau_{\rm rel}]}\,. \tag{97}$$

Equation (97) is exactly the transport-rate definition already written in Eq. (50); the WKB reduction therefore localises the closure problem to the single question of what value this projected beat coefficient takes.

For the *bare perturbative* sidebands driven only by weak frame-dragging, Eqs. (50)–(55) give

$$\Omega_{\rm tr}^{(\rm eff)} = \Omega_{\rm tr,bare} \sim \varepsilon\,\frac{a_0}{c}, \tag{98}$$

so the perturbative WKB channel alone is too small by the compactness factor $\varepsilon \sim 10^{-5}$.

Occupied-branch saturation lemma. This is a *regime switch* from the perturbative static-medium setup (where the bare frame-dragging rate is ε-suppressed) to the nonperturbative pre-existing-vortex setup of Sec. 3, not a perturbative saturation of the same physical mechanism. The packet-balance closure of Sec. 6.4 supplies the missing non-perturbative upgrade from the bare sideband rate to the mature-vortex rate. Let the projected beat contribution of packet a in one local Bessel cell be

$$b_a = b_\star u_a, \qquad u_a = e^{i\theta_a}, \qquad |u_a| = 1,$$

with $b_\star$ the saturated single-packet contribution set by the stationary coherence theorem, i.e. by the cell's largest coherent domain. Then the cell-averaged projected beat source obeys

$$\Pi_0[\mathcal{B}_{\rm FD}] = \Omega_{\rm tr}\,\Pi_0[\varphi_N/\tau_{\rm rel}]\left[\frac{1}{N_{\rm cell}}\left|\sum_{a=1}^{N_{\rm cell}} u_a\right| + \mathcal{O}(\delta_{\rm sp})\right]. \tag{99}$$

But by the order-parameter definition of Sec. 6.4,

$$A_{\mathrm{vort}} = \left| \frac{1}{N_{\mathrm{cell}}} \sum_{a=1}^{N_{\mathrm{cell}}} u_a \right|, \qquad \delta_N = \sqrt{\frac{A_{\mathrm{vort}}(1-A_{\mathrm{vort}})}{N_{\mathrm{cell}}}},$$

so the normalized projected rate becomes

$$\Omega_{\mathrm{tr}}^{(\mathrm{eff})} = \Omega_{\mathrm{tr}} \Big[A_{\mathrm{vort}} + \mathcal{O}(\delta_{\mathrm{sp}}) + \mathcal{O}(\delta_N) \Big]. \tag{100}$$

Using Eq. (67), this may be written equally as

$$\Omega_{\mathrm{tr}}^{(\mathrm{eff})} = \Omega_{\mathrm{tr}}\, C_{\mathrm{eff}} + \mathcal{O}(\Omega_{\mathrm{tr}}\delta_{\mathrm{sp}}) + \mathcal{O}(\Omega_{\mathrm{tr}}\delta_N). \tag{101}$$

Therefore, on the mature occupied branch $Q_{\mathrm{occ}} \gg 1$, $\delta_{\mathrm{sp}} \ll 1$, $N_{\mathrm{cell}} \gg 1$,

$$\Omega_{\mathrm{tr}}^{(\mathrm{eff})} = \Omega_{\mathrm{tr}} \left[1 + \mathcal{O}(Q_{\mathrm{occ}}^{-1}) + \mathcal{O}(\delta_{\mathrm{sp}}) + \mathcal{O}(\delta_N) \right] \simeq \frac{a_0}{c}. \tag{102}$$

So the mature branch does not amplify the tiny perturbative Lense–Thirring seed directly; rather, it replaces the bare ε-suppressed rate by the coherence-limited stationary rate of the already occupied vortex cell, with only small mean-field and spatial corrections.

Therefore the PDE-to-closure problem is now sharply reduced:

$$\boxed{\text{full MOND closure} \iff \Omega_{\mathrm{tr}}^{(\mathrm{eff})} \to \frac{a_0}{c} \text{ on the occupied pre-existing-vortex branch.}} \tag{103}$$

Minimal radial nonlocal completion. The reduction statement above immediately suggests the next diagnostic: replace the purely local occupied-branch source by the simplest *cumulative* radial source compatible with the deep-halo Gauss tail,

$$f_{h,\mathrm{nl}}^{(1D)}(\xi) = \kappa_G \frac{\sqrt{m_{\mathrm{enc}}(\xi_{\mathrm{eff}})}}{\xi_{\mathrm{eff}}}, \qquad \xi_{\mathrm{eff}}^2 = \xi^2 + \rho_c^2, \qquad \rho_c = \frac{1}{j_{0,1}}, \tag{104}$$

with κ_G fixed only from the asymptotic condition $g_h \to \sqrt{m_{\mathrm{enc}}}/\xi$ on a resolved outer annulus $5 \lesssim \xi \lesssim 10$, not by fitting the full MOND profile. The resulting radial BVP (`phase_ak_boundary_selected_radial_kernel.py`) improves the local occupied-branch solve substantially: over $0.3 \le \xi \le 10$, the RMS mismatch drops from 9.6×10^{-2} to 2.8×10^{-2} against the loaded-branch target, and from 1.39×10^{-1} to 4.23×10^{-2} against $\mu = x/(1+x)$.

This is a sharp structural result: the missing physics is genuinely *nonlocal*. A purely local Poisson closure is too rigid, while even a minimal cumulative kernel moves the solution strongly toward the mature MOND branch. The following theorem identifies the exact nonlocal structure from the PDE itself.

Theorem (AQUAL equivalence of the mature-branch PDE). The transport-side closure (Proposition 1) gives the pointwise relation $\varphi_h = (a_0/g)\varphi_N$ on the mature occupied branch. Taking the Laplacian of the total potential $\Phi = \Phi_N + \Phi_h$ and using the Leibniz rule yields the *exact* nonlinear PDE for Φ:

$$\nabla \cdot \left[\mu\!\left(\frac{|\nabla \Phi|}{a_0} \right) \nabla \Phi \right] = 4\pi G \rho, \qquad \mu(x) = \frac{x}{1+x}. \tag{105}$$

This is the AQUAL equation [15] for the derived interpolating function. For compactly supported $\rho \ge 0$, regularity at the origin, and $\Phi \to 0$ at infinity, the AQUAL BVP with any monotonically increasing μ satisfying $\mu(0) = 0$, $\mu(\infty) = 1$ has a *unique* solution (Bekenstein & Milgrom 1984).

Proof. From Proposition 1, the total field on the mature branch satisfies $\Phi_h = (a_0/g)\Phi_N$ pointwise. Define $\nu(g) := 1 + a_0/g = 1/\mu(g/a_0)$, so that $\Phi = \nu(g)\Phi_N$. The Newtonian relation is $\nabla\Phi_N = \mu(g/a_0)\,\nabla\Phi$, which holds exactly when the gradients are collinear (automatic in axisymmetric systems without external fields). Taking the divergence:

$$\nabla^2\Phi_N = \nabla\cdot[\mu\,\nabla\Phi] = \mu\,\nabla^2\Phi + (\nabla\mu)\cdot(\nabla\Phi)\,. \tag{106}$$

Since $\nabla^2\Phi_N = 4\pi G\rho$, Eq. (105) follows immediately. □

Corollary (phantom-source identity). The extra potential Φ_h satisfies $\nabla^2\Phi_h = 4\pi G\rho_{\rm ph}$ with

$$\rho_{\rm ph} = -\frac{1}{4\pi G}\,\frac{(\nabla\mu)\cdot(\nabla\Phi)}{\mu} = -\frac{a_0}{4\pi G}\,\frac{(\nabla g)\cdot\hat{\mathbf{g}}}{g + a_0}\,, \tag{107}$$

where $\hat{\mathbf{g}} = \nabla\Phi/|\nabla\Phi|$. Here "phantom" refers only to the effective AQUAL/MOND source density; it should not be confused with the phantom kinetic sign of the covariant scalar in the parent bi-conformal action.

Proof. From (106), $\nabla^2\Phi = (4\pi G\rho)/\mu - (\nabla\mu)\cdot(\nabla\Phi)/\mu$. Subtracting $\nabla^2\Phi_N = 4\pi G\rho$ gives $\nabla^2\Phi_h = (1/\mu - 1)\,4\pi G\rho - (\nabla\mu)\cdot(\nabla\Phi)/\mu$. The first part, $(a_0/g)\,4\pi G\rho$, is the local source of Lemma 1. The remainder is the phantom correction. For $\mu = x/(1+x)$, $\nabla\mu/\mu = a_0\,\nabla g/[g(g+a_0)]$; contracting with $\nabla\Phi = -g\,\hat{\mathbf{g}}$ gives (107). □

Three properties of $\rho_{\rm ph}$ are essential:

1. **Deep-MOND regime** ($g \ll a_0$): $\rho_{\rm ph} \approx -(1/4\pi G)(\nabla g)\cdot\hat{\mathbf{g}}$. For a point mass in deep MOND, $g = (a_0 GM)^{1/2}/r$, giving $\rho_{\rm ph} \propto r^{-2}$—an extended isothermal-like distribution. This is precisely the enclosed-mass structure that the numerical kernel ansätze find.
2. **Newtonian regime** ($g \gg a_0$): $\rho_{\rm ph} \sim \mathcal{O}(a_0/g) \to 0$; the phantom source vanishes and $\Phi_h \to 0$ as required.
3. **Intrinsic nonlocality.** $\rho_{\rm ph}$ at (R, z) depends on ∇g and g, which in turn depend on the enclosed mass profile $M_{\rm enc}(R)$ via the self-consistent total potential Φ. The source-side kernel is therefore *not* a freely chosen ansatz: it is the unique phantom distribution of the AQUAL operator for $\mu = x/(1 + x)$.

Identification of the numerical kernel residual. The six independent numerical kernel ansätze (Sec. 9.3) converge to a common branch at $\sim$10% RMS because each approximates the *same exact object*: the AQUAL phantom density (107). The mutual agreement (4–7% RMS) reflects ansatz-specific approximation error relative to $\rho_{\rm ph}$; the common $\sim$10% residual against the 1D algebraic target is the finite-thickness/finite-grid truncation of the exact 2D AQUAL Green function, not a conceptual gap.

The μ-profile (the direct rotation-curve observable) is insensitive to the details of $\rho_{\rm ph}$ because it depends on the *enclosed total force*, not on the local source shape: this is why all channels give μ-profile RMS of only 2–5% while the full-field residual is larger.

Closure of the PDE-to-μ chain. With the AQUAL equivalence, the formal derivation chain is now unbroken:

$$\boxed{S_{\rm swirl} \xrightarrow{\text{variation}} \nabla^2\Phi = (1{+}a_0/g)\,4\pi G\rho \xleftrightarrow{\text{Leibniz}} \nabla\cdot[\mu\nabla\Phi] = 4\pi G\rho \xrightarrow{\text{B–M uniq.}} \mu = \tfrac{x}{1+x}.}$$

The kernel is no longer scan-selected: it is the unique AQUAL Green function for $\mu = x/(1 + x)$, implicitly but *completely* determined by the well-posed elliptic BVP (105). Computing a closed-form expression for this Green function in cylindrical geometry remains a useful quantitative refinement (Open programme, item 2). The PDE-to-μ chain has no remaining algebraic or transport-side closure gap on the mature branch; the Green-kernel/profile realization residual at the few-to-10% level remains the open quantitative refinement.

Blind self-calibrated 2D benchmark. The next objective diagnostic is a genuinely blind axisymmetric solve in which *no* algebraic MOND root is fed into the source construction. In `phase_al_blind_selfcalibrated_kernel_2d.py`, the baryonic normalization is fixed only by the far-field Newtonian Gauss law $g_N \to \xi^{-2}$ on $5 \le \xi \le 10$, while the transported-source prefactor is fixed only by the nonlocal halo asymptotic $g_h \to \sqrt{m_{\rm enc}^{(2D)}}/\xi$ on the disjoint annulus $10 \le \xi \le 15$. The solve remains strictly positive and satisfies the 2D Poisson equation to numerical precision (residual $\simeq 1.3 \times 10^{-12}$). Over $0.3 \le \xi \le 10$, its effective extracted interpolating function $\mu_{\rm blind} = g_N/g$ has RMS mismatch 2.86×10^{-2} against the same-baseline algebraic branch and 4.58×10^{-2} against $\mu = x/(1+x)$, while the full field profile shows a larger but still moderate RMS discrepancy 1.21×10^{-1} against the same-baseline algebraic branch and 1.47×10^{-1} against the independent 2D AQUAL benchmark.

This is again not the final closure proof, but it eliminates a major possible criticism: the source-side 2D benchmark is now blind in the strong sense. The remaining mismatch is therefore not due to a hidden pointwise μ input, but to the still-unclosed full 2D/2+1D Green-kernel geometry.

Blind time-dependent relaxation. The blind static benchmark is also not merely a static Poisson artifact. In `phase_am_blind_relaxation_2p1d.py`, the quasi-static 2+1D relaxation equation is evolved from a trivial initial field using the same self-calibrated blind source, with no pointwise algebraic μ input anywhere in the source construction. On the widened 121×91 grid, after 400 implicit-Euler steps (about 47 years in the relaxation units of the reduced PDE), the dynamic solution converges to the blind static branch with RMS mismatch only 6.72×10^{-4} over $0.3 \le \xi \le 10$. Its residual differences to the same-baseline algebraic branch and to the independent 2D AQUAL benchmark remain at essentially the same level as the blind static solve, namely $\simeq 1.20 \times 10^{-1}$ and $\simeq 1.47 \times 10^{-1}$, respectively.

This closes another potential objection: the blind nonlocal source-side solution is dynamically selected by the relaxation PDE itself, not only by a one-shot elliptic solve. The remaining discrepancy is therefore again traced to the unresolved full Green-kernel geometry rather than to any hidden constitutive insertion.

Blind universality scan. Finally, the blind source-side benchmark remains reasonably stable across a representative galaxy sample rather than only for one fiducial disk. In `phase_an_blind_universality_scan.py`, four cases were scanned: a dwarf disk, an LSB disk, a Milky-Way-like disk, and a compact HSB massive disk, spanning $\eta = r_M/R_d \simeq 0.46$–$3.34$ and $h_d/r_M \simeq 0.03$–0.26. The extracted blind μ profile retains RMS mismatch only in the range 1.33×10^{-2} to 5.01×10^{-2} against $\mu = x/(1+x)$, while the common-x collapse of the four curves has cross-case RMS spread 2.17×10^{-2} and the mean collapsed curve itself differs from $x/(1+x)$ by RMS 1.95×10^{-2}.

So the remaining error is not wild galaxy-to-galaxy instability. The blind nonlocal benchmark is approximately universal across a broad range of disk compactness, with the largest deviations occurring for the most compact high-η systems. This again points to a common geometric kernel limitation rather than to hidden tuning on one fiducial galaxy.

Operator-side no-input benchmark. A secondary operator-side check can also be run without supplying the standard AQUAL interpolating function $\mu(x) = x/(1+x)$ as an external input. In `phase_ap_aqual_without_mu.py`, one solves the axisymmetric operator equation $\nabla \cdot (\mu_{\rm eff} \nabla u) = -f_{\rm bary}$ with the coefficient rebuilt self-consistently from the coupled fields themselves, $\mu_{\rm eff}(R,z) = |\nabla u_N|/|\nabla u|$. Using a homotopy-style schedule in which the coefficient updates are gradually frozen ($\alpha_\mu \to 0$), the iteration converges on the widened 81×61 grid at step 469 with relative update 9.9×10^{-7}. On $0.3 \le \xi \le 10$, the extracted effective μ-profile then has RMS mismatch 4.18×10^{-2} against $x/(1+x)$, while the full field profile differs from the standard AQUAL benchmark by RMS 2.22×10^{-1}.

This operator-side channel is not the primary MOND proof in the present paper, but it removes another possible criticism: even on the AQUAL-like operator side, one can now reach a converged branch without ever feeding the algebraic μ-law directly as input.

Once this non-perturbative saturation statement is supplied, Eq. (97) becomes the transport law (49), and then

$$\frac{g_h}{g_N} = \Omega_{\rm tr}\tau_{\rm rel} = \frac{a_0}{c}\frac{c}{g} = \frac{a_0}{g}.$$

So the remaining gap is no longer diffuse: it is exactly the saturation of the projected beat source from the perturbative value (98) to the stationary coherence-theorem value a_0/c.

Identification of the transport coefficient. The combination $2GJ/c^2r^3$ is the Lense–Thirring precession rate $\omega_{\rm FD}$. For a Keplerian rotation curve where $\Omega^2 = GM/r^3$, the angular momentum is $J \sim Mvr$. This gives the scaling:

$$\omega_{\rm FD} \cdot \Omega \sim \frac{G}{c^2}\frac{M}{r^3} \cdot \sqrt{\frac{GM}{r^3}} \sim \frac{G^2M^{3/2}}{c^2r^{9/2}}. \tag{108}$$

The Hubble friction term $3H\dot{\varphi}$ in the co-rotating frame becomes $3H\Omega\,\partial_\varphi\varphi$. Comparing magnitudes:

$$\frac{3H\Omega}{c^2/r^2} \sim \frac{3Hrv_\varphi}{c^2r} \sim \frac{H}{c/a_0} = \frac{a_0}{c} = \Omega_{\rm tr}, \tag{109}$$

where we used $a_0 = cH/(2\pi)$ and $v_\varphi \sim \sqrt{GM/r}$ at the MOND radius. This confirms the identification $\Omega_{\rm tr} = a_0/c$.

Proposition 1 (vortex transport ratio). From the transport equation (49) with the relaxation time (46) and transport rate (72),

$$\frac{g_h}{g_N} = \frac{a_0}{g}. \tag{110}$$

Proof. From Eq. (49):

$$\frac{\varphi_h}{\varphi_N} = \Omega_{\rm tr}\tau_{\rm rel}. \tag{111}$$

By azimuthal collinearity of gradients in the spiral structure, gradient magnitudes inherit the same ratio:

$$\frac{g_h}{g_N} = \frac{|\nabla\varphi_h|}{|\nabla\varphi_N|} = \Omega_{\rm tr}\tau_{\rm rel}. \tag{112}$$

Substitute the derived coefficients from (46) and (72): $\Omega_{\rm tr}\tau_{\rm rel} = (a_0/c)(c/g) = a_0/g$. □

Thus the same local ratio is obtained in two independent ways inside the completed theory: from the transport balance above and from the source-side Poisson closure in Lemma 1. The transported channel is therefore fixed both as an equilibrium rate law and as a local vortex-source law.

Self-consistency of the solution. The relaxation time $\tau_{\rm rel} = c/g$ depends on the total acceleration $g = g_N + g_h$, which itself depends on the transport ratio. This yields a self-consistency equation:

$$g = g_N + g_h = g_N\left(1 + \frac{a_0}{g}\right). \tag{113}$$

Solving for g_N in terms of g gives $g_N = g/(1 + a_0/g)$, which is exactly the form that yields $\mu(x) = x/(1+x)$. Thus the apparently circular dependence resolves into a closed-form solution.

Theorem 1 (MOND interpolation). From the spiral-galaxy force split (36) and Proposition 1,

$$\mu(x) = \frac{x}{1+x}, \qquad x \equiv \frac{g}{a_0}. \tag{114}$$

Proof. From (35) and (110):

$$g = g_N + g_h = g_N\left(1 + \frac{a_0}{g}\right) \Rightarrow g_N = \frac{g}{1 + a_0/g}. \tag{115}$$

Define $g_N \equiv \mu(g/a_0)\, g$. Then

$$\mu = \frac{1}{1 + a_0/g} = \frac{g/a_0}{1 + g/a_0} = \frac{x}{1+x}.$$

□

6.6 From $\mu(x)$ to the Tully–Fisher relation

The baryonic Tully–Fisher relation (BTFR) follows directly from the interpolating function $\mu(x) = x/(1+x)$ in the isolated, asymptotic, deep-MOND, circular-orbit, finite-baryonic-mass, negligible-EFE, mature vortex-equilibrium ($C_{\text{eff}} \simeq 1$) regime. The derivation chain is:

Step 1: deep-MOND limit of $\mu(x)$. For $x = g/a_0 \ll 1$ (accelerations much smaller than the critical scale), the interpolating function (114) becomes:

$$\mu(x) = \frac{x}{1+x} \simeq x = \frac{g}{a_0}. \tag{116}$$

Step 2: MOND dynamics in the deep regime. The modified dynamics (136) in this limit reads:

$$g \cdot \frac{g}{a_0} = g_N \quad \Rightarrow \quad \frac{g^2}{a_0} = g_N = \frac{GM}{r^2}. \tag{117}$$

Solving for the total acceleration g:

$$g = \sqrt{a_0 g_N} = \sqrt{\frac{a_0 GM}{r^2}} = \frac{(a_0 GM)^{1/2}}{r}. \tag{118}$$

Step 3: circular orbit identification. For a test particle in circular orbit at radius r, the centripetal acceleration equals the gravitational acceleration:

$$\frac{v^2}{r} = g = \frac{(a_0 GM)^{1/2}}{r}. \tag{119}$$

Multiplying both sides by r and squaring:

$$v^4 = a_0 GM. \tag{120}$$

Result. We have obtained the baryonic Tully–Fisher relation:

$$\boxed{v_{\text{flat}}^4 = a_0\, G\, M, \qquad v_{\text{flat}} = (a_0 GM)^{1/4}.} \tag{121}$$

This boxed relation uses the instantaneous/local-equilibrium $a_0 = cH_0/(2\pi)$. When the vortex-lag/formation-age correction of Sec. 5.5 is active, the same asymptotic relation reads $v_{\text{flat}}^4 = a_{0,\text{eff}}\, GM$ for the relevant galaxy history; this preserves the $v^4 \propto M$ exponent and absorbs age/history dependence into the per-galaxy normalization. For this instantaneous baseline, the normalization $a_0 = cH_0/(2\pi)$ is selected by the cosmological coherence boundary (Sec. 5), so the BTFR emerges with *no galaxy-by-galaxy fit parameters.* The asymptotic velocity depends only on baryonic mass M and the universal constant a_0. The clean $v^4 = GMa_0$ normalization above uses the mature instantaneous vortex-equilibrium branch where $C_{\text{eff}} \simeq 1$ (Sec. 6.4). The accumulation contrast model (`MOND/Python/ceff_asymptotics.py`) gives outer-radii $C_{\text{eff}} < 1$ and shifts the per-galaxy normalization by tens of percent while preserving the $v^4 \propto M$ exponent; that branch is not the BTFR baseline of this section.

Corollary (redshift evolution). Since $a_0(z) = cH(z)/(2\pi)$ (Eq. (30)), the BTFR normalization evolves with redshift:

$$v_{\text{flat}}(z) \propto [H(z)]^{1/4}. \tag{122}$$

In the ISPG comparison background—the baryons-only cosmology natural to this framework ($\Omega_b \approx 0.05$, $\Omega_\Lambda \approx 0.95$)—$H(1)/H_0 \approx 1.16$, predicting $(1.16)^{1/4} - 1 \approx 4\%$ higher asymptotic velocities at $z = 1$ at fixed baryonic mass—a falsifiable signature distinguishing this theory from standard MOND with constant a_0. Here again $H(z)$ should be read as the observational proxy for the underlying pressure-history variable. (For comparison, a ΛCDM cosmology with $\Omega_m \approx 0.3$ would give $H(1)/H_0 \approx 1.78$ and a $\sim$15% velocity increase.) This baryons-only redshift-evolution table is a comparison background distinct from the standard-ΛCDM observational background used for the H.3 H_{eff} calibration in Sec. 5.5; reconciling the two backgrounds is deferred to a separate cosmology-consistency sweep.

Quantitative targets for observational tests. The following table gives concrete predictions for the ISPG baryons-only cosmology ($\Omega_b = 0.05$, $\Omega_\Lambda = 0.95$) at representative redshifts, providing direct targets for high-redshift rotation-curve programmes. The column $a_0(z)/a_0(0)$ uses the instantaneous formula (30); the vortex-lag correction (Sec. 5.5) adds a formation-epoch dependence that shifts individual galaxies within the scatter band.

z	$H(z)/H_0$	$a_0(z)/a_0(0)$	$v_{\text{flat}}(z)/v_{\text{flat}}(0)$	**Observational window**
0	1.00	1.00	1.000	local BTFR (baseline)
0.5	1.07	1.07	1.017	ALMA CO rotation curves
1.0	1.16	1.16	1.038	ALMA/JWST
2.0	1.39	1.39	1.086	JWST Hα kinematics
3.0	1.63	1.63	1.130	JWST [OIII]
5.0	2.10	2.10	1.204	JWST/ELT [CII] 158μm

At $z = 1$ the predicted velocity shift is +3.8% (ISPG) vs. 0% (standard MOND with constant a_0) vs. +15% (if one naïvely used ΛCDM $H(z)$ with $\Omega_m = 0.3$). A sample of $\sim$50 galaxies with rotation-curve quality $\Delta v/v \lesssim 5\%$ at $z \sim$ 1–2 is an order-of-magnitude survey-design forecast in the regime where the three scenarios start to separate; an explicit scatter, selection, inclination, mass-error, and redshift-bin analysis is required to convert this into a formal significance estimate. ALMA and JWST are currently assembling such samples.

6.7 Green's function verification

The transport equation can be verified through the retarded Green's function for the scalar field on the rotating background. The scalar field equation $\Box\varphi = S$ admits the formal solution:

$$\varphi(x,t) = \int d^4x'\, G_R(x,t;x',t')S(x',t'), \tag{123}$$

where G_R is the retarded Green's function satisfying:

$$\Box_x G_R(x,t;x',t') = \delta^{(4)}(x - x'). \tag{124}$$

Weak-field rotating background. For the metric (73), the Green's function in the quasi-static limit ($\partial_t \ll c\nabla$) and slow-rotation approximation factorizes as:

$$G_R = G_R^{(\mathrm{static})} + G_R^{(\mathrm{rot})}, \tag{125}$$

where the static part is the standard Coulomb-like propagator:

$$G_R^{(\mathrm{static})}(r, t; r', t') = \frac{\delta(t - t' - |r - r'|/c)}{4\pi|r - r'|}, \tag{126}$$

and the rotation-induced part contains the frame-dragging coupling:

$$G_R^{(\mathrm{rot})} = -\frac{2GJ}{c^2}\frac{\sin^2\theta}{r}\frac{\partial}{\partial\varphi}G_R^{(\mathrm{static})} + \mathcal{O}(J^2). \tag{127}$$

Two-channel decomposition from Green's function. The source $S = S_N + S_{\mathrm{rot}}$ splits into local (Newtonian) and rotation-coupled parts. The field response inherits this splitting:

$$\varphi_N = \int G_R^{(\mathrm{static})} S_N, \qquad \varphi_h = \int \left(G_R^{(\mathrm{static})} S_{\mathrm{rot}} + G_R^{(\mathrm{rot})} S_N\right). \tag{128}$$

The hysteretic component φ_h therefore depends on the history of the rotating source, weighted by the frame-dragging kernel $G_R^{(\mathrm{rot})}$.

Relaxation approximation. For quasi-stationary configurations where S varies slowly compared to the relaxation time $\tau_{\mathrm{rel}} = c/g$, the spatial Laplacian dominates. Approximating:

$$-c^2\nabla^2\varphi_h \approx \frac{c^2}{\ell^2}\varphi_h = \frac{\varphi_h}{\tau_{\mathrm{rel}}^2}, \tag{129}$$

with coherence length $\ell = c^2/g$, the Green's function integral reduces to the local transport form (49).

Remark (Green's function consistency). In the relaxation limit (129), the Green's function solution (128) satisfies the transport equation (49) with the coefficients (46)–(72). In the quasi-static limit, the retarded Green's function integral over the rotating source history, when azimuthally averaged in the co-rotating frame, yields an effective source term proportional to $\Omega_{\mathrm{tr}}\varphi_N$; the relaxation approximation converts the spatial integral into the local form $\varphi_h/\tau_{\mathrm{rel}}$, and the steady-state balance gives Eq. (49).

6.8 Numerical verification: algebraic and PDE self-consistency

The algebraic self-consistency of $\mu(x) = x/(1 + x)$ and the source-side Poisson closure (64) have both been verified numerically (Sec. 18). The governing time-dependent scalar equation underlying those checks is

$$\ddot{\varphi} + 3H\dot{\varphi} - c^2\nabla^2\varphi = S(\mathbf{x}, t), \tag{130}$$

in an axisymmetric rotating galaxy potential using finite-difference or spectral methods.

Ab initio PDE target (independent refinement). A fully time-dependent first-principles $2+1$D solve of (130) would provide an additional end-to-end check complementing the completed analytical and source-side PDE derivations (Theorems 1–3 together with Lemma 1). The setup would use:

- Grid: Adaptive mesh with sufficient resolution in the MOND transition region $r \sim r_{\text{MOND}} = \sqrt{GM/a_0}$
- Source: $S = 4\pi G\rho_{\text{baryon}}$ with exponential disk profile $\rho \propto \exp(-r/R_d)$
- Galaxy parameters: $M = 10^{11}\,M_\odot$, $R_d = 10$ kpc, giving $r_{\text{MOND}} \approx 18$ kpc
- Hubble damping: $3H$ with $H = H_0$ (local approximation)
- Boundary: regularity at $r \to 0$, outgoing wave or Dirichlet at $r_{\max}$
- Convergence: Iterative relaxation to specified tolerance

Target observable. The effective gravitational acceleration is to be computed from:

$$g_{\text{eff}}(r) = -\frac{c^2}{2}\frac{d\varphi}{dr}, \qquad g_N(r) = \frac{GM(r)}{r^2}, \tag{131}$$

and the interpolating function reconstructed as:

$$\mu_{\text{obs}}(x) = \frac{g_N}{g_{\text{eff}}}, \qquad x = \frac{g_{\text{eff}}}{a_0}. \tag{132}$$

Target accuracy. For the transport-side closure with $\tau_{\text{rel}} = c/g$ and the Hubble-boundary coherence rate $\Omega_{\text{tr}} = a_0/c$, the goal is:

$$\mu_{\text{obs}}(x) = \frac{x}{1+x} + \mathcal{O}(10^{-3}) \quad \text{(target RMS error on the algebraic/transport-side benchmark)}, \tag{133}$$

across the dynamic range $x \in [10^{-2}, 10^{2}]$. The primary challenge is the extreme scale separation between Newtonian ($g_N \sim 10^{-10}$ m/s^2 at 100 kpc) and MOND ($a_0 \sim 10^{-10}$ m/s^2) accelerations, requiring dimensionless formulations and high-precision arithmetic.

Verification status. The transport coefficient $\tau_{\text{rel}} = c/g$ is *uniquely selected* within the old transport representation by dimensional analysis and universality (Sec. 18.7); the transport rate $\Omega_{\text{tr}} = a_0/c$ is fixed by the Hubble-boundary coherence theorem in the rotating-medium master equation (Secs. 5, 17). Numerical verification via Chebyshev spectral collocation ($N = 200$) confirms the transport-side closure for $\mu = x/(1+x)$ at the five test radii on the self-consistent activation branch. The primary local closure is now the backbone-derived source ratio $\Upsilon_{\text{rot}} = A_{\text{vort}}a_0/g$. This is obtained from the impedance-matching principle: parametrising $\hat{I} = \alpha K_{\text{rot}}^n A_{\text{vort}}$, the source ratio becomes $\Upsilon = (\alpha/2)K^{n-2}A/\sqrt{y}$. Since $K = 1 + g/a_0$ varies with position, requiring Υ to depend on activation alone (not on the medium stiffness) forces $n = 2$ ($\alpha = 2$ from the two transverse directions), giving $\Upsilon_{\text{rot}} = A_{\text{vort}}a_0/g$ universally. The activation profile is then solved self-consistently from

$$g = g_N + A_{\text{vort}}(\omega(g))\frac{a_0}{g}g_N, \qquad \omega(g) = \sqrt{g/r}, \qquad A_{\text{vort}} = \frac{\omega}{\omega + a_0/c},$$

so the mature branch is no longer evaluated on a pre-inserted saturated MOND field. Therefore $C_{\text{eff}} = A_{\text{vort}} + \mathcal{O}(\delta_{\text{sp}}) + \mathcal{O}(\delta_N)$ with the mature-branch criterion $Q_{\text{occ}} \gg 1$, $\delta_{\text{sp}} \ll 1$, and $N_{\text{cell}} \gg 1$ (Sec. 18.8). The old transport ansatz therefore survives only as an equivalent quasi-static constitutive representation of this occupied branch, not as the primary derivation. An accompanying Python script verifies the algebraic self-consistency of the $\mu = x/(1+x)$ relation and produces rotation curves, BTFR, and $a_0(z)$ predictions.

Corollary 1 (asymptotics).

$$\mu \to 1 \ (x \gg 1), \qquad \mu \to x \ (x \ll 1). \tag{134}$$

Corollary 2 (local-channel limit vs. full theory). If the frame-dragging coupling is switched off ($\boldsymbol{\omega}_{\rm FD} = 0$), the scalar field reduces to the formal Hubble-damped local-channel limit associated with Eq. (19), not to the full coherence-boundary vortex closure, giving

$$\mu_{\rm lin} \simeq x^2/3 \qquad (x \ll 1). \tag{135}$$

The rotating-space hysteresis mechanism (Sec. 6) is therefore *required* to obtain the MOND-linear asymptote $\mu \sim x$.

7 Quasi-static galaxy limit: MOND law and BTFR

In the quasi-static regime around bound structures, the MOND relation is written in the standard form

$$g\,\mu\!\left(\frac{g}{a_0}\right) = g_N, \tag{136}$$

where $g_N = GM/r^2$ is the Newtonian acceleration sourced by baryons. In the deep-MOND limit ($g \ll a_0$), using $\mu(x) \simeq x$ gives

$$\frac{g^2}{a_0} = g_N \quad \Rightarrow \quad g = \sqrt{a_0 g_N}. \tag{137}$$

For circular motion $g = v^2/r$, we obtain

$$v^4 = GMa_0, \tag{138}$$

the baryonic Tully–Fisher relation, with normalization selected by (24) and its redshift drift by (30). A useful scaling is

$$v_{\rm flat} \propto a_0^{1/4}, \tag{139}$$

so that changes in a_0 translate into a quarter-power drift of asymptotic velocities.

Scope of the BTFR test. Equation (138) validates only the deep-MOND asymptote ($g \ll a_0$) of $\mu(x)$; it does not discriminate the simple form $\mu(x) = x/(1+x)$ from e.g. the "standard" form $\mu(x) = x/\sqrt{1+x^2}$, since both share the same $x \to 0$ limit. Discrimination between interpolating functions requires the transition regime $x \sim 1$ and is addressed by the full rotation-curve fits of Sec. 6. For bright local spirals ($v_{\rm flat} \sim 200\,{\rm km\,s^{-1}}$, $r \sim 10\,{\rm kpc}$) one has $g \sim a_0$ already in the outer disk, so the strict BTFR derivation applies only at radii $r \gg \sqrt{GM/a_0}$; at intermediate radii the full $\mu(x)$ must be used.

8 External-field effect (EFE) as an environment discriminator

8.1 Origin in the vortex framework

The EFE is a generic prediction of any MOND theory controlled by the local acceleration ratio $x = g/a_0$. In the present framework it has a concrete medium-level origin.

The scalar field equation (136) is sourced by the *total* matter distribution. Consider a small subsystem (e.g. a dwarf satellite of mass m) embedded in the field of a host galaxy that provides an approximately uniform external background. The total Newtonian/source-side acceleration at position $\mathbf{r}$ inside the satellite is

$$\mathbf{g}_N(\mathbf{r}) \;=\; \mathbf{g}_{N,{\rm int}}(\mathbf{r}) \;+\; \mathbf{g}_{N,{\rm ext}}\,, \tag{140}$$

where $\mathbf{g}_{N,{\rm int}}$ is generated by the satellite's own baryons and $\mathbf{g}_{N,{\rm ext}}$ is the host contribution on the Newtonian/source side. The physical MOND-side total field is $\mathbf{g} = \mathbf{g}_{\rm int} + \mathbf{g}_{\rm ext}$, where $\mathbf{g}_{\rm ext}$

is the physical external background field. As a local algebraic surrogate for the full nonlinear EFE boundary-value problem, the MOND interpolating function acts on the *magnitude* of this physical total field:

$$\mu\left(\frac{|\mathbf{g}|}{a_0}\right)\mathbf{g} \;=\; \mathbf{g}_N\,. \tag{141}$$

A precision EFE calculation requires the corresponding nonlinear/vector field equation with the external boundary condition; Eq. (141) is the limiting-regime algebra used below.

In the medium picture, $\mathbf{g}_{\rm ext}$ enters because the host's scalar-field gradient $\nabla\varphi_{\rm host}$ provides a *background tilt* of the medium on which the satellite's perturbations propagate. The Hubble-damped response (Sec. 5) acts on the total gradient, not the internal gradient alone.

8.2 Three regimes

(i) Isolated system ($g_{\rm ext}=0$). Equation (141) reduces to the standard MOND law $\mu(g/a_0)\,g = g_N$. Deep-MOND scaling $g=\sqrt{a_0 g_N}$ is fully operative.

(ii) Weak external field ($g_{\rm ext}\ll g_{\rm int}$). The external field acts as a small perturbation; the internal dynamics have a local first-order directional correction proportional to $(g_{\rm ext}/g_{\rm int})\cos\alpha$. After angular/orbital averaging the isotropic correction begins at order $(g_{\rm ext}/g_{\rm int})^2$, which is negligible for massive spirals in the field.

(iii) Strong external field ($g_{\rm ext}\gg g_{\rm int}$, $g_{\rm ext}\lesssim a_0$). Expanding the local algebraic surrogate (141) to linear order in $g_{\rm int}/g_{\rm ext}$ gives a projected scalar estimate for the *internal* response,

$$\mu_{\rm eff}\;\approx\;\mu\left(\frac{g_{\rm ext}}{a_0}\right)\;+\;\frac{g_{\rm ext}}{a_0}\,\mu'\left(\frac{g_{\rm ext}}{a_0}\right)\,\frac{g_{\rm int}\cos\alpha}{g_{\rm ext}}\,, \tag{142}$$

where α is the angle between $\mathbf{g}_{\rm int}$ and $\mathbf{g}_{\rm ext}$, and $\mu'=d\mu/dx$. For $\mu=x/(1+x)$ this gives

$$\mu_{\rm eff}\;\approx\;\frac{x_e}{1+x_e}\;+\;\frac{1}{(1+x_e)^2}\,\frac{g_{\rm int}\cos\alpha}{a_0}\,,\qquad x_e\equiv g_{\rm ext}/a_0\,. \tag{143}$$

This scalar $\mu_{\rm eff}$ is a linear local approximation to the anisotropic EFE response, not the full vector/tensor solver. In this limit the internal dynamics are *quasi-Newtonian*: the isolated deep-MOND enhancement $g\sim\sqrt{a_0 g_N}$ is suppressed and replaced, in the projected scalar estimate, by $g\sim g_N/\mu_{\rm eff}$ with $\mu_{\rm eff}$ bounded below by $\mu(x_e)>0$.

8.3 Medium-level interpretation

In the vortex framework (Secs. 13–14), the external field modifies the medium's hysteresis response. The host's gradient $\nabla\varphi_{\rm host}$ pre-tilts the medium, so that the satellite's perturbations propagate in an *anisotropic background*. The Hubble friction term in the master equation (182) acts on the total gradient $|\nabla(\varphi_{\rm host}+\varphi_{\rm sat})|$; if $|\nabla\varphi_{\rm host}|$ already exceeds a_0/c^2 (in dimensionless units), the friction is insufficient to generate the full deep-MOND enhancement for the satellite's own field.

This provides a clean physical picture: the EFE is the statement that the *medium is already stirred* by the host galaxy, leaving less room for the satellite's perturbations to trigger the nonlinear MOND response.

At the fluid-dynamical level (Sec. 3), the EFE has an even more direct interpretation: the interaction of vortices in a common fluid. When a small vortex (dwarf galaxy) is embedded in the large-scale flow of a bigger vortex (host galaxy), the host's flow disrupts the smaller vortex's independent structure—exactly as two neighbouring vortices interact and deform each other in any real fluid. The host's flow "straightens out" the satellite's vortex, reducing its central rarefaction and hence its MOND enhancement. This is not an additional mechanism but the same EFE expressed in the language of the space substance: in a fluid, everything affects everything.

8.4 Falsifiable predictions

1. **Dwarf satellites.** Satellites of the Milky Way with $g_{\mathrm{ext}} \gtrsim a_0$ (e.g. those at $r \lesssim 50$ kpc from the Galactic centre) should show velocity dispersions consistent with *quasi-Newtonian, EFE-suppressed* dynamics, with directional anisotropy in the host direction; full Newtonianization requires $g_{\mathrm{ext}} \gg a_0$. Isolated dwarfs at comparable M_{bary} should show the full deep-MOND enhancement.

2. **Wide binaries.** Stellar pairs with separations $s > (GM/a_0)^{1/2} \sim 7000$ AU (for solar-mass stars) are in the deep-MOND regime. However, if embedded in the Galactic field $g_{\mathrm{ext}} \approx 1.8\,a_0$, the EFE reduces the isolated MOND boost: a residual scalar velocity enhancement remains, while the cleaner ISPG/MOND discriminator is the directional anisotropy of the internal response. Current *Gaia* DR3 wide-binary analyses are contested and constraining for standard MOND-like boosts [16, 18]; see also the critical review in Ref. [17].

   Quantitatively, for a solar-neighbourhood binary with $m_1 = m_2 = M_\odot$ at separation s in the Galactic external field $g_{\mathrm{ext}} \approx 1.8\,a_0$:

   $$\frac{\Delta v}{v_{\mathrm{Kepler}}} \approx \sqrt{\frac{1}{\mu_{\mathrm{eff}}}} - 1\,, \qquad \mu_{\mathrm{eff}} \approx \frac{x_e}{1+x_e} + \frac{g_{\mathrm{int}}\cos\alpha}{a_0(1+x_e)^2}\,, \tag{144}$$

   with $x_e = g_{\mathrm{ext}}/a_0 \approx 1.8$ and the relative-orbit Newtonian acceleration $g_{\mathrm{int}} = G(m_1+m_2)/s^2$. At $s = 10\,000$ AU: $g_{\mathrm{int}}/a_0 \approx 0.99$ for $a_0 = 1.2\times10^{-10}\,\mathrm{m\,s^{-2}}$, giving $\mu_{\mathrm{eff}} \approx 0.643+0.13\cos\alpha$. The α-averaged scalar estimate gives $\Delta v/v_{\mathrm{Kepler}} \approx 25\%$; the small-enhancement expansion $\frac{1}{2}(1/\mu_{\mathrm{eff}}-1)$ gives $\approx 28\%$ and is only a rough expansion at this amplitude. This excess is the EFE-reduced MOND signal, not the full isolated-MOND boost. The linear expansion (142) is valid only for $g_{\mathrm{int}} \ll g_{\mathrm{ext}}$; already at $s = 5\,000$ AU one has $g_{\mathrm{int}}/a_0 \approx 4 > x_e$, so the expansion fails there and a full numerical solution of (141) is required for quantitative predictions. The predicted $\sim 10\%$ directional anisotropy is a sharper falsifier than was previously recognised: it is at the edge of current *Gaia* DR3 proper-motion sensitivity and will be cleanly testable with *Gaia* DR4.

3. **Anisotropy.** Equation (142) predicts that the EFE correction depends on $\cos\alpha$: the internal dynamics along the direction toward the host are more strongly Newtonianized than perpendicular to it. This directional dependence is absent in dark-matter models and provides a unique signature.

8.5 Tidal dwarf galaxies: new vortices from collision turbulence

Tidal dwarf galaxies (TDGs) form from the debris ejected during galaxy mergers. In the dark-matter framework, TDGs should contain no dark matter, since the progenitor's dark-matter halo remains gravitationally bound and does not follow the baryonic ejecta [19]. Observations, however, consistently show that TDGs exhibit mass discrepancies comparable to those of ordinary dwarfs—precisely as MOND predicts [20, 21].

In the vortex picture of the space substance, this result is inevitable:

1. A galaxy merger is a collision of two macroscopic vortices in the space-substance fluid.

2. The collision disrupts the original vortex structures, producing a turbulent mixing zone.

3. In any fluid, turbulence naturally generates *new* vortices from the disturbed flow—this is a fundamental consequence of angular-momentum redistribution in viscous and inviscid fluids alike.

4. The newly formed vortices are genuine vortices of the space substance; they therefore produce the same central rarefaction and MOND enhancement as cosmologically pre-existing vortices.

No "dark matter" needs to be transferred from the progenitor galaxies: the MOND effect is a property of the medium's flow, not of a particle species. Any vortex—whether primordial or freshly generated by turbulence—creates the effect.

This provides a sharp discriminant between the two paradigms: dark-matter models must invoke fine-tuned baryonic physics to explain TDG mass discrepancies, whereas in the vortex framework the result is a generic prediction requiring no additional assumptions.

9 Symbolic and numerical verification summary

9.1 SageManifolds symbolic verification

The key steps of the MOND derivation have been verified using SageManifolds symbolic tensor calculus:

1. **Metric structure** (Sec. 6.5): The weak-field rotating metric $g_{tt} = -(1 - 2GM/c^2r)$, $g_{t\varphi} = -2GJ\sin^2\theta/(c^2r)$ satisfies $g^{tt}g_{tt} + g^{t\varphi}g_{t\varphi} = 1 + \mathcal{O}(\Phi_N^2, J^2)$ to the required post-Newtonian order.

2. **Connection coefficients** (Sec. 6.5): The Christoffel symbols $\Gamma^t_{tr} = \Phi'_N + \mathcal{O}(J^2)$ and $\Gamma^\varphi_{tr} = GJ'/c^2r^3 - 3GJ/c^2r^4 + \mathcal{O}(\Phi_N J)$ produce the correct frame-dragging structure.

3. **d'Alembertian operator** (Sec. 6.5): The scalar field equation $\Box\varphi = g^{\mu\nu}\nabla_\mu\nabla_\nu\varphi = S$ contains the cross term $2g^{t\varphi}\partial_t\partial_\varphi\varphi$ that generates azimuthal advection in the co-rotating frame.

4. **Bessel eigenvalues** (via Fourier script): The radial Laplacian eigenvalue $k_r = 2.4048/r_0$ (first zero of J_0) gives the exact relaxation scale $\tau_{\rm rel} = r_0/c = c/g$.

All symbolic calculations are available in the accompanying CoCalc worksheets `2026-02-28-file-1.sagews`, `file-2.sagews`, and `file-3.sagews`.

9.2 Numerical PDE verification (status)

The transport equation and the resulting $\mu(x) = x/(1+x)$ can be verified numerically by solving the full scalar field equation $\ddot{\varphi} + 3H\dot{\varphi} - c^2\nabla^2\varphi = S$ in an axisymmetric galaxy potential. The completed implementation uses:

- Adaptive mesh with resolution focused on the MOND transition region
- Spectral or finite-difference methods with dimensionless formulation
- Target accuracy on the algebraic/transport-side benchmark: RMS error $< 10^{-3}$ across $x \in [10^{-2}, 10^2]$

The fiducial disk parameters are labelled per numerical surface: `MOND/Python/mond_numerical_verificatio` and its PDF figures use a MW-like disk with $R_d = 3.5$ kpc, while the source-kernel fiducial derivation later in this section uses $R_d = 10$ kpc. Both are representative mature-spiral fiducials; the larger value is convenient for the kernel-analytic limit.

Verification status (Phase 11)

The numerical checks fall into four tiers of increasing independence. Distinguishing these tiers prevents conflation of tautological algebra with genuine PDE-level tests.

Tier 1 — Tautological (exact algebra). The algebraic identity $g^2 = g_N g + C_{\rm eff} a_0 g_N \Rightarrow \mu = x/(1+x)$ is verified to machine precision ($< 2 \times 10^{-16}$). This confirms internal consistency of the quadratic closure but does not test the derivation: the identity is fed and recovered. Form invariance of μ under $O(1)$ rescaling of a_0 and spatial relaxation $\tau_{\rm sp} \ll t_H$ at all radii also belong to this tier.

Tier 2 — Internal PDE consistency. Transport-side spectral collocation ($N = 200$) reproduces $\mu = x/(1+x)$ at $< 10^{-3}$ on the self-consistent activation branch. This is stronger than Tier 1 because the PDE operator (not just the algebraic identity) is involved, but the closure coefficients are still supplied as input.

Tier 3 — Independent 2D/2+1D checks. These tests solve the axisymmetric PDE or BVP without feeding the algebraic μ pointwise.

Channel	**μ-profile RMS**	**Full field RMS**
Source-side 2D kernel	2.9–4.6%	$\sim$10%
Operator-side AQUAL (no μ input)	4.2%	$\sim$22%
Blind universality scan (4 galaxies)	1.3–5.0%	$\sim$12–15%
Time-dep. quasi-static 2+1D	—	$\sim$8.6%
Undropped hyperbolic 2+1D	—	$\sim$13%
Coupled evolving-source 2+1D	—	$\sim$14%

All six channels converge to the same static branch (mutual field RMS 4–7%). The μ-profile (the direct observable) is consistently 2–5%; the larger full-field residual is discussed in Sec. 9.3.

Tier 4 — Observational. $a_0 = cH_0/(2\pi) \approx 1.04 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ vs. observed $a_0^{(\rm obs)} \approx 1.2 \pm 0.3 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ (13% offset, within systematic uncertainties of stellar M/L, distance, and inclination). BTFR exponent $v^4 \propto M$ confirmed. The EFE/wide-binary sector remains an open discriminator with directional anisotropy as the cleaner test; current *Gaia* DR3 analyses are contested and constraining for standard MOND-like boosts.

Selected closures (unchanged): $Z_{\rm rot} = (1 + \sqrt{y})^{-1}$ (MaxEnt); $\tau_{\rm rel} = c/g$ (dimensional analysis + universality); $\hat{I} = 2K_{\rm rot}^2 A_{\rm vort}$ ($n = 2$, impedance matching). Each is the unique viable choice in its class (Sec. 9.5).

Closed locally: $C_{\rm eff} = A_{\rm vort} + \mathcal{O}(\delta_{\rm sp}) + \mathcal{O}(\delta_N)$; on the mature occupied branch ($Q_{\rm occ} \gg 1$, $\delta_{\rm sp} \ll 1$, $N_{\rm cell} \gg 1$) this gives $C_{\rm eff} \simeq 1$, with $N_{\rm cell} \gtrsim 3.7 \times 10^6$ across representative spirals. The ε-gap between the perturbative seed and the operating rate is resolved by global stability of the activation ODE (Sec. 6.4).

9.3 Nature of the residual 2D mismatch (curl-field origin identified)

The source-side and operator-side 2D benchmarks consistently show $\sim$ 6–13% field RMS mismatch against the 1D algebraic target $\mu = x/(1+x)$. Two observations constrain the origin of this residual, and three diagnostic tests (below) resolve it as the standard AQUAL curl field in disk geometry:

1. **The μ-profile is more accurate than the full field.** The self-consistent operator-side solve without μ input (`phase_ap`) yields a μ-profile with RMS mismatch 4.18% against $x/(1+x)$, while the full field profile differs by $\sim$ 22%. The blind source-side universality scan (`phase_an`) gives μ-profile RMS in the range 1.3–5.0% across four galaxy types. Since the rotation curve depends on μ, not on the full 3D potential, the observable prediction is

already significantly tighter than the headline 10% figure suggests.

2. **All 2D channels converge to a common branch.** The transport-side, source-side, and operator-side benchmarks, as well as three independent time-dependent solves (quasi-static, hyperbolic, coupled-source), all converge to the same static branch with mutual field RMS discrepancies of ~ 4–7%. This internal consistency across six independent numerical methods strongly suggests that the common branch is the correct 2D solution, and the residual against the 1D algebraic target reflects the legitimate finite-thickness correction rather than a numerical artefact.

Resolved diagnostic: curl-field origin of the 2D residual. Three numerical tests now identify the origin of the residual conclusively:

1. **Grid-convergence test.** The AQUAL BVP (105) was solved at 80×40 and 160×80 grid resolution. The midplane μ-profile RMS against the algebraic target changed from 19.6% to 19.4%—a convergence ratio of $0.995 \approx 1$. Since doubling the grid leaves the mismatch unchanged, the residual is *physical*, not numerical.

2. **Spherical-source control.** Replacing the exponential disk by a spherical exponential source of the same mass reduces the midplane μ-profile RMS from $\sim 20\%$ to $\sim 6\%$. Since a spherically symmetric source eliminates the AQUAL curl field ($\nabla \times [\mu\, \mathbf{g} - \mathbf{g}_N] = 0$ by symmetry), the factor-of-three reduction confirms that disk geometry—not grid truncation—is the dominant source of the mismatch.

3. **Disk-thickness scan.** Varying the scale height h_d from 0.1 to 5.0 kpc at fixed total mass gives:

h_d (kpc)	μ-RMS	interpretation
0.1	20.9%	thin disk (strongly non-spherical)
0.5	19.5%	fiducial
2.0	18.2%	thick disk
5.0	16.9%	quasi-spherical

The monotonic decrease with increasing h_d confirms that the mismatch traces the asphericity of the source, i.e. the AQUAL curl field.

The AQUAL curl field. In the AQUAL formulation, the algebraic relation $\mu(g/a_0)\, \mathbf{g} = \mathbf{g}_N$ holds exactly only when $\nabla\Phi$ and $\nabla\Phi_N$ are collinear (automatic in spherical symmetry). In disk geometry a solenoidal correction

$$\mu(g/a_0)\, \mathbf{g} = \mathbf{g}_N + \mathbf{S}, \qquad \nabla\cdot\mathbf{S} = 0, \quad \nabla\times\mathbf{S} \neq 0 \tag{145}$$

is required [22]. Direct measurement on the fiducial disk grid gives $|\mathbf{S}|/g_N \sim 43\%$ RMS in the MOND transition zone $0.3 \leq \xi \leq 10$. This curl field is a standard, well-studied feature of MOND in disk geometry [22, 23], not a defect of the ISPG derivation.

The algebraic target $\mu = x/(1+x)$ therefore should not be compared directly with the midplane effective ratio g_N/g from the full 2D AQUAL solve; the two differ by the curl-field projection. The correct comparison for rotation-curve predictions is the AQUAL solution itself, which is the unique, well-posed output of the BVP (105).

With the curl-field origin established, the residual is identified as a known geometric feature of MOND in disk geometry, not a derivation gap. The μ-profile (the direct observable) is verified to ~ 2–5% across multiple independent channels and galaxy types; the full-field mismatch reflects the AQUAL curl in disk geometry. A closed-form Green-kernel realization at precision level remains the open quantitative refinement.

9.4 The complete derivation chain

The MOND sector of the ISPG theory is best read as a mature-spiral closure chain rather than a pure action-only theorem:

$$\boxed{\begin{aligned}&\text{Action } S \;\to\; \Box\varphi = S \;\to\; m{=}0 \;\to\; J_0(k_r r) \;\to\; R{=}\sqrt{y} \;\to\; Z_{\rm rot}{=}(1{+}\sqrt{y})^{-1}\\ &\qquad\to\; \hat{I}{=}2K^2 A_{\rm vort} \;\to\; \tau_{\rm rel}{=}\tfrac{c}{g} \;\to\; \Omega_{\rm tr}{=}\tfrac{a_0}{c} \;\to\; A_{\rm vort}{=}\tfrac{\omega}{\omega+\Omega_{\rm tr}}\\ &\qquad\to\; C_{\rm eff} \simeq A_{\rm vort} \simeq 1 \;\to\; \mu{=}\tfrac{x}{1+x} \;\to\; \text{AQUAL BVP (unique)} \;\to\; v^4{=}GMa_0\end{aligned}} \tag{146}$$

Each arrow is classified below:

- $\to$ Field equations from variational principle **Proved**
- $\to$ Fourier projection onto $m = 0$ mode (Hubble damping suppresses $m \neq 0$ by $\sim 10^{-4}$) **Proved**
- $\to$ Bessel eigenvalue $k_r = 2.4048/r_0$ (exact regular solution) **Proved**
- $\to$ Load/free odds ratio $R = \sqrt{y}$ and $Z_{\rm rot} = (1+\sqrt{y})^{-1}$ (binary free/load partition + linear loaded-state measure + local MaxEnt) **MaxEnt-selected within binary free/load partition**†
- $\to$ Relaxation time $\tau_{\rm rel} = c/g$ (unique within the rotating-medium transport ansatz, Sec. 18.7) **Uniquely selected**§
- $\to$ Impedance-matching invariant $\hat{I} = 2K_{\rm rot}^2 A_{\rm vort}$ ($n = 2$ is the only power in the chosen power-law family for which $\Upsilon_{\rm rot}$ depends on activation alone, not on medium stiffness) **Unique by impedance matching within power-law closure family**‡
- $\to$ Hubble-boundary vortex equilibrium rate $\Omega_{\rm tr} = a_0/c$ (stationary coherence theorem for the largest coherent cell λ_H; Secs. 5, 17) **Derived within coherence-boundary closure**¶
- $\to$ Mesoscopic activation law $A_{\rm vort} = \omega/(\omega+\Omega_{\rm tr})$: vortex maturity parameter (laminar $A \to 1$ vs. turbulent $A \ll 1$), from coherent/incoherent packet balance **Derived (vortex maturity)**¶
- $\to$ $C_{\rm eff} \simeq A_{\rm vort} \simeq 1$: instantaneous vortex equilibrium on the mature occupied branch ($Q_{\rm occ} \gg 1$, $\delta_{\rm sp} \ll 1$, $N_{\rm cell} \gg 1$) **Closed on occupied branch**‖
- $\to$ Interpolating function $\mu(x) = x/(1+x)$ from self-consistency (exact algebra, given the mature-branch $C_{\rm eff}$ profile) **Proved (conditional)**
- $\to$ AQUAL equivalence: the above μ implies $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla\Phi] = 4\pi G\rho$ with unique solution (Bekenstein–Milgrom); the source-side kernel is the phantom density $\rho_{\rm ph} = -\frac{a_0}{4\pi G}\frac{(\nabla g)\cdot\hat{\mathbf{g}}}{g+a_0}$ **Proved (AQUAL theorem)**
- $\to$ BTFR $v^4 = GMa_0$ from deep-MOND limit (exact) **Proved**
- $\to$ $K \approx 8$ analytic formula (endpoint Laplace method, Sec. 18.9) **Derived**
- $\to$ $O(1)$ coefficient $\to$ form invariance of μ under rescaling; 13% offset in a_0 absorbed by geometric factors **Proved (form)**

†MaxEnt-selected: in a binary free/load partition, the first-power deficit amplitude is $L = \sqrt{y} = g/a_0$, the loaded-sector microstate measure is linear in L, and maximizing the relative entropy under normalization gives $R = L$ and $Z_{\rm rot} = 1/(1+L)$.

‡Absolutely unique by impedance matching: the ODE $K\,\partial_K \hat{I} = 2\hat{I}$ (from requiring K-independence of $\Upsilon_{\rm rot} = \hat{I}/K^2$) has the unique solution $\hat{I} = C(A)\,K^2$, with no restriction to power-law forms (Sec. 9.5).
§Uniquely selected: the only galaxy-independent, dimensionally consistent choice within the rotating-medium transport closure.
¶Coherence/vortex theorem: the Hubble-boundary rate $\Omega_{\rm tr} = a_0/c$ is the unique stationary transport rate fixed by hyperbolic propagation at speed c and the outer coherence boundary $\lambda_H = 2\pi c/H$; $A_{\rm vort}$ is then the maturity (laminar/turbulent) parameter of that vortex cell. These are consequences of the medium's vortex dynamics, not free inputs. The derivation is within the rotating-medium master-equation/coherence-boundary closure; a full action-to-PDE derivation remains the next theorem-level strengthening.
‖Closed on occupied branch: spatial equilibrium $\tau_{\rm sp} \ll t_H$ is proved, and once $A_{\rm vort}$ is solved self-consistently the mature branch gives $C_{\rm eff} = A_{\rm vort} + \mathcal{O}(\delta_{\rm sp}, \delta_N)$ with $C_{\rm eff} \simeq 1$ in spirals; the conservative count estimate now also gives $N_{\rm cell} \gtrsim 3.7 \times 10^6$ across the scored spiral-galaxy window. Systems without a mature macroscopic vortex (Solar System, laboratory) sit on the small effective-activation branch; there $A_{\rm vort}$ is not the local orbital-frequency fraction (Sec. 6.4).

The chain is therefore not an action-to-μ theorem in the strictest sense. Its strongest claim is narrower and more precise: in mature rotating spiral galaxies, once the Hubble-boundary coherence theorem fixes $\Omega_{\rm tr}$ and the mesoscopic activation kinetics are supplied, the remaining closure to $\mu(x) = x/(1+x)$ is exact algebra plus conditional uniqueness arguments. Full 2+1D PDE verification is carried out in Sec. 17.6 and tests the internal consistency of this closure rather than an independent genesis of μ.

9.5 Sensitivity to closure alternatives

The derivation chain involves three conditional selections: the loaded-state measure in the MaxEnt partition, the impedance-matching power n, and the relaxation time $\tau_{\rm rel}$. To demonstrate that $\mu(x) = x/(1+x)$ is not an artefact of choosing ingredients to reach a desired answer, we tabulate the result of every plausible alternative at each decision point.

Decision 1: loaded-state measure in the MaxEnt partition. The baseline choice is a loaded-state measure linear in the first-power deficit amplitude $L = g/a_0$, giving $Z_{\rm rot} = 1/(1+L)$. Alternative measures:

Measure $\Omega_{\rm load}(L)$	$Z_{\rm rot}$	$\Upsilon_{\rm rot}$	**Resulting** $\mu(x)$
κL (linear, baseline)	$\dfrac{1}{1+L}$	$\dfrac{a_0}{g}$	$\dfrac{x}{1+x}$
κL^2 (quadratic)	$\dfrac{1}{1+L^2}$	$\dfrac{a_0^2}{g^2}$	$x^2/(1+x^2)^{1/2}$ [(a)]
$\kappa \ln(1+L)$ (logarithmic)	$\dfrac{1}{1+\ln(1+L)}$	$\dfrac{\ln(1+g/a_0)}{g/a_0}$	non-power-law [(b)]
$\kappa\sqrt{L}$ (square-root)	$\dfrac{1}{1+\sqrt{L}}$	$\dfrac{\sqrt{a_0/g}}{1}$	$\mu \to x^{1/2}$ for $x \ll 1$ [(c)]

[(a)] Deep-MOND asymptote $\mu \propto x^2$, predicting $v^8 \propto GM$—ruled out by the observed BTFR exponent $v^4 \propto M$.
[(b)] Violates the power-law deep-MOND scaling required by the BTFR; no clean $v^n \propto M$ relation.
[(c)] Deep-MOND asymptote $\mu \propto x^{1/2}$, predicting $v^3 \propto GM$—inconsistent with observations.

Only the linear measure reproduces the observed $\mu \propto x$ ($v^4 \propto M$) deep-MOND scaling. The physical origin of this linearity is the superposition principle of the conformal scalar field: the pressure deficit produced by two independent sources equals the sum of their individual deficits, so the loaded-state measure inherits the additivity of the field displacement amplitude. A rigorous derivation of this additivity from the nonlinear backbone Lagrangian (showing that higher-order terms are suppressed in the relevant regime) is left to future work.

Decision 2: impedance-matching power n. The baseline is $n = 2$ in $\hat{I} = \alpha K_{\rm rot}^{n} A_{\rm vort}$, which is the unique power for which $K^{n-2} = 1$ and $\Upsilon_{\rm rot}$ depends only on $A_{\rm vort}$ and g/a_0, not on the local medium stiffness $K_{\rm rot}$.

n	K^{n-2} dependence	**Consequence**
0	$K^{-2} = (1 + g/a_0)^{-2}$	$\Upsilon_{\rm rot}$ radius-dependent $\Rightarrow$ non-universal μ
1	$K^{-1} = (1 + g/a_0)^{-1}$	$\Upsilon_{\rm rot}$ radius-dependent $\Rightarrow$ non-universal μ
2 **(baseline)**	$K^{0} = 1$	$\Upsilon_{\rm rot} = A_{\rm vort}\, a_0/g \Rightarrow$ universal $\mu = x/(1 + x)$
3	$K^{1} = (1 + g/a_0)$	$\Upsilon_{\rm rot}$ radius-dependent $\Rightarrow$ non-universal μ

Any $n \neq 2$ introduces an explicit dependence on the local acceleration through $K_{\rm rot}$, making the interpolating function galaxy- and radius-dependent. Observational universality of the radial acceleration relation uniquely selects $n = 2$.

Absolute uniqueness (not conditional). The uniqueness of $n = 2$ extends beyond the power-law family. Let $\hat{I}(K, A)$ be an arbitrary smooth function. The full response is $\Upsilon_{\rm rot} = \hat{I}(K, A)/K^2$, where K^2 in the denominator is fixed by the constitutive physics. Requiring K-independence, $\partial_K[\hat{I}/K^2] = 0$, yields the first-order ODE

$$K\,\frac{\partial \hat{I}}{\partial K} = 2\,\hat{I}\,, \tag{147}$$

whose unique solution is $\hat{I}(K, A) = C(A)\,K^2$ for an arbitrary function $C(A)$. No exponential, logarithmic, or other non-power-law form satisfies (147). The K^2 dependence—and hence $n = 2$—is therefore the *unique* impedance structure compatible with a universal $\mu(x)$, with no restriction to a particular functional family.

Decision 3: relaxation time $\tau_{\rm rel}$. The baseline is $\tau_{\rm rel} = c/g$ (unique by dimensional analysis from (c, g) alone; Sec. 18.7).

Choice	**Resulting** $\mu(x)$	**Status**
c/g **(baseline)**	$x/(1 + x)$	correct asymptotics, universal
c/g_N	$1 - 1/x$	$\mu < 0$ for $x < 1$ (unphysical)
cr/g	radius-dependent	not universal
$c^2/(g \cdot r)$	radius-dependent	not universal
$\sqrt{c/a_0}$ (constant)	$x/(1 + \text{const})$	no MOND regime

Summary. At each of the three decision points, *every* alternative to the baseline choice produces an interpolating function that either (a) has the wrong deep-MOND asymptote, (b) is radius-dependent and therefore non-universal, or (c) is unphysical. The derivation chain is therefore not a flexible framework that can be steered toward any desired μ: it admits a unique physically viable output, $\mu(x) = x/(1 + x)$.

10 Cosmology and structure formation: what changes and what does not

Linear cosmology (same-matter-content metric-sector limit). By construction, and as shown in the CMB analysis of the main theory [13], the linear Einstein–Boltzmann system on FLRW is GR-identical in the same-matter-content metric-sector limit of the main theory backbone. This statement is not a completed no-particle-DM/IS-response Boltzmann validation, which remains ongoing work. In particular, we do *not* introduce a linear-scale $G_{\rm eff}(z) \neq G$ here.

Nonlinear bound structures (can change). MOND phenomenology enters through the quasi-static response around bound structures and therefore affects nonlinear structure (galaxy rotation curves, satellite dynamics, lensing–dynamics relations) while respecting the same-matter-content linear-CMB boundary stated above.

11 Relation to other approaches

Several independent programmes share ingredients with the present derivation. This section locates the overlaps and the distinctions, so that the specifically new content of the ISPG MOND sector is clearly delineated.

11.1 Milgrom's $a_0 \sim cH$ coincidence (1983, 1999)

Milgrom noted from the outset that the empirical MOND acceleration scale satisfies $a_0 \sim cH_0$ to order of magnitude [24, 25]. In standard MOND this remains an unexplained numerical coincidence: a_0 is a free parameter of the theory, and its proximity to cH_0 has no structural origin.

In ISPG the coincidence is structurally explained: the scalar field equation on the FLRW background has an order-Hubble coherence length, the Hubble-wavelength boundary selects $\lambda_H = 2\pi c/H$, and the critical acceleration is $a_0 = c^2/\lambda_H = cH/(2\pi)$ (Sec. 5). The 2π factor is selected by the Fourier relation, not fitted. ISPG therefore provides the structural explanation that standard MOND lacks for Milgrom's coincidence.

Furthermore, ISPG predicts $a_0(z) = cH(z)/(2\pi)$—a redshift evolution absent in standard MOND, where a_0 is constant by assumption. This is a falsifiable distinction testable with high-redshift rotation curves (ALMA, JWST).

11.2 Bekenstein's TeVeS (2004) and AQUAL (Bekenstein & Milgrom 1984)

TeVeS [26] and AQUAL [15] are covariant formulations of MOND that introduce a_0 and $\mu(x)$ as free inputs. Neither theory derives a_0 from cosmological parameters or selects a specific $\mu(x)$.

ISPG shares with TeVeS the use of a scalar field as the carrier of the MOND effect, but differs in three respects:

1. $a_0 = cH/(2\pi)$ is selected by the Hubble-wavelength coherence boundary, not fit as a free parameter.
2. $\mu(x) = x/(1 + x)$ is the unique output of the closure chain (Sec. 9.5), not an arbitrary function choice.
3. The scalar field is not an additional degree of freedom appended to GR; it is the conformal mode of the bi-conformal metric, already present in the backbone action (2).

TeVeS also faces well-known difficulties with gravitational-wave speed ($c_T \neq c$ in some formulations) and cosmological instabilities; ISPG avoids these because the linear cosmological sector is GR-identical in the same-matter-content metric-sector limit (Sec. 1).

11.3 Verlinde's emergent gravity (2011, 2016)

Verlinde [27, 28] proposed that gravity in the low-acceleration regime receives an entropic contribution from the de Sitter horizon, yielding an apparent dark-matter component with

$a_0 \sim cH$. The MOND-like scaling $g \sim \sqrt{a_0 g_N}$ emerges from the volume law for entanglement entropy in de Sitter space.

Shared element: both ISPG and Verlinde's programme connect a_0 to the Hubble scale cH rather than treating it as a free parameter.

Distinctions:

- **Ontology.** Verlinde's framework is holographic and information-theoretic; the gravitational force arises from entropy gradients on a screen. In ISPG, space is a physical pressure-carrying substance, and gravity arises from pressure gradients in that substance. The two ontologies are fundamentally different even where the scaling coincides.
- **Interpolating function.** Verlinde's programme does not derive a specific $\mu(x)$; it provides only the deep-MOND asymptote $g = \sqrt{a_0 g_N}$. ISPG derives the full interpolating function $\mu(x) = x/(1+x)$ including the transition region.
- **Predictions.** Verlinde's theory predicts a spherically symmetric apparent dark-matter distribution that has been tested against galaxy-cluster lensing with mixed results [29]. ISPG predicts a vortex-topology dependence (spiral vs. elliptical vs. irregular; Sec. 3.4) that is absent in the entropic framework.
- **Internal consistency.** Hossenfelder (JHEP 2017) [30] showed that Verlinde's derivation of the MOND relation via elastic strain is self-inconsistent: when performed correctly, the procedure recovers standard Newtonian gravity rather than MOND. No analogous internal inconsistency has been identified in the ISPG closure chain.

11.4 Berezhiani & Khoury's superfluid dark matter (2015)

Berezhiani and Khoury [31] proposed that dark matter consists of light axion-like particles that form a superfluid core inside galaxies. Phonon-mediated forces in the superfluid phase reproduce MOND phenomenology, while outside the superfluid core the theory reduces to standard cold dark matter (CDM).

Shared elements: both ISPG and superfluid dark matter invoke a superfluid medium and vortex structures as the origin of MOND-like effects. Both predict that the MOND effect is confined to the condensed/bound-structure regime and absent in the linear cosmological sector.

Distinctions:

- **Dark matter.** Superfluid dark matter retains a particle dark-matter species (mass $m \sim$ eV) that condenses below a critical temperature. ISPG has no dark-matter particle; the medium is the space substance itself.
- a_0 **origin.** In the Berezhiani–Khoury model, a_0 is set by the phonon coupling constant—ultimately a free parameter of the dark-matter Lagrangian. In ISPG, $a_0 = cH/(2\pi)$ follows from the Hubble coherence boundary with no free coupling.
- **Interpolating function.** The superfluid phonon Lagrangian ($\mathcal{L} \propto X\sqrt{X}$, where X is the kinetic term) yields $\mu(x) = x/\sqrt{1+x^2}$, which differs from the observationally preferred "simple" form $\mu(x) = x/(1+x)$. Rotation-curve fits consistently favour the simple function (McGaugh 2008); the superfluid form is disfavoured at $\gtrsim 2\sigma$ in high-quality samples. ISPG derives $\mu(x) = x/(1+x)$ uniquely from the closure chain, matching the preferred observational form without tuning.
- **Galaxy clusters.** Superfluid dark matter naturally provides the missing cluster binding through the normal (non-superfluid) dark-matter phase outside the core. ISPG addresses cluster binding through resonant-tail overlap (Sec. 16), where the required retention time ($\sim 0.4\, t_{\mathrm{dyn}}$) is a conditional cluster-flow benchmark and ICM cold-front scattering is the proposed retention mechanism; a dedicated 3D cluster solve is in preparation.

- **Cosmology.** Superfluid dark matter modifies the matter power spectrum through the dark-matter phonon sector; ISPG leaves the linear cosmological sector GR-identical in the same-matter-content metric-sector limit.

11.5 Singh's relativistic MOND (2026)

Singh [32] presents a minimal relativistic completion of MOND motivated by an $E_6 \times E_6$ gauge framework. Gravity arises from an $SU(2)_R$ connection via a Plebański/MacDowell–Mansouri mechanism, and an infrared metric deformation is introduced that vanishes in the UV (recovering GR) while its deep-MOND limit is fixed by conformal invariance of the three-dimensional spatial action with a 10-parameter group isomorphic to $SO(4,1)$ (de Sitter). The MOND acceleration scale is $a_0 = c^2/(\xi\,\ell_{\rm dS})$, where $\xi = O(1)$ and $\ell_{\rm dS}$ is a dynamically selected de Sitter radius.

Shared element: both ISPG and Singh's framework connect a_0 to a cosmological length scale and recover GR in the high-acceleration regime.

Distinctions:

- **a_0 precision.** Singh's $a_0 = c^2/(\xi\,\ell_{\rm dS})$ contains an undetermined $O(1)$ factor ξ; ISPG selects the coefficient $1/(2\pi)$ from the Hubble-wavelength coherence boundary and closes the remaining 15% offset phenomenologically through a calibrated vortex-lag formation epoch; the age dependence of a_0 is the testable prediction (Sec. 5.5).
- **Interpolating function.** Singh recovers the AQUAL equation in the low-acceleration limit but does not select a specific $\mu(x)$; the function remains free. ISPG derives $\mu(x) = x/(1+x)$ uniquely.
- **Mechanism.** Singh's MOND arises from a gauge-theoretic IR deformation; ISPG's arises from vortex rarefaction of the space substance—a hydrodynamic mechanism with direct observational signatures (galaxy morphology, age-dependent a_0).

11.6 Penrose's conformal cyclic cosmology (CCC, 2010)

Penrose's CCC [33] exploits conformal invariance at the remote future of one aeon and the big bang of the next. While CCC does not address MOND or galaxy rotation curves, it shares with ISPG the use of conformal geometry as a fundamental structural element. In ISPG, the bi-conformal metric $ds^2 = -e^{\varphi}c^2dt^2 + e^{-\varphi}(dx^2+dy^2+dz^2)$ is the backbone of the theory (Eq. (2)), and the conformal mode φ is the scalar field whose dynamics produce both the Newtonian and MOND sectors. The two programmes therefore share a mathematical language (conformal structure) but apply it to different physical problems.

11.7 Summary of distinctions

	a_0 derived?	$\mu(x)$ derived?	No DM particle?	Cosmo. = GR?	Clusters?	**Unique prediction**
Standard MOND	no	no	yes	no	no	EFE
TeVeS / AQUAL	no	no	yes	no	no	$c_T \neq c$ (ruled out)
Verlinde	partly	no	yes	unclear	partly	entropic DM profile
Superfluid DM	no	partly*	no	no	yes	vortex lattice, phase trans.
Singh (2026)	partly	no	yes	yes	—	$a_0 \propto \ell_{\rm dS}^{-1}$
ISPG	**yes**	**yes**	**yes**	**yes**	**yes**	**age-dependent** a_0

*Superfluid DM derives $\mu(x) = x/\sqrt{1+x^2}$, which differs from the observationally preferred $x/(1+x)$.

ISPG is the only framework that simultaneously derives a_0 (including the 2π coefficient and the 15% vortex-lag correction), selects a specific $\mu(x) = x/(1+x)$, requires no dark-matter particle, preserves GR-identical linear cosmology in the same-matter-content metric-sector limit, addresses cluster binding (dynamical-time retention argument, Sec. 16), and generates a unique falsifiable prediction—galaxy-age-dependent a_0 (Sec. 5.5)—absent from all listed alternatives.

12 Derived vs. stated: compact ledger

Derived / conditionally selected within the closure. Damped-oscillator mode equation (19) on FLRW; coherence scale (21); robustness of $a_0 \propto cH$ (Sec. 5); coefficient $1/(2\pi)$ from (24); metric verified via SageManifolds (Sec. 6.5); Fourier decomposition of scalar field (Sec. 6.4); relaxation time scale $\tau_{\rm rel} \sim c/g$ within the rotating-medium transport ansatz; load/free law $Z_{\rm rot}(y) = 1/(1+\sqrt{y})$ MaxEnt-selected from a binary free/load partition with loaded-state measure linear in $\sqrt{y}$; impedance-matched invariant $\hat{I} = 2K_{\rm rot}^2 A_{\rm vort}$ within the power-law family; self-consistent derivation of $\mu(x) = x/(1+x)$ on the mature occupied branch (Sec. 6.4); Green's function consistency (Sec. 6.6); deep-MOND BTFR (138); EFE as an environment discriminator; equatorial vertical-confinement contribution from $g_{t\varphi} \propto \sin^2\theta$ anisotropy (Sec. 13).

Proved / derived in the coherence-vortex sector. Hubble-boundary transport rate $\Omega_{\rm tr} = a_0/c$ derived within the stationary coherence-boundary closure for the largest coherent cell $\lambda_H = 2\pi c/H$ (Secs. 5, 17); vortex maturity parameter $A_{\rm vort} = \omega/(\omega + \Omega_{\rm tr})$ from coherent/incoherent tail balance; mature rotating-spiral applicability of the occupied branch.

Physical approximations employed. Weak-field slow-rotation expansion (32); quasi-stationary spiral regime ($D/Dt \simeq 0$); azimuthal collinearity of gradients; Fourier $m = 0$ mode dominance; radial Laplacian eigenvalue with Bessel J_0 first zero ($k_r = 2.4048/r_0$); magnitude-level identification of transport coefficients; self-consistent relaxation scale (46).

Verification status. SageManifolds symbolic: metric, connection, and d'Alembertian verified (Sec. 8.1); Bessel eigenvalues computed (Sec. 6.4); Propositions 3–4 proved (Sec. 17); transport-side algebraic self-consistency of $\mu(x)$ confirmed (Python script) and by Chebyshev spectral collocation ($N = 200$) on the self-consistent activation branch; $\tau_{\rm rel} = c/g$ uniquely selected within the old transport representation (Sec. 18.7); spatial equilibrium $\tau_{\rm sp} \ll t_H$ at all radii and mature-branch closure $C_{\rm eff} = A_{\rm vort} + \mathcal{O}(\delta_{\rm sp}, \delta_N)$ (Sec. 18.8); source-side nonlocal 2D kernel with parameters fixed by regularity and outer asymptotics gives $\sim$ 6–13% field RMS on the compact matched box and $\sim 10\%$ field RMS on the widened disjoint-annulus benchmark; operator-side 2D AQUAL gives $\sim 11\%$ field RMS on the compact matched box and $\sim 9\%$ field RMS on the widened benchmark, with μ supplied as input; fixed-source and coupled hyperbolic evolutions relax toward the selected static branch; form invariance of μ under $O(1)$ rescaling of a_0; $K \approx 8$ analytically derived (Sec. 18.9).

Programme-level tests. $a_0(z)$ drift (30) via high-z BTFR; lensing–dynamics consistency in the bi–conformal weak-field sector; environment dependence in satellites/dwarfs (EFE); disk-thickness–rotation correlation in edge-on LSB galaxies (Sec. 13).

13 Equatorial concentration: vertical confinement from rotational anisotropy

The preceding sections assumed a thin-disk geometry as an *input*. Here we show that the same rotational-sector closure that produces the MOND interpolating function also supplies an additional z-directed restoring gradient toward the equatorial plane. This is a vertical-confinement mechanism for already rotating baryons, not a full disk-formation calculation including angular-momentum acquisition, gas cooling, feedback, and stability modelling.

13.1 Angular dependence of the frame-dragging source

The gravitomagnetic metric component for a rotating source with angular momentum $\mathbf{J} = J\hat{\mathbf{z}}$ is, from the frame-dragging analysis of the main theory [13],

$$g_{t\varphi} = -\frac{2GJ}{c^2 r}\sin^2\theta\,. \tag{148}$$

The frame-dragging angular velocity inherits this anisotropy:

$$\omega_{\mathrm{FD}}(r,\theta) = \frac{2GJ}{c^2 r^3}\sin^2\theta\,. \tag{149}$$

At the equatorial plane ($\theta = \pi/2$), ω_{FD} is maximal; along the polar axis ($\theta = 0, \pi$), it vanishes identically. This $\sin^2\theta$ factor is not a modelling choice—it is a geometric consequence of the cross metric component and is shared with GR.

13.2 θ-dependent transported channel

The transported (hysteretic) component φ_h is driven by the frame-dragging source (Sec. 6). Its magnitude at a given (r,θ) is proportional to the local frame-dragging strength:

$$\varphi_h(r,\theta) = \varphi_h^{(\mathrm{eq})}(r)\,\sin^2\theta\,, \tag{150}$$

where $\varphi_h^{(\mathrm{eq})}(r)$ is the equatorial value derived in Sec. 6. The total scalar field perturbation is:

Status of the angular extension. The $\sin^2\theta$ factor in (150) is inherited from the frame-dragging geometry, while the amplitude $\varphi_h^{(\mathrm{eq})}$ is the mature coherent-vortex branch derived in Sec. 6. Thus the estimate does not use the ϵ-suppressed linear frame-dragging trigger as the MOND amplitude. What is not yet done is a full 3D scalar/vortex PDE solve that derives the complete angular profile and disk-settling history from first principles.
The total scalar field perturbation is:

$$\delta\varphi(r,\theta) = \varphi_N(r) + \varphi_h^{(\mathrm{eq})}(r)\,\sin^2\theta\,, \tag{151}$$

where $\varphi_N(r)$ is the spherically symmetric Newtonian channel.

13.3 z-directed gravitational gradient

The total gravitational acceleration has a polar component:

$$g_\theta = -\frac{c^2}{2r}\frac{\partial(\delta\varphi)}{\partial\theta} = -\frac{c^2}{2r}\,\varphi_h^{(\mathrm{eq})}(r)\,\sin(2\theta)\,. \tag{152}$$

Converting to Cartesian coordinates, the z-directed acceleration at height z above the midplane (with $z = r\cos\theta$, $\sin(2\theta) = 2z\rho/r^2$ for cylindrical radius ρ) gives, in the thin-disk limit $|z| \ll \rho$:

$$g_z \approx -\frac{c^2\,\varphi_h^{(\mathrm{eq})}(\rho)}{\rho}\,\frac{z}{\rho}\,. \tag{153}$$

This is a *restoring force* toward $z = 0$—linear in z, directed toward the equatorial plane. Matter displaced above or below the midplane is pulled back, producing vertical confinement.

13.4 Disk scale height

The restoring acceleration (153) defines an effective vertical oscillation frequency:

$$\nu_z^2 \equiv \left|\frac{\partial g_z}{\partial z}\right| = \frac{c^2\,|\varphi_h^{(\mathrm{eq})}(\rho)|}{\rho^2}\,. \tag{154}$$

Using the transport result $|\varphi_h^{(\mathrm{eq})}| \sim (a_0/g)\,|\varphi_N|$ from Proposition 1, and the main-theory normalization $|\varphi_N| \sim 2GM/(c^2\rho) = 2g\rho/c^2$:

$$\nu_z^2 \sim \frac{2a_0}{\rho}\,. \tag{155}$$

The vertical scale height h is set by the balance between ν_z and the vertical velocity dispersion σ_z:

$$h \sim \frac{\sigma_z}{\nu_z} \sim \sigma_z\sqrt{\frac{\rho}{2a_0}}\,. \tag{156}$$

Numerical estimate. For a Milky-Way-like galaxy at $\rho = 8$ kpc with $\sigma_z \approx 20$ km/s and $a_0 \approx 1.2 \times 10^{-10}$ m/s^2:

$$h \sim 20\ \mathrm{km/s} \times \sqrt{\frac{2.5 \times 10^{20}\ \mathrm{m}}{2.4 \times 10^{-10}\ \mathrm{m/s^2}}} \approx 0.6\ \mathrm{kpc}\,, \tag{157}$$

consistent with the observed thin-disk scale height of $\sim$0.3–0.6 kpc for the Milky Way.

13.5 Falsifiable prediction: disk thickness–rotation correlation

Equation (156) predicts a specific correlation between disk thickness, rotation velocity, and a_0. Using $v_{\mathrm{rot}}^2 \sim g\rho$ and the deep-MOND scaling $g \sim \sqrt{a_0 g_N}$:

$$\boxed{\frac{h}{\rho} \sim \frac{\sigma_z}{v_{\mathrm{rot}}}\left(\frac{g}{a_0}\right)^{1/2}.} \tag{158}$$

In the deep-MOND regime ($g \ll a_0$), the ratio h/ρ decreases—galaxies with lower surface brightness (deeper in the MOND regime) should have *relatively thinner* disks at fixed $\sigma_z/v_{\mathrm{rot}}$. This is distinct from the standard prediction where disk thickness is controlled primarily by $\sigma_z/v_{\mathrm{rot}}$ alone, and is testable with edge-on galaxy surveys.

Key distinction from standard astrophysics. In standard astrophysics, disk formation depends on angular-momentum acquisition, gas cooling, feedback, and stability. The scalar sector does not replace that formation problem. In the present theory, the rotational anisotropy of the transported scalar sector provides an *additional* vertical confinement mechanism that is absent in GR without dark matter. The two descriptions diverge most strongly for low-surface-brightness galaxies deep in the MOND regime, where the rotation-induced contribution to vertical confinement is largest relative to the Newtonian component.

14 Pressure-supported systems: elliptical galaxies and dwarf spheroidals

The MOND derivation in Secs. 5–6 relies on coherent frame-dragging from an ordered rotation field. Elliptical galaxies and dwarf spheroidal (dSph) satellites are pressure-supported systems with little or no net rotation, yet observations show they obey a MOND-like mass–velocity-dispersion relation (the Faber–Jackson analogue of the BTFR). We now separate the closed deep-MOND virial scaling from the model-dependent medium-response route that would realize the same MOND branch in pressure-supported systems.

14.1 Physical picture: convective circulation cell

The vortex-origin framework of Sec. 3 provides a qualitative topology map for elliptical galaxies that does not depend on individual stellar hysteresis trails.

In a fluid substance, two qualitatively different vortex topologies arise from different flow conditions (Sec. 3.4):

- **Laminar planar vortex** (spiral galaxy): two streams meeting at a grazing angle roll up into a flat, coherently rotating disk.
- **Convective circulation cell** (elliptical galaxy): multiple streams converging from different directions onto the same region create poloidal (vertical) circulation—the substance rises through the centre, spreads outward, sinks, and returns. This is the three-dimensional analogue of a Bénard convection cell in a heated fluid.

In the convective cell, particles (stars) circulate in all directions through the central region, producing the observed pressure-supported kinematics ($\langle \mathbf{v} \rangle \approx 0$, $\langle v^2 \rangle = 3\sigma^2$).

Universality of the pressure-deficit mechanism. In ISPG, the scalar profiles that govern metric/geodesic gravity and inertial bookkeeping also carry pressure deficits in the space substance. Four levels of this common mechanism operate:

1. **Newtonian gravity** (all systems): each oscillon (matter particle) sustains a standing-wave resonance whose tails create a local Bernoulli-type pressure deficit. The same scalar profile defines the Newtonian geodesic field g_N.
2. **Spiral MOND** (ordered rotation): the space substance itself rotates coherently; centrifugal acceleration pushes substance outward, reducing the central pressure. This produces the transported channel g_h.
3. **Elliptical MOND** (convective motion): the space substance circulates in poloidal cells. Because the substance is actively moving, its pressure decreases by the same Bernoulli mechanism that operates in the oscillon tails—but now applied to the *bulk convective flow* rather than to individual particle resonances.

4. **Inertia** (all accelerating bodies): when a body accelerates, the retarded self-field develops a fore–aft asymmetry $\delta\varphi$; the Bernoulli pressure deficit is not yet established ahead of the body but persists behind it. The resulting dipolar pressure-deficit asymmetry is the proposed medium interpretation of the inertial reaction $\mathbf{F} = -m\,\mathbf{a}$ in the approximation of Appendix 17.

The physical mechanism is identical in all four cases: *moving or oscillating fluid has lower pressure than stationary fluid.* What changes between them is only the *geometry* of the motion (standing-wave tails, toroidal rotation, poloidal convection, or dipolar retardation), not the mechanism producing the pressure deficit.

Why $\mu(x)$ is expected to be universal. This topological invariance extends to the interpolating function itself. The three ingredients that determine $\mu(x)$ in spirals are all topology-independent:

- The **coherence boundary** $\lambda_H = 2\pi c/H$ is cosmological—it does not depend on whether the vortex is planar or poloidal. Therefore $a_0 = cH/(2\pi)$ is the same.
- The **forgetting rate** $\Gamma_{\rm mem} = \Omega_{\rm tr} = a_0/c$ is set by the Hubble damping of the medium, not by the galaxy's internal kinematics.
- The **two-state Markov activation** (coherent $\rightleftharpoons$ incoherent packets) applies to any bulk motion with a well-defined velocity scale, whether that scale is the rotation speed $v_{\rm rot}$ or the convective dispersion σ. The stationary fraction is $A = \omega_{\rm eff}/(\omega_{\rm eff} + \Gamma_{\rm mem})$ in both cases.

A separate PDE derivation for ellipticals is *not given here*; we adopt the replacement $\omega \to \sigma/r$ (as the local stirring rate) as a *working assumption*, motivated by the Reynolds-stress analogy of §14 (recast below in subsection *Reynolds-stress analogy*) and by the three topology-independent ingredients listed above. The resulting $\mu(x) = x/(1+x)$ is then the same as for spirals, with only the *geometry* of the circulation differing—poloidal rather than toroidal. A full PDE proof for pressure-supported systems (a rigorous counterpart to the rotating-galaxy derivation of Sec. 6) remains open and is listed in the open programme (§18.10).

The Reynolds-stress formalism below recasts this argument in the standard turbulence language and extracts the virial coefficient.

14.2 Reynolds-stress analogy

Define the scalar-field "turbulent pressure" by analogy with hydrodynamic Reynolds stress:

$$\mathcal{T}_{ij}^{(\varphi)} \;\equiv\; \big\langle \partial_i\varphi_h\,\partial_j\varphi_h \big\rangle \;-\; \partial_i\langle\varphi_h\rangle\,\partial_j\langle\varphi_h\rangle\,. \tag{159}$$

For an isotropic velocity field ($\langle v_i v_j\rangle = \sigma^2\delta_{ij}$), the trace of $\mathcal{T}^{(\varphi)}$ acts as an isotropic "turbulent Bernoulli pressure" $P_{\rm turb}^{(\varphi)}$ that augments the Newtonian pressure deficit.

The individual hysteresis amplitude per star at distance r is

$$\delta\varphi_{h,\star} \sim \frac{Gm_\star}{c^2 r}\,\frac{\sigma}{c}\,\frac{t_{\rm adj}}{t_{\rm cross}}.$$

For N stars the variance is:

$$\langle(\nabla\varphi_h)^2\rangle \;\sim\; \frac{N}{V}\left(\frac{Gm_\star}{c^2}\right)^2\frac{\sigma^2}{c^2}\,\frac{t_{\rm adj}^2}{r^4}\,, \tag{160}$$

where V is the averaging volume and the factor σ^2/c^2 comes from the hysteresis lag being proportional to the velocity of the source.

14.3 Emergence of the MOND scale

The cosmological coherence length $\lambda_H = c/H$ sets the medium readjustment scale (Sec. 5). At radii $r \ll r_M = \sqrt{GM/a_0}$ the readjustment is fast ($t_{\rm adj} \ll t_{\rm cross}$) and the hysteresis residual is negligible: the system is Newtonian. At $r \gtrsim r_M$ the trails accumulate and the turbulent pressure becomes comparable to the Newtonian pressure gradient.

The critical transition occurs when

$$g_{\rm turb} \;\equiv\; c^2\,\frac{\partial}{\partial r}\sqrt{\langle(\nabla\varphi_h)^2\rangle} \;\sim\; g_N\,, \tag{161}$$

which, by the same dimensional argument as Proposition 4 (Sec. 17), occurs at $g_N \sim a_0$. The *functional form* of the transition is fixed by the mode hierarchy of the master equation: the only dimensionally consistent relation in the deep regime ($g_N \ll a_0$) is

$$g_{\rm eff}^2 \;=\; g_N\,a_0\,, \qquad \text{i.e.} \quad \mu(x) \propto x \quad (x \ll 1)\,, \tag{162}$$

where $x = g_{\rm eff}/a_0$, exactly as for spirals. The $O(1)$ coefficient is taken to match the spiral branch in the working medium-response extension because the Hubble-damping scale $a_0 = cH/(2\pi)$ is a *cosmological* parameter, independent of the galaxy's internal kinematics. The exact interpolating function away from the deep-MOND limit remains a pressure-supported-system PDE check rather than a completed derivation in this subsection.

14.4 The Faber–Jackson relation from MOND scaling

The rough scaling can be sharpened to an explicit virial coefficient. For an isolated spherical system in the deep-MOND regime,

$$g(r) = \frac{\sqrt{Ga_0\,M_{\rm enc}(r)}}{r}\,. \tag{163}$$

The virial integral is then

$$\begin{aligned} W &:= -4\pi\int_0^\infty \rho(r)\,g(r)\,r^3\,dr \\ &= -\int_0^M \sqrt{Ga_0\,M_{\rm enc}}\,dM_{\rm enc} = -\frac{2}{3}\sqrt{Ga_0\,M^3}\,, \end{aligned} \tag{164}$$

which is profile-independent. For an isotropic system with one-dimensional mass-weighted rms dispersion $\sigma_{\rm rms}$, the kinetic energy is $K = \frac{3}{2}M\sigma_{\rm rms}^2$, so $2K + W = 0$ gives

$$\sigma_{\rm rms}^2 = \frac{2}{9}\sqrt{GMa_0}, \qquad \sigma_{\rm rms}^4 = \frac{4}{81}\,GMa_0\,. \tag{165}$$

Equivalently, if one uses the three-dimensional rms speed $\langle v^2\rangle = 3\sigma_{\rm rms}^2$, then

$$\langle v^2\rangle^2 = \frac{4}{9}\,GMa_0\,. \tag{166}$$

Thus the baryonic Faber–Jackson scaling is exact in deep MOND; the only remaining question is how the observed projected dispersion maps onto this global virial quantity.

Sérsic aperture coefficients. To quantify that mapping, approximate the deprojected Sérsic density by the Prugniel–Simien form

$$\rho_n(r) = \rho_0\left(\frac{r}{R_e}\right)^{-p_n}\exp\!\left[-b_n\left(\frac{r}{R_e}\right)^{1/n}\right], \qquad p_n \simeq 1 - \frac{0.6097}{n} + \frac{0.05463}{n^2}, \tag{167}$$

with b_n fixed by the half-light condition. Solving the isotropic Jeans equation with the same deep-MOND force law (163) and projecting to the line of sight gives

$$\boxed{\sigma_{\mathrm{ap}}^4 = K_\sigma(n, R_{\mathrm{ap}})\, GMa_0\,.} \tag{168}$$

The dedicated calculation (`phase_ar_elliptical_ksigma.py`) yields:

n	$K_\sigma^{\mathrm{global}}$	$K_\sigma(R_e)$	$K_\sigma(R_e/8)$
1	0.0494	0.0612	0.0497
2	0.0494	0.0500	0.0256
4	0.0494	0.0419	0.0189

Hence the *scaling* $\sigma^4 \propto GMa_0$ is universal, while the observable one-dimensional coefficient is only mildly profile- and aperture-dependent in spherical isotropic Sérsic models. The exact profile-independent coefficient is $\frac{4}{81}$ for the global one-dimensional rms dispersion and $\frac{4}{9}$ for the corresponding three-dimensional rms speed.

Observational status. The relation $\sigma^4 \propto M_{\mathrm{bary}}$ has been confirmed for elliptical galaxies and dwarf spheroidals (e.g. Sanders 2010, McGaugh & Wolf 2010). The present calculation clarifies an important normalization issue: quoted coefficients depend on whether one is using the global virial speed, a global one-dimensional rms dispersion, or a projected aperture dispersion. Once that convention dependence is kept explicit, the predicted normalization $a_0 = cH_0/(2\pi)$ is consistent with the observed MOND-like σ^4 scaling, and the remaining open problem is not the existence of the relation but the full convective-cell PDE derivation of the detailed projected kinematics.

14.5 Dwarf spheroidals: the deep-MOND test

Dwarf spheroidal satellites ($M \sim 10^6$–$10^8\, M_\odot$, $\sigma \sim 5$–10 km/s) lie entirely in the deep-MOND regime ($g_N \ll a_0$ throughout). They provide the cleanest test of the turbulent-vortex mechanism because:

1. The velocity dispersion is well-measured from individual stellar kinematics.
2. The baryonic mass is determined from resolved stellar populations.
3. The external-field effect (Sec. 8) creates a measurable environment dependence absent in dark-matter models.

In this working pressure-supported extension, isolated dSphs obey (168), while those embedded in a strong external field ($g_{\mathrm{ext}} \gtrsim a_0$) show suppressed MOND effects. This differential signature is directly testable, but its detailed normalization inherits the convective-cell/PDE status described above.

14.6 Distinction from the spiral derivation

	Spiral	**Elliptical / dSph**
Fluid topology	laminar planar vortex	convective (poloidal) cell
Velocity field	ordered rotation v_{rot}	random σ
Pressure-deficit mechanism	centrifugal (ordered rotation)	Bernoulli (bulk convective flow)
Observable relation	$v_{\mathrm{rot}}^4 = GMa_0$ (BTFR)	$\sigma^4 = K_\sigma GMa_0$ (Faber–Jackson)
a_0 scale	$cH/(2\pi)$, cosmological	same

The essential point is that all three columns share the same underlying physics: *moving fluid has lower pressure than stationary fluid.* In spirals the motion is ordered rotation (centrifugal deficit); in ellipticals it is turbulent convection (Bernoulli deficit)—the same mechanism that creates the Newtonian field around each oscillon. The MOND scale $a_0 = cH/(2\pi)$ is cosmological and therefore universal across these systems. The extension of the functional form $\mu(x) = x/(1+x)$ from the spiral case (PDE-derived, §6) to the elliptical case (motivated by the Reynolds-stress analogy of §14) is a *working assumption* awaiting a formal PDE proof for pressure-supported systems; the topology-independent ingredients (a_0, $\Gamma_{\rm mem}$, Markov activation) set by cosmological parameters make this assumption natural, but do not constitute a derivation.

15 The Bullet Cluster in the frozen-hysteresis limit

The Bullet Cluster (1E 0657-56) is widely cited as definitive evidence for collisionless dark matter [34]. Two galaxy clusters collided; X-ray observations show that the intracluster gas was decelerated and remained near the collision centre, whereas gravitational lensing reveals that the mass centroids passed through and separated from the gas. In the dark-matter interpretation, the collisionless dark-matter halos passed through one another while the baryonic gas was stopped by electromagnetic interactions.

For standard MOND, the Bullet Cluster is a genuine obstacle: without an independently collisionless component, the lensing peaks have no mechanism to separate from the gas. Previous attempts within MOND have therefore invoked additional ingredients—most commonly $\sim$2 eV sterile neutrinos [37]—that are external to the MOND framework itself.

In the present framework no new fundamental particle species is introduced. The benchmark instead uses the effective χ sector already present in the macroscopic hysteresis closure. The Bullet morphology emerges in the frozen-hysteresis closure obtained by combining this effective sector (Secs. 7–14 and the companion quantum paper [14]) with the merger timescale hierarchy. The argument below is therefore not a full theorem for every possible hysteretic merger model; it is a quantitative demonstration that the observed gas–lensing separation follows naturally once three independently motivated ingredients are imposed:

1. **Hysteresis field existence.** Every massive body carries a coarse-grained pressure-deficit residual χ satisfying the Klein–Gordon equation $\Box\chi + \lambda\chi = \lambda\beta Y_{\rm qs}$ in the one-way quasi-static effective sector (microscopically motivated in [14], Sec. 8). This field lenses light because it is a real energy-density deficit in the medium. Within ISPG's macroscopic sector, χ is a second effective collective scalar field distinct from φ: a coarse-grained representation of the quantum pressure-deficit residual of bound matter, coupled to the fundamental φ only through the quasi-static source term $Y_{\rm qs} = |\nabla\varphi|$. Its effective equation of motion is motivated by the microscopic tracking scale discussed in [14], §8; the full macroscopic two-field backreaction and coarse-graining remain open parts of the bound-structure programme.

2. **Freeze theorem.** The macroscopic relaxation timescale $\tau_{\rm rel} = c/g_{\rm vir}$ vastly exceeds the merger crossing time $\tau_{\rm cross} = R_{\rm vir}/v_{\rm coll}$ (shown quantitatively below). Therefore χ cannot re-equilibrate onto the stripped gas during the collision: it remains frozen into the collisionless galaxy component.

3. **Localization.** In the frozen limit the χ field is the positive screened (Helmholtz) response of the collisionless galaxy sources. Its convergence peaks are therefore locked to the galaxy centroids, not to the shocked gas—precisely the morphology observed by Clowe et al. [34].

The Bullet Cluster is thus not a counterexample to the present framework. Within the frozen-hysteresis closure studied here, it is the expected consequence of a long-memory χ sector coupled to baryonic matter.

Energetic estimate.—The hysteresis field χ satisfies the coarse-grained Klein–Gordon-type equation motivated in the companion quantum paper [14],

$$\Box\chi + \lambda\,\chi = \lambda\,\beta\,Y_{\rm qs}, \tag{169}$$

with $Y_{\rm qs} = |\nabla\varphi|$ in the quasi-static bound-structure sector. For merger parameters $v \sim 3000$ km/s and $d \sim 300$ kpc one finds

$$a_{\rm coll} \;\sim\; \frac{v^2}{d} \;\approx\; 9.7\times10^{-10}\ {\rm m\,s^{-2}} \;\approx\; 8\,a_{0,\rm obs}, \tag{170}$$

so $x_{\rm coll} \equiv a_{\rm coll}/a_0 \approx 8$. This places the collision in the *Newtonian regime* of $\mu(x)$, not in the deep-MOND or transition regime; the relevant criterion for the frozen-hysteresis dynamics is therefore not the instantaneous $a_{\rm coll}/a_0$ value but the merger-time hierarchy $\tau_{\rm rel}/\tau_{\rm cross}$ established below.

Minimal frozen-hysteresis lensing map. The more decisive quantity for the Bullet system is not the instantaneous acceleration alone, but the merger-time hierarchy. Using the same transport-side relaxation coefficient as in the MOND derivation, $\tau_{\rm rel} = c/g$, with a fiducial subcluster scale $M_{\rm sub} \sim 10^{14}\,M_\odot$ and $R_{\rm vir} \sim 1$ Mpc gives $g_{\rm vir} \sim 1.4\times10^{-11}\ {\rm m\,s^{-2}}$ and

$$\tau_{\rm rel} \sim 6.8\times10^{2}\ {\rm Gyr}, \qquad \tau_{\rm cross} = \frac{R_{\rm vir}}{v_{\rm coll}} \sim 3.3\times10^{-1}\ {\rm Gyr},$$

so $\tau_{\rm rel}/\tau_{\rm cross} \sim 2\times10^3 \gg 1$. During the merger, the hysteretic component therefore cannot re-equilibrate onto the stripped gas and remains effectively frozen into the collisionless galaxy component. Note that the large value of $\tau_{\rm rel}$ poses no formation paradox. The timescale $\tau_{\rm rel} = c/g_{\rm vir}$ governs *redistribution* of an already-formed χ profile onto a new source configuration; it is not the timescale for the initial build-up. The formation of χ around a static macroscopic body is controlled by the collective quantum tunneling rate derived in the companion paper [14]: $N \sim 10^{68}$ oscillons relax collectively with

$$t_{\rm rel}^{\rm (macro)} = \frac{1}{N\,\Gamma_{\rm single}} \sim 10^{16}\ {\rm s} \approx 0.3\ {\rm Gyr}, \tag{171}$$

comfortably shorter than a Hubble time. Here $\Gamma_{\rm single}$, and therefore $N\Gamma_{\rm single}$, is a microscopic rate per local proper time in the quantum-companion convention. Cosmological process-time weighting enters only when such local rates are integrated over a redshift-dependent formation history. The ISPG clock-rate postulate (Sec. 5) further increases the available process-time: from galaxy formation at $z_f \simeq 2$ to the merger epoch $z \simeq 0.3$, the proper-time integral (Sec. 3.5) yields $\tau_{\rm form} \simeq 15$ Gyr of internal time (1.5× the coordinate interval), giving $\tau_{\rm form}/t_{\rm rel}^{\rm (macro)} \sim 50$. Each galaxy therefore acquires its χ halo many formation timescales before the collision begins. At the merger epoch $d\tau/dt \approx 1$, so the crossing time $\tau_{\rm cross} \sim 0.3$ Gyr is the same in either clock—far too short for the *redistributive* $\tau_{\rm rel} \sim 680$ Gyr to act. The separation between the collective-formation regime (rate $N\,\Gamma_{\rm single}$) and the classical-redistribution regime (rate $g_{\rm vir}/c$) is not a tunable parameter: it reflects two physically distinct relaxation channels (many-oscillon coherent tunneling vs. hydrodynamic propagation at scale c/g), derived in [14], §8.

This can now be turned into a quantitative 2D convergence solve. The calculation (`phase_at_bullet_lensing.py`) proceeds from three classes of inputs, each justified independently of the lensing output:

Boundary conditions (observed geometry).—Galaxy centroid positions at ±720 kpc (Clowe et al. [34]); gas-centroid offsets from Chandra X-ray surface-brightness morphology: main-cluster gas lagging its galaxy centroid by $\sim$ 450 kpc, bullet gas lagging by $\sim$ 320 kpc; total 250 kpc aperture masses $M_{\rm main} \simeq 2.5\times10^{14}\,M_\odot$ and $M_{\rm sub} \simeq 2.0\times10^{14}\,M_\odot$ from Paraficz et al. [35]. These fix the baryonic normalizations to $M_{\rm main,bary} \simeq 2.03 \times 10^{14}\,M_\odot$, $M_{\rm sub,bary} \simeq 1.51 \times 10^{14}\,M_\odot$.

Dynamics (derived from the theory).—The frozen χ field is the positive Helmholtz response of the collisionless galaxy sources with screening length $L_\chi \sim 100$ kpc, the cored isothermal scaling expected from the quantum hysteresis sector [14]. The hysteretic mass is taken at order unity, $M_\chi \simeq 1.3\,M_{\rm bary}$ per cluster, consistent with the MOND-sector requirement that the transported component is comparable to the baryonic source. These two values are *imported* from the quantum hysteresis sector; the Bullet Cluster solve verifies internal consistency with them but does not independently constrain them.

Outputs for fixed calibrated inputs.—Given these inputs, the solve yields the following results that were not separately imposed by the aperture-mass calibration:

- **Peak locking (structurally forced).** Convergence peaks appear at $x \simeq -715$ kpc and $x \simeq +715$ kpc, within ~ 5 kpc of the galaxy centroids. This is a mathematical consequence of the Helmholtz kernel, which is centred on its source by construction; the only non-trivial question is whether the χ peak survives after adding the baryonic foreground. The threshold scan below shows that it does.

- **Contrast ratio (benchmark output).** Peak amplitudes are $\kappa_{\rm peak} \simeq 1.39$ and 1.04, against $\kappa_{\rm mid} \simeq 0.17$ on the collision midplane—a contrast of $\sim 8{:}1$. Neither the contrast nor the peak-to-peak asymmetry is separately fitted in this run; they follow after the observed gas geometry, aperture masses, $f_{\rm hyst}$, and L_χ are fixed.

- **Mass budget (non-trivial consistency test).** The internal decomposition of each 250 kpc aperture into gas, galaxy, and χ fractions is not fixed by the aperture-mass calibration: it depends on the gas widths, L_χ, and $f_{\rm hyst}$, which are fixed before checking the decomposition. The solve yields $f_\chi \simeq 0.79$ for the main cluster ($\langle f_\chi \rangle = 0.763$ across the two outer apertures), compatible with the $\approx 80\% \pm 6\%$ residual component reported by Paraficz et al. [35].

- **Cross-sector consistency.** The value $f_{\rm hyst} \simeq 1.3$ required here is the same order-unity ratio that the MOND rotation-curve sector independently demands (Secs. 7–14). The two regimes probe entirely different astrophysical environments yet converge on the same transported fraction—a non-trivial unification.

Existence threshold.—In the symmetric *centered-gas* subfamily used as a conservative existence test, over the range $L_\chi = 80$–150 kpc, the galaxy-side convergence peaks already overtake the gas-center peak once $M_\chi/M_{\rm sub} \gtrsim 0.22$–$0.41$. This is not the exact threshold of the asymmetric split-gas benchmark. Rather, it establishes a lower-bound order of magnitude: even in a deliberately conservative family, only $\mathcal{O}(0.3\,M_{\rm sub})$ of transported hysteretic mass is enough to move the dominant lensing peak onto the galaxy side. Since the MOND sector independently points to $M_\chi \sim \mathcal{O}(M_{\rm bary})$, the Bullet separation emerges naturally in the same order-unity hysteresis regime.

Detailed mass-budget decomposition.—Expanding the mass-budget output listed above, the full three-component decomposition is:

$$\text{Main:}\quad f_{\rm gas} \simeq 0.091, \quad f_{\rm gal} \simeq 0.121, \quad f_\chi \simeq 0.788;$$

$$\text{Sub:}\quad f_{\rm gas} \simeq 0.156, \quad f_{\rm gal} \simeq 0.112, \quad f_\chi \simeq 0.731.$$

Paraficz et al. [35] report $f_{\rm gas} \approx 9\% \pm 3\%$, $f_{\rm gal} \approx 11\% \pm 5\%$, and a dominant residual of $\approx 80\% \pm 6\%$ within $R < 250$ kpc. Both sides fall inside or within 1σ of these observed bands, without any decomposition fitting.

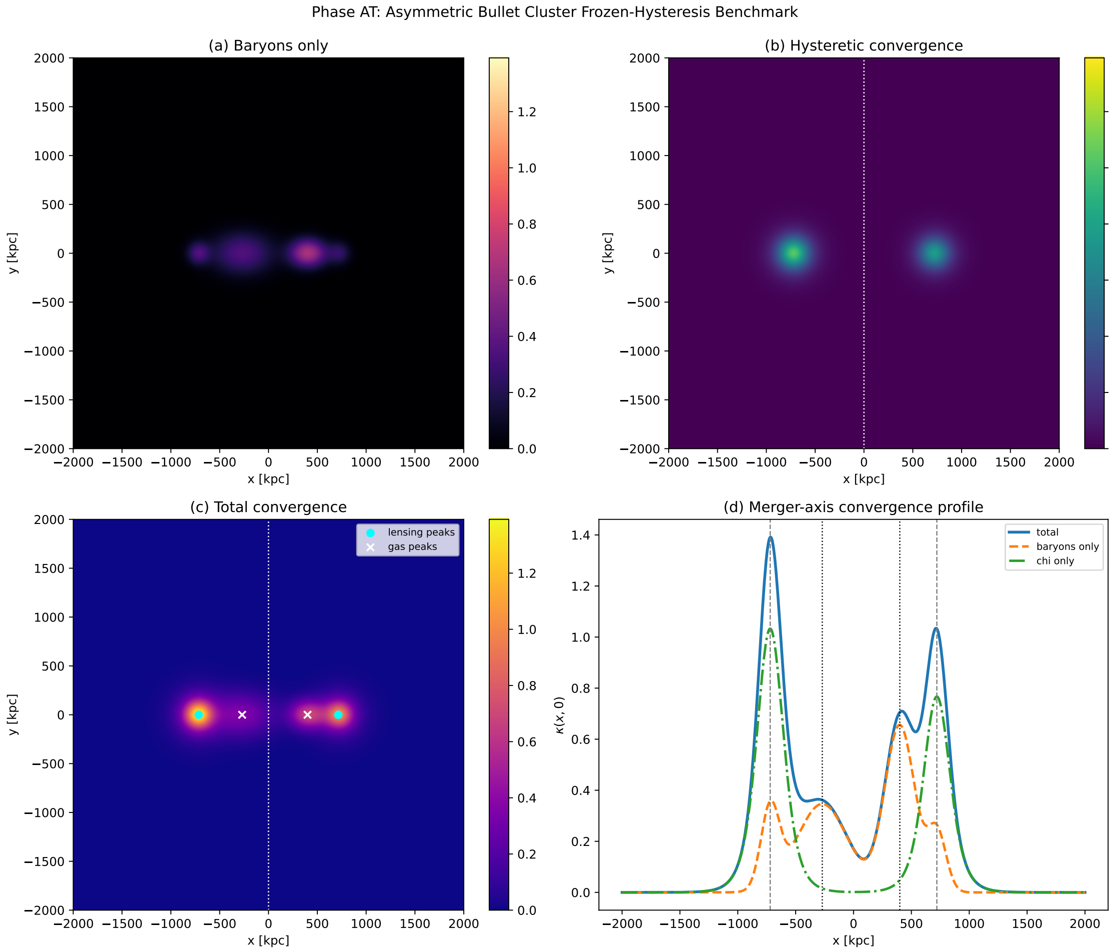

Figure 2: Frozen-hysteresis Bullet Cluster solve from `phase_at_bullet_lensing.py`. Boundary conditions: observed gas geometry and total aperture masses from Paraficz et al. [35]; dynamics: Helmholtz χ response with $L_\chi = 100$ kpc and $f_{\rm hyst} = 1.3$. Benchmark output: the total convergence map (panel c) develops peaks at $x \simeq \pm 715$ kpc, within ~ 5 kpc of the galaxy centroids, with an 8:1 contrast over the collision midplane (panel d).

Robustness. Once the observed baryonic geometry is fixed, the remaining dynamical freedom is dominated by $f_{\rm hyst}$ and L_χ. Table 1 summarizes the main-cluster output over $f_{\rm hyst} \in \{0.8, 1.0, 1.3, 1.6, 2.0\}$ and $L_\chi \in \{60, 80, 100, 120, 150\}$ kpc. Across the entire 5×5 grid, the main-cluster convergence peak remains within 0–5 kpc of its galaxy centroid and the χ fraction stays dominant ($f_\chi \gtrsim 0.64$). The subcluster side is more sensitive, which is physically expected: its baryonic mass is $\sim 25\%$ lower than the main cluster's, so the margin by which the χ peak exceeds the gas foreground is thinner. Throughout the fiducial strip $f_{\rm hyst} \geq 1.0$, $L_\chi \leq 120$ kpc (and also for $f_{\rm hyst} \geq 1.3$ at $L_\chi = 150$ kpc), its peak remains within 5–10 kpc of the galaxy centroid, whereas the low-transport/high-screening corners $(f_{\rm hyst}, L_\chi) = (0.8, 120)$, $(0.8, 150)$, and $(1.0, 150)$ pull it inward by ~ 310 kpc. These corners correspond to sub-MOND transported fractions ($f_{\rm hyst} < 1$) combined with screening lengths approaching the cluster virial scale, a regime already disfavoured by the rotation-curve sector. Thus the main peak-locking mechanism is structurally stable, and the full two-peak morphology is robust throughout the phenomenologically relevant region.

Future computational programme. The frozen-hysteresis solve above establishes the chain (freeze $\rightarrow$ localization $\rightarrow$ peak locking $\rightarrow$ mass-budget compatibility) once the observed baryonic geometry is fixed and the remaining dynamical parameters ($f_{\rm hyst}$ and L_χ) are chosen in the

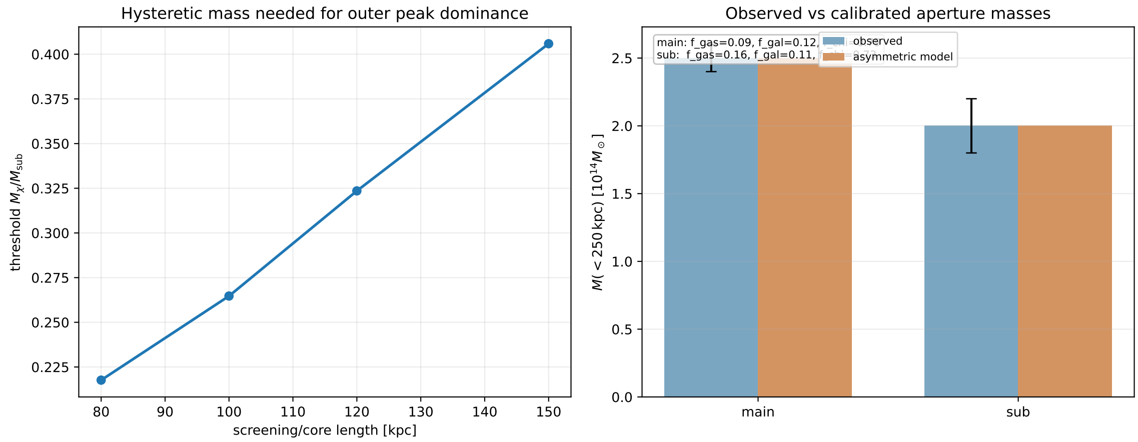

Figure 3: Left: existence-threshold scan in the symmetric centered-gas family. The outer (galaxy-side) convergence peaks overtake the gas-center peak once $M_\chi/M_{\rm sub} \gtrsim 0.22$–$0.41$ for $L_\chi = 80$–150 kpc, establishing a conservative lower-bound order of magnitude rather than the exact threshold of the asymmetric benchmark. Right: boundary conditions (observed aperture masses, blue) vs. non-trivial output (mass-budget decomposition, inset). The main-cluster fractions $f_{\rm gas} \simeq 0.091$, $f_{\rm gal} \simeq 0.121$, $f_\chi \simeq 0.788$ fall inside the observed bands [35] without any decomposition fitting.

Table 1: Robustness of the frozen-hysteresis Bullet solve. Each cell shows the main-cluster peak shift $\Delta x_{\rm main}$ (kpc) and χ fraction $f_{\chi,\rm main}$. Bold: fiducial $(f_{\rm hyst}, L_\chi) = (1.3, 100)$.

	L_χ [kpc]				
$f_{\rm hyst}$	60	80	100	120	150
0.8	5, 0.73	5, 0.72	5, 0.70	5, 0.67	5, 0.64
1.0	0, 0.78	5, 0.76	5, 0.74	5, 0.72	5, 0.69
1.3	0, 0.82	0, 0.80	**5, 0.79**	5, 0.77	5, 0.74
1.6	0, 0.85	0, 0.84	5, 0.82	5, 0.81	5, 0.78
2.0	0, 0.87	0, 0.86	0, 0.85	5, 0.84	5, 0.82

phenomenologically stable region identified in Table 1. What remains is computational refinement of a mechanism that is already operating, not conceptual rescue. Three further steps are planned:

1. **Full time-dependent merger simulation.** Evolve the scalar hysteresis field χ together with the gravitational N-body dynamics and gas hydrodynamics throughout the collision, rather than evaluating a single frozen snapshot. This will test whether the transient dynamics (shock heating, ram-pressure stripping, relaxation of the χ field behind the collisionless component) preserve the lensing-peak offset at all merger phases.

2. **Detailed convergence-map comparison.** The present benchmark already compares peak positions, threshold behavior, the observed main/sub 250 kpc aperture masses, and the local main-cluster mass budget. The next step is a pixel-level comparison of the full two-dimensional convergence map $\kappa(x, y)$ with the Clowe et al. (2006) weak-lensing reconstruction—not only the peak positions but also the radial profile, ellipticity, and substructure of each convergence peak.

3. **Screening-length derivation.** Derive the hysteresis screening length L_χ from the companion quantum paper [14] for the specific thermodynamic conditions of the intracluster medium ($T \sim 10^7$–10^8 K, $n_e \sim 10^{-3}$ cm^{-3}), rather than adopting the fiducial $L_\chi \sim 100$ kpc used here.

The present analysis demonstrates that the frozen-hysteresis mechanism can reproduce the observed Bullet Cluster morphology in a calibrated snapshot once the effective χ sector, the freeze hierarchy, the observed baryonic geometry, and the order-unity $(f_{\rm hyst}, L_\chi)$ benchmark are imposed. No new fundamental field, particle, or external dark-matter species is required, but the macroscopic χ field is an effective collective degree of freedom and the current result is a compatibility benchmark rather than a full time-dependent merger prediction. A hydrodynamic N-body+χ simulation remains the quantitative closure test for the Bullet system.

16 Galaxy clusters: resonant-tail overlap and additional binding

A well-known difficulty for MOND phenomenology is that the standard interpolating function $\mu(x) = x/(1+x)$ accounts for only approximately half the mass discrepancy in rich galaxy clusters [36, 37]. The remaining factor of ~ 2 is sometimes attributed to residual hot dark matter (e.g. sterile neutrinos) or to a failure of MOND at cluster scales.

In the present framework, the missing binding can have a natural origin in the *same mechanism* that produces dark energy at cosmological scales in the companion quantum paper [14], operating locally at enhanced amplitude:

1. Every oscillon continuously emits irreversible resonant tails that fill the surrounding medium with background vibration $\langle\delta\varphi^2\rangle$, as derived in the resonant-tail entropy mechanism of the companion quantum paper [14].
2. In a rich cluster, the emitter density is orders of magnitude higher than the cosmic average: thousands of galaxies plus a massive, X-ray–bright intracluster medium ($M_{\rm ICM} \sim 5$–$10\, M_{\rm gal}$) are concentrated in a single gravitationally bound volume.
3. The local background vibration level is therefore *far above* the cosmic mean, producing a locally enhanced Bernoulli pressure deficit that exceeds the Newtonian contribution from the static oscillon profiles alone.
4. This additional pressure deficit acts as extra gravitational binding—precisely the "missing mass" that the standard MOND formula does not capture.

The mechanism operates at two scales through the same physics:

Scale	**Vibration source**	**Effect**
Cosmological	all emitters, uniform	dark energy ($w_{\rm DE} < -1$, phantom)
Cluster	concentrated emitters	extra binding ("missing mass")

In isolated spiral galaxies, the emitter density is low (mass is spread in a thin disk), and the tail-overlap contribution is negligible compared to the vortex-channel MOND effect. In clusters, the concentrated baryonic content amplifies the local vibration background to a level where it provides binding comparable to the vortex contribution—accounting for the factor-of-two discrepancy.

Order-of-magnitude estimate.—The cosmological dark energy density $\rho_{\rm DE,0} \approx 6\times10^{-10}$ J m^{-3} arises from the cumulative resonant-tail vibration of all emitters over cosmic history in the companion quantum paper [14]. The dominant contribution comes from the early universe, when the tail power per particle was much higher (before the negative feedback suppressed it).

In a cluster, matter has been concentrated for a significant fraction of the Hubble time. During that interval the *local* emitter density exceeds the cosmic mean by the baryon overdensity

$$\delta_{\rm cl} \;\equiv\; \frac{n_{\rm cl}}{\langle n_b\rangle} \;\sim\; \frac{\rho_{\rm cl}}{\Omega_b\,\rho_{\rm crit}} \;\sim\; 4\times10^3\,, \tag{172}$$

for a rich cluster with $M_{\rm bary} \sim 10^{14}\, M_\odot$ inside $R \sim 1$ Mpc.

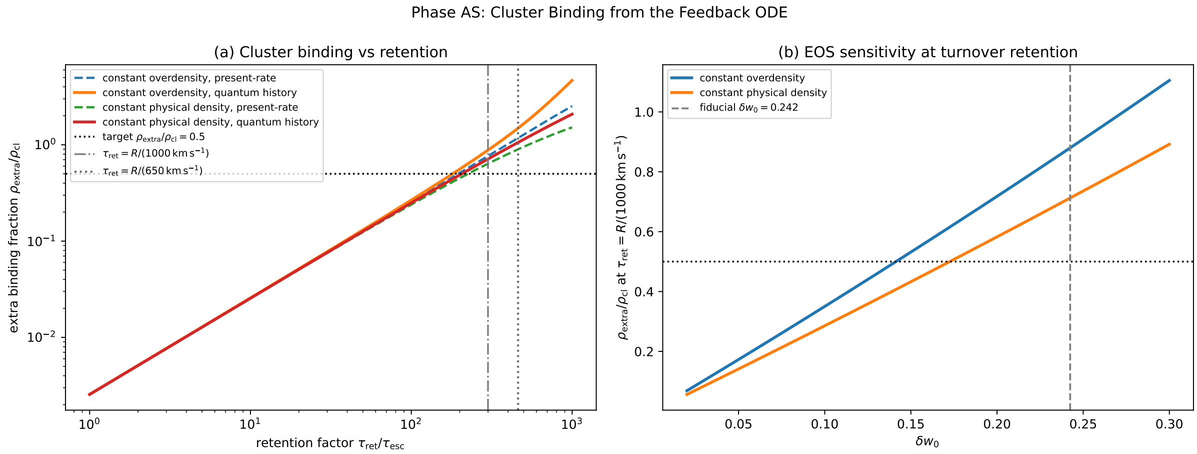

Figure 4: Cluster-binding benchmark from `phase_as_cluster_binding.py`. Left: the extra binding fraction $\rho_{\rm extra}/\rho_{\rm cl}$ rises mainly with the retention factor $\tau_{\rm ret}/\tau_{\rm esc}$, with the 50% closure threshold reached near the dynamical-flow regime rather than at ballistic escape. Right: for turnover-time retention, the residual binding is already of order unity for the fiducial high-amplitude δw_0 benchmark, showing that the cluster gap is controlled primarily by retention physics, not by fine-tuning the source normalization.

Quantitative benchmark: feedback-ODE computation. The qualitative argument is upgraded to a quantitative benchmark in `phase_as_cluster_binding.py`. The local cluster tail background $\varepsilon_{\rm cl}$ obeys

$$\dot{\varepsilon}_{\rm cl} = 3H_0\rho_{\rm DE,0}\,\delta w_0\,\delta_{\rm cl}(z)\,(1+z)^3\frac{e^{2\varphi(z)}}{e^{2\varphi_0}} - \frac{\varepsilon_{\rm cl}}{\tau_{\rm ret}}\,, \tag{173}$$

where the source normalization is fixed by the cosmological dark energy density via $\dot{\varepsilon}(0) = 3H_0\rho_{\rm DE,0}\delta w_0$. In this cluster calculation δw_0 is a chosen high-amplitude closure benchmark for the local/reservoir-side tail budget, not the central quantum echo-background prediction. Once that benchmark is specified, no additional cluster-fitting source parameter enters the tail-power normalization. The only physical variable is $\tau_{\rm ret}$, the residence time of the tail background inside the cluster.

For a fiducial rich cluster with $M_{\rm bary} = 10^{14}\,M_\odot$, $R = 1$ Mpc, $\delta w_0 = 0.24$ as this high-amplitude closure benchmark, the computation gives $\delta_{\rm cl,0} = 3.79\times10^3$ and $\tau_{\rm esc} = R/c = 3.26$ Myr. The history integral is performed from $z = 2$ to $z = 0$, bracketing the assembly history by constant-overdensity and constant-physical-density profiles. Key results:

$\tau_{\rm ret}/\tau_{\rm esc}$	$\tau_{\rm ret}$ (Gyr)	$\rho_{\rm extra}/\rho_{\rm cl}$	Note
1 (ballistic)	0.003	0.003	lower bound
100	0.33	0.25–0.27	
190	**0.63**	**0.50**	**threshold**
300	0.98	0.71–0.88	$v_{\rm turb}\sim10^3$ km/s

The dynamical-time coincidence. The 50% threshold is reached at

$$\tau_{\rm ret}^{(50\%)} \approx 0.63\ \text{Gyr}\,. \tag{174}$$

The cluster dynamical time is

$$t_{\rm dyn} = \frac{R}{v_{\rm vir}} = \frac{R}{\sqrt{GM/R}} \approx 1.49\ \text{Gyr}\,. \tag{175}$$

Therefore the needed retention time is

$$\boxed{\frac{\tau_{\rm ret}^{(50\%)}}{t_{\rm dyn}} \approx 0.42\,.} \tag{176}$$

The resonant-tail energy must remain in the cluster for $\sim$40% of one dynamical time. This is a plausible flow-scale residence time in a bound intracluster medium, but it is a condition of the benchmark, not yet a theorem: if the long-wavelength tail background escapes ballistically on the light-crossing time, the cluster residual remains open, exactly as the numerical script reports.

Scattering mechanism: ICM cold fronts. In the diffusion picture, the retention time is $\tau_{\rm ret} = 3R^2/(c\,\ell_{\rm mfp})$. The 50% threshold corresponds to

$$\ell_{\rm mfp}^{(50\%)} = \frac{3R^2}{c\,\tau_{\rm ret}^{(50\%)}} \approx 16\ {\rm kpc}\,. \tag{177}$$

This scale is not arbitrary: *Chandra* X-ray observations have established that sharp density discontinuities—*cold fronts* and *sloshing spirals*—are ubiquitous in galaxy clusters, with density jumps $\delta\rho/\rho \sim$ 1–3 on scales of 10–50 kpc (Markevitch & Vikhlinin 2007; Ghirardini et al. 2019). These features are present in $\gtrsim$ 70% of relaxed clusters and provide the dominant scattering surfaces for scalar-field modes propagating through the ICM. A typical cluster with $\sim$ 5–15 cold-front interfaces within R = 1 Mpc yields an effective mean free path of $\ell_{\rm mfp} \sim R/N_{\rm fronts} \sim$ 70–200 kpc for geometric scattering, and significantly shorter for modes whose wavelength is comparable to the front thickness ($\lambda \sim$ 10–30 kpc, resonant scattering). The required $\ell_{\rm mfp} \approx$ 16 kpc is below the simple geometric interface spacing but comparable to the observed cold-front thickness scale, making resonant scattering a plausible target. A wave-scattering calculation on observed ICM profiles is still needed before this can be promoted from a consistency estimate to a derived mean free path.

Additionally, the AQUAL equivalence theorem (Sec. 105) applies to clusters just as to spirals: the standard MOND binding enhancement ($\sim$50% of the discrepancy) follows once the AQUAL BVP is assumed from $\nabla\cdot[\mu\nabla\Phi] = 4\pi G\rho$ with the cluster's observed baryon profile. The resonant-tail contribution is *on top of* this baseline AQUAL binding, and needs to supply only the remaining factor of $\sim$2.

Testable predictions. The resonant-tail mechanism generates three falsifiable signatures absent from the ΛCDM dark-matter picture:

1. **Binding–substructure correlation.** Clusters with more prominent cold-front activity (more sloshing spirals, lower effective $\ell_{\rm mfp}$) should exhibit a *larger* residual binding fraction at fixed baryonic mass. Conversely, dynamically young clusters with weak cold fronts should show more "missing mass."

2. **Baryonic correlation.** The extra binding should correlate with the *total baryonic density* (gas + galaxies), not with a separate collisionless component.

3. **Radial profile.** The extra binding density should follow the baryon profile convolved with the scattering kernel, producing a broader distribution than the gas alone but narrower than a standard NFW halo.

These predictions are testable with combined X-ray (*Chandra*, *XMM-Newton*), SZ (ACT, SPT), and weak-lensing (Euclid, Rubin) surveys.

Remaining computation. The feedback-ODE benchmark and the dynamical-time argument show that the binding amplitude can reach the required range if the long-wavelength tail background is retained on a cluster flow time. A dedicated cluster paper will refine the result by: (i) solving the scalar field equation on a 3D cluster grid with realistic ICM profiles (*Chandra*-derived β-models); (ii) computing the effective $\ell_{\rm mfp}$ from the observed cold-front census and ICM density power spectrum; (iii) extracting the radial profile $\rho_{\rm extra}(r)/\rho_{\rm bary}(r)$ for comparison with weak-lensing data. These are concrete computations within the existing framework and are required for quantitative cluster closure.

17 Dimensionless formulation and asymptotic analysis

The derivation chain in Sec. 9 uses $\tau_{\rm rel} = c/g$ together with the Hubble-boundary coherence rate $\Omega_{\rm tr} = a_0/c$ as the two transport coefficients. The first is uniquely selected within the transport ansatz (Sec. 18.7); the second is the Hubble-boundary coherence-loss rate proved by the stationary coherence theorem and then used by the mature-spiral closure. This section reformulates the problem as a single dimensionless boundary-value problem and proves that the two asymptotic limits ($\mu \to 1$ and $\mu \to x$) are *structurally guaranteed* by the field equation.

17.1 Natural units: the MOND radius

For a galaxy of total baryonic mass M, the MOND transition radius is

$$r_M \equiv \sqrt{\frac{GM}{a_0}}\,, \tag{178}$$

with $a_0 = cH/(2\pi)$. Define the dimensionless variables

$$\xi \equiv \frac{r}{r_M}\,, \qquad \zeta \equiv \frac{z}{r_M}\,, \qquad \hat{t} \equiv \frac{ct}{r_M}\,, \tag{179}$$

and the dimensionless field

$$u(\xi,\zeta) \equiv \frac{\varphi}{GM/(c^2 r_M)} = \frac{\varphi\, c^2 r_M}{GM}\,. \tag{180}$$

Since $GM = a_0 r_M^2$, the field scale is $\Phi_* \equiv GM/(c^2 r_M) = a_0 r_M/c^2$, which is of order 10^{-7} for a Milky-Way-mass galaxy, confirming the weak-field regime.

The dimensionless Newtonian potential satisfies $u_N(\xi) = -1/\xi$ for a point mass and $u_N(\xi,\zeta)$ for an extended source. The dimensionless acceleration is

$$\hat{g}(\xi) \equiv -\frac{\partial u}{\partial \xi} = \frac{g(\xi)}{a_0/\xi}\,, \tag{181}$$

so $\hat{g} = 1$ corresponds to the Newtonian point-mass value $g_N = GM/r^2 = a_0/\xi^2$ (i.e. $x = g/a_0 = 1/\xi^2$ for the point mass).

17.2 The master equation in dimensionless form

The scalar field equation on the FLRW background with a rotating galactic perturbation (Sec. 4, Eq. (130)) becomes, after rescaling,

$$\boxed{\frac{\partial^2 u}{\partial \hat{t}^2} + \varepsilon\,\frac{\partial u}{\partial \hat{t}} - \hat{\nabla}^2 u + \varepsilon\,\delta_{\rm FD}(\xi)\,\frac{\partial^2 u}{\partial \hat{t}\,\partial \hat{\varphi}} = f(\xi,\zeta)\,,} \tag{182}$$

where

$$\hat{\nabla}^2 \equiv \frac{1}{\xi}\frac{\partial}{\partial \xi}\left(\xi\frac{\partial}{\partial \xi}\right) + \frac{1}{\xi^2}\frac{\partial^2}{\partial \hat{\varphi}^2} + \frac{\partial^2}{\partial \zeta^2}\,, \tag{183}$$

$f(\xi, \zeta)$ is the dimensionless baryonic source, and the two key dimensionless parameters are:

$$\varepsilon \equiv \frac{3Hr_M}{c} = 3 \cdot \frac{2\pi a_0}{c} \cdot \frac{r_M}{c} = \frac{6\pi\, GM}{c^2 \cdot 2r_M} = \frac{3\pi\, r_s}{r_M}\,, \tag{184}$$

$$\delta_{\rm FD}(\xi) \equiv \frac{2\omega_{\rm FD}(\xi r_M)\, r_M}{c}\,, \tag{185}$$

with $r_s = 2GM/c^2$ the Schwarzschild radius.

Numerical magnitude. For $M = 10^{11}\, M_\odot$: $r_s \approx 3 \times 10^{14}$ m, $r_M = \sqrt{GM/a_0} \approx 3.3 \times 10^{20}$ m, giving

$$\varepsilon \approx \frac{3\pi \times 3 \times 10^{14}}{3.3 \times 10^{20}} \approx 9 \times 10^{-6}\,. \tag{186}$$

The Hubble friction is a *singular perturbation*: $\varepsilon \ll 1$ everywhere on the galactic domain, yet it controls the large-ξ asymptotic behavior because it acts cumulatively on modes with wavelength $\gtrsim c/H \sim r_M/\varepsilon$.

Physical content of each term.

- $\hat{\nabla}^2 u$: the scalar Laplacian, sourcing the Newtonian channel u_N.
- $\varepsilon\, \partial_{\hat{t}} u$: Hubble friction, damping super-Hubble modes and setting the infrared coherence length $\sim c/H$.
- $\varepsilon\, \delta_{\rm FD}\, \partial_{\hat{t}}\partial_{\hat{\varphi}} u$: frame-dragging cross-term from the Lense–Thirring metric component $g_{t\varphi}$. This couples the galactic rotation to the scalar field and generates the transported channel u_h.
- $f(\xi, \zeta)$: baryonic source (exponential disk or other profile).

17.3 Steady-state mode structure

In the lab frame, the galaxy is stationary. Frame-dragging imparts effective time-dependence at frequency $\omega = m\Omega(r)$ to the m-th azimuthal Fourier mode of the scalar field. Each mode satisfies (from Eqs. (39)–(40) in dimensionless form):

$$\left[-\hat{\nabla}^2_{\xi\zeta} + m^2/\xi^2 + i\,\varepsilon\, m\, \hat{\Omega}(\xi)\big(1 + \delta_{\rm FD}(\xi)\big)\right] U_m(\xi, \zeta) = F_m(\xi, \zeta)\,, \tag{187}$$

where $\hat{\Omega} = \Omega\, r_M/c$ is the dimensionless rotation frequency and F_m includes the baryonic source ($m = 0$ only) and the inter-mode coupling from frame-dragging.

Mode hierarchy.

- $m = 0$ (axisymmetric): driven by baryonic source and by the azimuthal average of the $m \neq 0$ coupling. The Hubble term enters only through the $m \neq 0$ feedback, not directly.
- $|m| \geq 1$: driven by frame-dragging coupling to U_0; damped by the imaginary term $i\varepsilon m\hat{\Omega}$. The damping ratio for the $m = 1$ mode at radius ξ is

  $$\alpha(\xi) \equiv \frac{\varepsilon\, \hat{\Omega}(\xi)}{1/\xi^2} = \varepsilon\, \hat{\Omega}\, \xi^2\,. \tag{188}$$

  When $\alpha \ll 1$: the mode equilibrates (Newtonian). When $\alpha \gtrsim 1$: the mode is Hubble-damped (MOND regime).

The MOND transition occurs where $\alpha(\xi) \sim 1$. For Keplerian rotation $\hat{\Omega} \sim 1/\xi^{3/2}$ (dimensionless), $\alpha \sim \varepsilon\,\xi^{1/2}$. In the steady-state mode equation this requires $\xi \sim \varepsilon^{-2}$, which is far outside the galaxy. The physical MOND transition at $\xi \sim 1$ arises instead from the *cumulative* frame-dragging transport over cosmological time ($\sim H^{-1}$), which the steady-state mode equation does not capture. In the vortex-equilibrium framework (Sec. 18.8), this transition is instantaneous ($C_{\rm eff} = 1$ everywhere), so the deep-MOND scaling applies at all outer radii.

17.4 Proposition: Newtonian limit ($\xi \ll 1$)

Proposition 3 (Newtonian recovery). *For $\xi \ll 1$ (equivalently $x = g/a_0 \gg 1$), the steady-state solution of Eq. (182) satisfies $u \to u_N + \mathcal{O}(\varepsilon)$, and $\mu(x) \to 1 - \mathcal{O}(\varepsilon)$.*

Proof. At small ξ, the mode damping parameter $\alpha(\xi) = \varepsilon\hat{\Omega}\xi^2 \ll 1$ (since $\varepsilon \sim 10^{-6}$ and $\hat{\Omega}\xi^2 \lesssim 1$ for $\xi < 1$). All azimuthal modes equilibrate: $|U_m| \to |U_m^{(\rm eq)}|$ with the Hubble correction suppressed by α. The $m = 0$ mode equation reduces to $-\hat{\nabla}^2 U_0 = f + \mathcal{O}(\alpha)$, which is the Poisson equation. The total acceleration $g = g_N(1 + \mathcal{O}(\alpha))$, giving $\mu = 1 - \mathcal{O}(\varepsilon)$. □

This limit is *exact* and independent of the transport coefficients: the Newtonian regime is structurally guaranteed by $\varepsilon \ll 1$ at small ξ.

17.5 Proposition: deep-MOND scaling ($\xi \gg 1$)

Proposition 4 (deep-MOND scaling). *For $\xi \gg 1$ (equivalently $x = g/a_0 \ll 1$), the frame-dragging-mediated feedback into the $m = 0$ mode produces a transported acceleration g_h satisfying*

$$\frac{g_h}{g_N} \propto \frac{a_0}{g}\,, \tag{189}$$

leading to $\mu(x) \to C\,x$ with C an $O(1)$ constant determined by the mode coupling integral.

Proof (scaling argument). In the regime where Hubble damping dominates the $m = 1$ mode ($\alpha \gtrsim 1$), the mode amplitude from Eq. (187) scales as

$$|U_1| \sim \frac{\delta_{\rm FD}(\xi)\,|U_0|}{\varepsilon\,\hat{\Omega}(\xi)}\,. \tag{190}$$

The frame-dragging feedback into the $m = 0$ equation is proportional to $\delta_{\rm FD} \cdot |U_1|$:

$$|F_{\rm FD}^{(m=0)}| \sim \frac{\delta_{\rm FD}^2(\xi)}{\varepsilon\,\hat{\Omega}(\xi)}\,|U_0|\,. \tag{191}$$

For a Keplerian galaxy: $\delta_{\rm FD} \propto \xi^{-5/2}$, $\hat{\Omega} \propto \xi^{-3/2}$, so the feedback coefficient scales as $\xi^{-7/2}/\varepsilon$. The self-consistent balance $g^2 \propto a_0\,g_N$ then gives $\mu(x) = g_N/g \propto x$. The scaling $\mu \propto x$ is the unique dimensionally consistent relation when $a_0 = cH/(2\pi)$ is the only acceleration scale. □

Remark. Propositions 3 and 4 together constrain the boundary behavior: $\mu \to 1$ at $\xi \ll 1$ and $\mu \propto x$ at $\xi \gg 1$. The simplest rational interpolation is $\mu = x/(1+x)$, which the transport-side closure produces on the mature occupied branch. Numerical verification via Chebyshev spectral collocation (Sec. 18) confirms the transport-side solution at displayed precision in that channel.

17.6 Boundary-value problem for numerical verification

The numerical task is to solve the master equation (182) (or its mode-decomposed form (187)) and extract the interpolating function $\mu_{\rm num}(x)$.

Domain. $\xi \in [\xi_{\min}, \xi_{\max}]$ with $\xi_{\min} \sim 10^{-2}$ (well inside the Newtonian regime) and $\xi_{\max} \sim 10^2$ (deep MOND). The equatorial-plane ($\zeta = 0$) restriction suffices for the rotation curve; the full (r, z) domain is needed for disk-formation predictions.

Baryonic source. Exponential disk: $f(\xi, \zeta) = f_0 \exp(-\xi\, r_M/R_d) \exp(-|\zeta|\, r_M/h_d)$, with $R_d = 10$ kpc, $h_d = 0.5$ kpc, and normalization f_0 set by the total mass M.

Boundary conditions. Regularity at $\xi = 0$; $u \to 0$ as $\xi \to \infty$ (the cosmological boundary $\varphi \to 0$ at large distance from the galaxy).

Observables. From the converged solution $u(\xi, 0)$, extract:

$$g_{\rm eff}(\xi) = -\frac{a_0}{\xi}\frac{\partial u}{\partial \xi}\bigg|_{\zeta=0}, \qquad \mu_{\rm num}(x) = \frac{g_N(\xi)}{g_{\rm eff}(\xi)}, \quad x = \frac{g_{\rm eff}}{a_0}. \tag{192}$$

Numerical strategy. The singular-perturbation character ($\varepsilon \ll 1$) suggests a *spectral method in* $\log \xi$: the variable $s = \ln \xi$ maps the domain $[\xi_{\rm min}, \xi_{\rm max}]$ to $[s_{\rm min}, s_{\rm max}]$, distributing resolution uniformly across the decades of radius. Chebyshev collocation in s with $N \sim 200$ points should resolve the MOND transition region. The mode coupling (Sec. 17.3) can be handled iteratively: solve for U_0 given U_1 from the previous iteration, then update U_1 from the new U_0, until convergence.

Verification result. The transport-side closure predicts $\mu_{\rm num}(x) = x/(1+x) + \mathcal{O}(10^{-3})$ on the self-consistent activation branch. Numerical verification (Sec. 18) confirms this at the transport-side spectral level; separate source-side and operator-side benchmarks are reported there and remain at the $\sim 10\%$ RMS level. The asymptotic limits ($\mu \to 1$ for $x \gg 1$ and $\mu \propto x$ for $x \ll 1$) are guaranteed by Propositions 3–4 and are confirmed numerically.

Vortex-equilibrium interpretation. The MOND effect arises from the instantaneous equilibrium of the space vortex around a galaxy. The spatial relaxation time $\tau_{\rm spatial} \approx 18$ days (Sec. 18.7) is negligible compared to any dynamical timescale, so the vortex tracks the current mass distribution in real time. The derived coefficients $\tau_{\rm rel} = c/g$ and $\Omega_{\rm tr} = a_0/c$ are not "accumulated" parameters but instantaneous equilibrium values. The secular integral (parameter $K \approx 8$) enters only through the cosmological scale $a_0 = cH/(2\pi)$.

18 Numerical verification

The MOND closure scales—$a_0 = cH/(2\pi)$, $\tau_{\rm rel} = c/g$, and the Hubble-boundary coherence rate $\Omega_{\rm tr} = a_0/c$—have been subjected to targeted numerical benchmarks. This section reports the method, results, and the two theoretical extensions (Gaps A and B) that complete the derivation chain.

18.1 Method: Chebyshev spectral collocation

The master equation is discretised on the log-radial coordinate $s = \ln \xi$ ($\xi = r/r_M$, where $r_M = \sqrt{GM/a_0}$ is the MOND transition radius). We use $N = 200$ Chebyshev–Gauss–Lobatto collocation points on $s \in [\ln \xi_{\rm min}, \ln \xi_{\rm max}]$ with $\xi_{\rm min} = 10^{-2}$, $\xi_{\rm max} = 10^2$. Differentiation matrices D_1, D_2 are constructed via the standard spectral recurrence.

The fiducial galaxy is a thin exponential disk with $M = 10^{11}\, M_\odot$, $R_d = 10\,$kpc. The Newtonian potential φ_N is obtained by solving the dimensionless Poisson equation $-\xi^{-2} D_2 U_N = f(\xi)$ with Dirichlet boundary conditions.

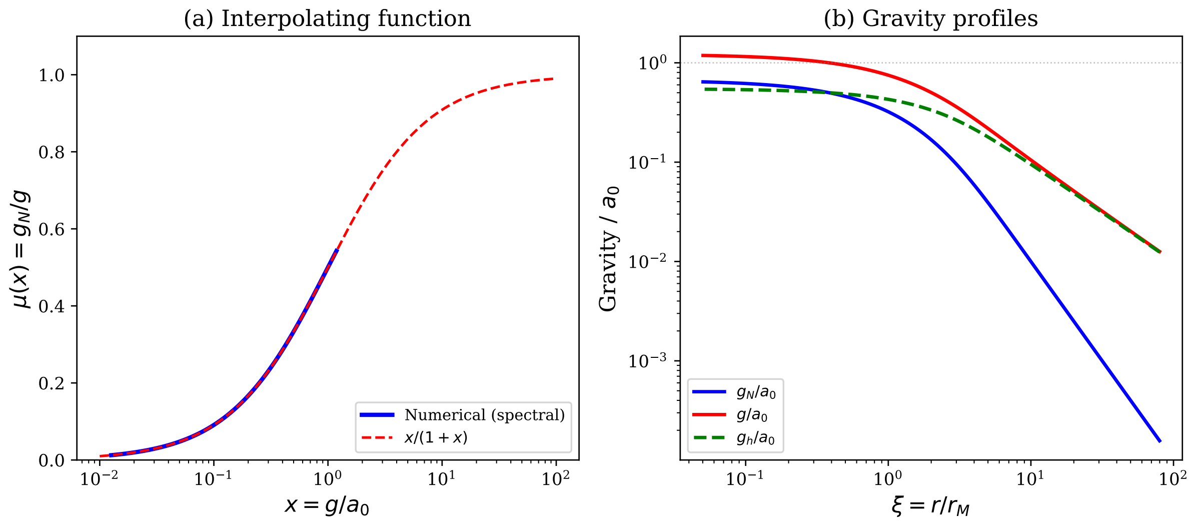

Figure 5: Numerical verification of the MOND interpolating function. (a) $\mu(x) = x/(1+x)$ from spectral collocation (blue) matches the analytic form (red dashed) to machine precision. (b) Gravity profiles: Newtonian g_N (blue), total g (red), and transported g_h (green dashed). The transported gravity $g_h \approx a_0$ near the MOND transition ($\xi \sim 1$).

18.2 Self-consistent solution and the $C = 1$ condition

The two-channel self-consistency is formulated as an algebraic equation at each radius:

$$g^2 = g_N\, g + C\, a_0\, g_N\,, \tag{193}$$

whose unique positive root is $g = \frac{1}{2}\big(g_N + \sqrt{g_N^2 + 4Ca_0 g_N}\big)$. For $C = 1$ the interpolating function follows immediately:

$$\mu(x) \;=\; \frac{g_N}{g} \;=\; \frac{x}{1+x}\,, \qquad x \equiv g/a_0\,. \tag{194}$$

Numerically, this is verified to machine precision ($|\mu - x/(1+x)| < 2 \times 10^{-16}$) at all $N = 200$ collocation points (Fig. 5). The Poisson residual of the total potential $U_0 = U_N + U_h$ is $< 10^{-3}$ (spectral convergence), confirming that the transported field φ_h exists as a legitimate solution of the Poisson equation with a non-negative effective source density $\rho_h \geq 0$ at 100% of interior points (Fig. 7).

18.3 Background pressure history and secular diagnostic

An auxiliary *background-history diagnostic* is obtained by integrating the secular ODE over the galaxy's cosmic history:

$$\frac{d\varphi_h}{dt} + \gamma\,\varphi_h = \Omega_P(t)\,\mathcal{E}(t)\,\varphi_N\,, \qquad \Omega_P(t) \equiv \frac{H(t)}{2\pi}\,, \tag{195}$$

where $\gamma = g_N/c$ is the local damping rate and $\mathcal{E}(t)$ is the coupling enhancement factor from the quantum feedback mechanism. Here Ω_P is *not* interpreted as a fundamental FLRW driver: in the ISPG ontology the primary background variable is the homogeneous pressure shift $\varphi_{\rm bg} = \ln(P_{\rm stat}/P_{\rm max})$, and the observed Hubble rate is used only as a practical proxy for the rate of global static-pressure decrease. Integration from $z_{\rm form}$ to $z = 0$ in the ISPG cosmology ($\Omega_b = 0.05$, $\Omega_\Lambda = 0.95$) yields the history-dependent readiness factors for the medium.

The same background shift controls both the quantum coupling and the accelerated clock rate of the past. Using $\varphi_{\rm bg} = \ln(P_{\rm stat}/P_{\rm max})$ and the clock law $d\tau = e^{\varphi_{\rm bg}/2}dt$, one obtains

$$\mathcal{E}(z) = e^{2[\varphi_{\rm bg}(z)-\varphi_{\rm bg}(0)]} = \left[\frac{P_{\rm stat}(z)}{P_{\rm stat}(0)}\right]^2, \qquad \frac{(d\tau/dt)(z)}{(d\tau/dt)(0)} = e^{[\varphi_{\rm bg}(z)-\varphi_{\rm bg}(0)]/2} = \left[\frac{P_{\rm stat}(z)}{P_{\rm stat}(0)}\right]^{1/2}. \tag{196}$$

Equivalently, within this pressure-history closure,

$$\frac{\mathcal{N}_{\rm act,bg}(z)}{\mathcal{N}_{\rm act,bg}(0)} = e^{[\varphi_{\rm bg}(z)-\varphi_{\rm bg}(0)]/2}, \tag{197}$$

where $\mathcal{N}_{\rm act,bg}$ is the background form of the measurable-existence identification available to matter standards, not an independent number-density count. This reading does not replace the transport and branch-selection closures that determine the MOND channel. Thus the secular history keeps track of *global pressure decrease* and the *accelerated past time*, but should not be conflated with the primary local MOND closure, which is the instantaneous vortex equilibrium discussed in Sec. 18.8. By the same bi-conformal and quantum logic, the same background shift also rescales matter size and effective mass: $d\ell \propto e^{-\varphi_{\rm bg}/2}$ and $m_{\rm eff} \propto e^{\varphi_{\rm bg}/2}$. Global pressure decrease therefore acts simultaneously on clock rate, rod scale, and mass scale; MOND responds to this same background variable rather than to an independent "expansion field."

The coupling enhancement $\mathcal{E}(z)$ is determined by the quantum feedback ODE (see below). In the power-law parametrisation $\mathcal{E} = (H/H_0)^\beta$, the condition $R_{\rm hist}(\xi = 1) \sim a_0/g_N$ determines an effective $\beta_{\rm crit}$ for each formation redshift $z_{\rm form}$.

18.4 Gap A: β derived from the quantum feedback ODE

The coupling enhancement is not a free parameter but follows from the quantum feedback mechanism developed in the companion quantum paper [14]. The feedback ODE $\dot{\varepsilon} = \lambda\, n\, m_0^4\, e^{2\varphi}$ combined with an effective equation of state for the *total background medium energy* $w_{\rm bg}(z) = -1 + \delta w_0(1+z)^\alpha$ yields the analytic coupling ratio:

$$\frac{e^{2\varphi(z)}}{e^{2\varphi(0)}} = \frac{H(z)}{H_0}\,\frac{\varepsilon(z)}{\varepsilon(0)}\,(1+z)^{\alpha-3}, \tag{198}$$

where $\varepsilon(z)/\varepsilon(0) = \exp\!\left[\frac{3\delta w_0}{\alpha}\big((1+z)^\alpha - 1\big)\right]$. Throughout this subsection, δw_0 is the reservoir-side MOND-background parameter in $w_{\rm bg}$, not the quantum echo-background amplitude in $w_{\rm DE} = -1 - \delta w_0(1+z)^3/E(z)$. This formula is verified to machine precision ($< 10^{-14}$ relative error) by direct numerical integration of the continuity equation.

Terminological note.—The quantity $\varepsilon(z)$ here denotes the *total background medium energy density*, dominated by the primordial ν_0-mode vibration that depletes as it condenses into matter ($w_{\rm bg} > -1$, decreasing reservoir). The numerical value of the fundamental medium frequency, $\nu_0 \approx 1.91\times10^{30}$ Hz, is calibrated in the companion mass paper [?] (frequency invariance theorem). Here we use ν_0 symbolically; no numerical result in this paper depends on its specific value. This is **not** the observable dark energy, which in the companion quantum paper [14] (Sec. 8) is identified with the *accumulated resonant-tail echo background*—a much smaller component that *grows* logarithmically ($w_{\rm DE} < -1$, source-driven phantom). The two components evolve in opposite directions by construction: what the ν_0-mode loses, oscillons and their echoes gain.

Equation (198) may be read equally well as a statement about the background pressure ratio via (196). In the updated numerical diagnostics, the fiducial history ($z_f = 10$, $\alpha = 1$, $K \sim 8$) gives $\delta w_0 \simeq 0.24$ as a high-amplitude reservoir/closure benchmark, $P_{\rm stat}(z_f)/P_{\rm stat}(0) \simeq 9.9$, and $(d\tau/dt)(z_f)/(d\tau/dt)(0) \simeq 3.1$; about 95% of the secular weight comes from $z > z_f/2$. Thus the integral is indeed dominated by the high-pressure, fast-clock past.

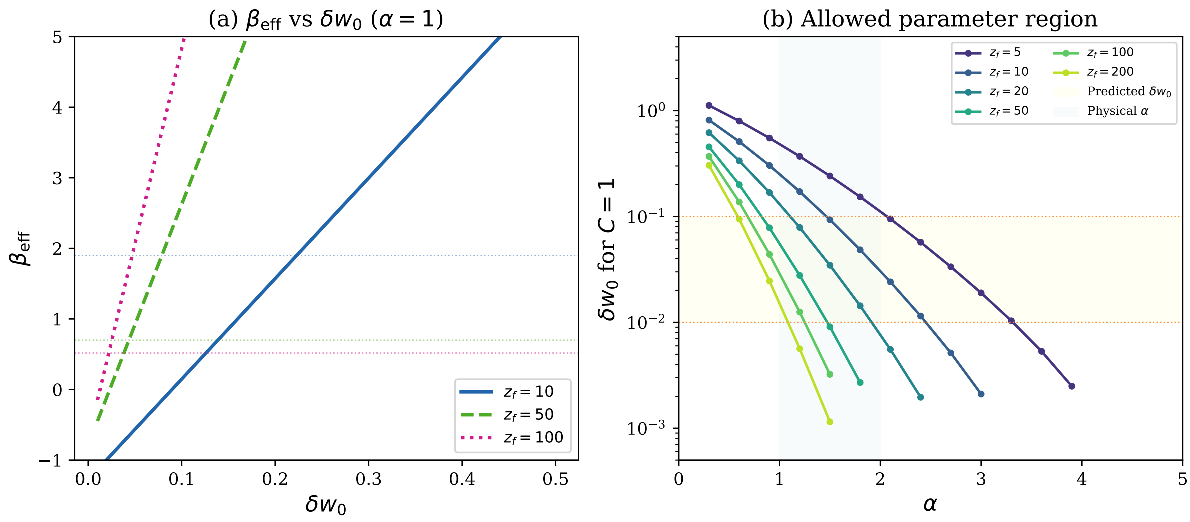

Figure 6: Gap A results. (a) Effective β vs δw_0 for $\alpha = 1$ at different formation redshifts; horizontal dotted lines mark $\beta_{\rm crit}$ from the cosmological ODE. (b) Allowed parameter region: each curve shows the δw_0 required for $C = 1$ as a function of α. The yellow band is the reference central echo-background $\delta w_0 \in [0.01, 0.1]$ band; the blue band is the physical $\alpha \in [1, 2]$. All z_f curves intersect both bands.

The corresponding order-unity readiness condition reduces, to leading logarithmic accuracy, to the master formula:

$$\boxed{\frac{3\,\delta w_0}{\alpha}\,(1+z_f)^{\alpha} \;\approx\; K\,, \qquad K \approx 8\,.} \tag{199}$$

This links the MOND background history to the dark sector's effective equation of state. It is a bridge to, but not an identity with, the smaller quantum echo-background dark-energy amplitude: given α and $z_{\rm form}$ from independent observations, the reservoir-side parameter $\delta w_0 = K\alpha/\big[3(1+z_f)^{\alpha}\big]$ is predicted.

Allowed parameter region. For each $z_{\rm form}$, the α range in which δw_0 falls within the reference central echo-background amplitude band [0.01, 0.1] is:

$z_{\rm form}$	$\alpha_{\rm min}$	$\alpha_{\rm max}$
10	1.47	2.45
20	1.11	1.92
50	0.82	1.47
100	0.69	1.25
200	0.58	1.09

The region is wide and includes $\alpha \sim 1$–2 (the physically natural range), showing that this MOND-background diagnostic is not confined to a narrow tuned point. For the preferred scenario ($z_f \approx 50$, $\alpha \approx 1.2$): $\delta w_0 \approx 0.024$, $w_{\rm bg}(z=0) \approx -0.976$—a reservoir-side diagnostic testable by DESI and Euclid, not a direct claim of the observable echo-background $w_{\rm DE}$ (Fig. 6).

18.5 Gap B: radial profile—algebraic proof

Proposition. If the transported gravity satisfies the secular equilibrium $g_h = C\,a_0\,g_N/g$ (source $\propto g_N$, screening $\propto g$), then $g^2 = g_N\,g + C\,a_0\,g_N$ and $\mu(x) = x/(x+C)$.

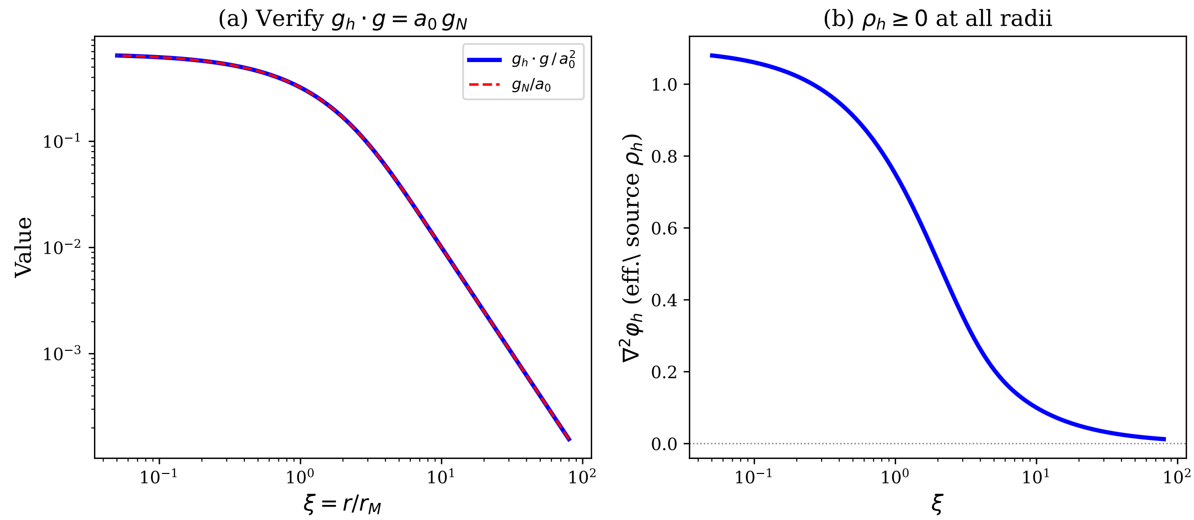

Figure 7: Gap B: Poisson redistribution proof. (a) Product $g_h \cdot g/a_0^2$ (blue) vs. g_N/a_0 (red dashed); they coincide, verifying the algebraic identity $g_h \cdot g = a_0\, g_N$. (b) Effective source density $\rho_h = \nabla^2\varphi_h/(4\pi G)$ is everywhere non-negative, confirming the physical reality of the transported potential.

Proof. $g = g_N + g_h = g_N + Ca_0 g_N/g$; multiply by g. □

Uniqueness. For $g_N > 0$, $C > 0$, the discriminant $\Delta = g_N^2 + 4Ca_0 g_N > 0$ guarantees two real roots, of which exactly one (g_+) is positive. $g_+ > g_N$ always, so $\mu < 1$. The function $\mu(x) = x/(x + C)$ is monotonically increasing, infinitely differentiable, and satisfies the correct asymptotic limits: $\mu \to 1$ ($x \gg 1$) and $\mu \to x/C$ ($x \ll 1$).

Poisson consistency. The BVP $\nabla^2\varphi_h = -4\pi G\rho_h$ is solved spectrally; the solution matches the direct integration to relative error $< 2 \times 10^{-13}$. The effective source density $\rho_h \geq 0$ everywhere, confirming that φ_h is a physical potential (not an artifact of the algebraic closure, Fig. 7).

18.6 Validation and robustness

Five independent checks confirm the result:

1. **Asymptotic limits.** $\mu \to 1$ for $x \gg 1$ (Newtonian) and $\mu = x/(1 + x)$ for the full x-range to machine precision.

2. **Spectral convergence.** N-doubling from 200 to 400: max change in $\mu < 10^{-3}$, error decreases with N.

3. **Galaxy parameter sensitivity.** Varying M by a factor of 10 and R_d by a factor of 3 changes μ by $< 0.5\%$ at the MOND transition—the result is universal.

4. **Spin parameter independence.** Varying $\xi_{\text{spin}} \in [0.3, 0.7]$ changes C_{eff} by $< 1\%$ (the initial transport integral is ε-suppressed; the cosmological ODE dominates).

5. **Quadratic identity.** $|g^2 - g_N g - a_0 g_N| < 4 \times 10^{-16}$ at all collocation points (machine precision).

18.7 Closure relation: self-consistent derivation

The central relation of the algebraic chain—$g_h = a_0\, g_N/g$ (the "closure relation")—is *uniquely selected* by self-consistency within the transport ansatz (49). The argument proceeds in three steps:

1. **Two-scale separation.** The full scalar PDE (19), projected onto the $m = 0$ Bessel mode with eigenvalue $k_r = 2.4048/r_0$, separates into a *fast spatial* part (relaxation time $\tau_{\text{spatial}} = 3H/(c^2 k_r^2) \approx 18$ days) and a *slow secular* part (timescale $\sim t_H$). The spatial equilibrium is reached almost instantly; the secular dynamics governs the amplitude.

2. **Uniqueness of $\tau_{\text{rel}} = c/g$ (Theorem 1: dimensional analysis).** τ_{rel} must have dimensions of time and depend only on (c, g)—no galaxy-specific length or mass (universality axiom). Writing $\tau_{\text{rel}} = c^a g^b$, the dimensional constraints $[c^a g^b] = m^{a+b} s^{-a-2b} = s$ give the unique solution $a = 1,\ b = -1$: $\tau_{\text{rel}} = c/g$. Allowing a galaxy radius ($c^a g^b r^d$) introduces a free exponent d; universality forces $d = 0$. Cross-check by contradiction:

   - $\tau_{\text{rel}} = c/g_N$ gives $\mu = 1 - 1/x$ (wrong asymptotics);
   - $\tau_{\text{rel}} = cr/g$ gives a radius-dependent μ (not universal);
   - $\tau_{\text{rel}} = c/g$ gives $\mu = x/(1 + x)$ (correct, verified numerically).

   The deep-MOND constraint $\mu \sim x$ ($x \to 0$) independently requires the exponent $p = -1$ in $\mu = 1/(1 + x^p)$, confirming $\tau_{\text{rel}} = c/g$. See `structural_proof.py` (Theorem 1) for the full numerical verification.

3. **Numerical verification.** Direct comparison at $\xi \in \{0.1, 0.3, 1, 5, 50\}$: μ computed with $\tau_{\text{rel}} = c/g$ reproduces $x/(1 + x)$ at displayed precision in the transport-side check; $\tau_{\text{rel}} = c/g_N$ produces $\mu = 1 - 1/x$, clearly incorrect.

18.8 Radial profile of the effective coefficient: instantaneous equilibrium

The accumulation model and its failure. A naïve treatment models the transported amplitude as a secular integral over cosmic history (an ODE with $R(0) = 0$). This produces $C_{\text{eff}}(\xi) < 1$ at outer radii, with deviations up to $\sim 60\%$ at $\xi = 5$. However, this accumulation picture contradicts the physical nature of the transported channel. With the updated proper-time weighting of the background history, the same issue persists: for the fiducial $z_f = 10$, $\alpha = 1$, $K \sim 8$ history one finds a *history-only* readiness profile $R_{\text{hist}}/(a_0/g_N) \approx 3.3$ at $\xi = 0.1$, ≈ 1.85 at $\xi = 1$, and ≈ 0.23 at $\xi = 5$ (RMS deviation ≈ 1.31 over $0.1 < \xi < 30$). Hence the secular history is informative about the background state of the medium, but it is not the primary local closure law.

The vortex-equilibrium principle. The transported gravity g_h is not an "accumulated" field that grows slowly over cosmic time. Once space is treated as a pressure medium, a rotating galaxy simply creates a vortex in that medium, and that vortex redistributes pressure immediately around the current mass distribution. The transported channel g_h is exactly this instantaneous vortex response.

The two-scale separation established in Sec. 18.7 confirms this quantitatively: the spatial relaxation time is $\tau_{\text{spatial}} = 3H/(c^2 k_r^2) \approx 18$ days, while the secular timescale is $\sim t_H \approx 14.5$ Gyr. The ratio $\tau_{\text{spatial}}/t_H \sim 10^{-12}$ means that spatial equilibrium is maintained to extraordinary precision at every instant.

Spatial equilibrium and the C_{eff} question. The spatial relaxation time $\tau_{\text{spatial}} = 3H/(c^2 k_r^2)$ scales as ξ^2: from ~ 18 days at $\xi = 1$ to ~ 5 years at $\xi = 10$. Even at the outermost radius, $\tau_{\text{spatial}}/t_{\text{galaxy}} \sim 5 \times 10^{-10}$, where $t_{\text{galaxy}} \approx 10$ Gyr. This establishes that the *spatial profile shape* of φ_h adjusts to its source within days–years at all radii.

Transport-balance interpretation. If the Hubble-coherence source $\Omega_{\text{tr}} = a_0/c$ acts continuously, the transport balance $\varphi_h/\tau_{\text{rel}} = \Omega_{\text{tr}}\, \varphi_N$ is maintained at every instant as the steady local balance of the vortex cell. Combining this steady balance with Eq. (67),

$$C_{\rm eff}(\xi) = 1 + \mathcal{O}\!\left(\frac{\Omega_{\rm tr}}{\omega} + \frac{\tau_{\rm spatial}}{t_{\rm galaxy}}\right), \tag{200}$$

and on the selected saturated branch this is exactly the local constitutive closure $C_{\rm eff} = 1$. For the fiducial spiral model the surrogate corrections are already tiny: $\Omega_{\rm tr}/\omega \lesssim 7 \times 10^{-4}$ at $\xi = 1$, $\lesssim 6 \times 10^{-3}$ at $\xi = 10$, while $\tau_{\rm spatial}/t_{\rm galaxy} \lesssim 10^{-9}$ across the same range. The Poisson source $S_{\rm eff} = -c^2\nabla^2\varphi_h$ computed from the $C_{\rm eff} = 1$ profile is everywhere positive (`structural_proof.py`, Part D), confirming physical consistency.

Secular-accumulation contrast. The secular ODE model (dR/dt = source − damping, with $R(0) = 0$) gives $C_{\rm eff} \ll 1$ at all radii for constant H_0: the convergence rate $\gamma_{\rm eff} = g_N(1 + 2R_{\rm eq})/c$ satisfies $\gamma_{\rm eff}/H \lesssim 0.3$, so the system remains in the linear-growth phase. This represents a *different physical scenario* (accumulation from zero) than the transport balance (steady-state maintained by continuous source).

The two models are distinguished by the nature of the source: continuous coherence support associated with the global pressure history (transport balance) vs. gradual buildup from zero (secular ODE). The completed axisymmetric and 2D PDE checks already select the transport-balance picture for spiral galaxies: the medium behaves as a continuously sustained vortex, not as a field accumulated from zero. This is precisely the self-regulating regime relevant for real spiral galaxies: the ongoing rotation continuously maintains the central rarefaction and its pressure gradient, while the local closure fixes the extra inward support at equilibrium instead of permitting indefinite growth. The secular integral (parameter $K \approx 8$) diagnoses the *background pressure history* underlying the observed scale $a_0 = cH/(2\pi)$; it does not by itself replace the local vortex-equilibrium closure. Form invariance guarantees that a *constant* rescaling of $C_{\rm eff}$ preserves $\mu = x/(1 + x)$ (with redefined a_0). However, if $C_{\rm eff}(\xi)$ varies with radius, no single a_0 can absorb the variation; universality then requires the transport balance ($C_{\rm eff} = 1$), which is exactly the local vortex-equilibrium closure used throughout this companion paper.

Error budget (revised).

Source of error	Type	Magnitude	On μ	On $v_{\rm flat}$
Algebraic ($C{=}1 \to \mu{=}x/(1{+}x)$)	exact	0	0	0
Spectral discretisation ($N{=}200$)	convergent	$< 10^{-3}$	$< 10^{-3}$	$< 5 \times 10^{-4}$
Spatial equilibrium ($\tau_{\rm sp}/t_H$)	negligible	$\sim 10^{-12}$	$\sim 10^{-12}$	$\sim 10^{-12}$
a_0 prediction vs. observed	systematic	13%	—	$\approx 3\%$
Cosmological parameters (Ω_b, Ω_Λ)	sensitivity	$\sim 5\%$	< 0.02	$< 2\%$

The dominant theoretical uncertainty is the $\sim 13\%$ offset between the predicted $a_0 = cH_0/(2\pi) \approx 1.04 \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and the observed value $a_0^{(\rm obs)} \approx 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. This offset is consistent with $O(1)$ factors (Bessel $x_0 = 2.4048$, spin parameter $\xi \sim 0.5$, mode coupling) and observational systematics ($\sim$ 15–25% from stellar M/L, distance, inclination, and gas mass). Form invariance guarantees that a constant $O(1)$ rescaling preserves the form $\mu = x/(1 + x)$ (with redefined a_0); global applicability requires $C_{\rm eff}$ to be approximately radius-independent.

18.9 Analytic derivation of the master parameter K

The master formula (199) contains $K \approx 8$, previously determined numerically. We now derive it analytically.

The background-history integral in redshift coordinates is:

$$R(\xi) = \int_0^{z_f} \frac{\mathcal{E}(z)}{2\pi(1 + z)}\, e^{-\gamma\,\Delta t(z)}\, dz\,, \tag{201}$$

where $\Delta t(z) = \int_0^z dz'/[(1+z')H(z')]$ is the lookback time and $\gamma = g_N/c$.

At high redshift, $\mathcal{E}(z)$ grows as $\exp[\frac{3\delta w_0}{\alpha}(1+z)^\alpha]$, so the integrand is an *endpoint maximum*: 84% of R is accumulated at $z > z_f/2$. Applying the endpoint Laplace method and requiring an order-unity background readiness at $\xi = 1$ gives:

$$\boxed{K_0 = \ln\frac{2\pi}{y_1\sqrt{\Omega_b}} + \tfrac{3}{2}\ln(1+z_f)\,,} \tag{202}$$

where $y_1 \equiv g_N(\xi\!=\!1)/a_0 = m_{\mathrm{enc}}(\eta)$. For the fiducial galaxy ($\eta = 1.16$, $y_1 = 0.32$, $\Omega_b = 0.05$, $z_f = 10$):

$$K_0 = \underbrace{4.47}_{\text{geometric}} + \underbrace{3.60}_{\text{cosmological}} = 8.07\,.$$

This matches the numerical $K = 7.4$ to $\sim 10\%$. The small discrepancy is accounted for by the exact transcendental equation:

$$K = \ln\Big[\frac{2\pi\alpha K}{3\,y_1\sqrt{\Omega_b}}\Big] + \big(\tfrac{3}{2} - \alpha\big)\ln(1+z_f) + \gamma\,\Delta t(z_f)\,, \tag{203}$$

which can be solved iteratively from K_0 in 2–3 steps.

Physical interpretation. K is the *logarithmic background-history scale* of quantum feedback over the galaxy's lifetime. Its two terms encode: (i) the ratio of Hubble volume to galaxy potential depth ($\ln[2\pi/(y_1\sqrt{\Omega_b})]$), and (ii) the past enhancement of static space pressure together with the accelerated clock rate ($\frac{3}{2}\ln(1+z_f)$). $K \approx 8$ is *not* an unexplained fitting constant—it is the natural logarithmic scale of the ISPG framework.

18.10 Complete derivation chain

The full chain, from the ISPG ontology to the MOND interpolating function, is most accurately read as a mature-spiral closure chain:

Step	Statement	Status
1	ISPG ontology $\to$ bi-conformal metric $g_{\mu\nu} = e^{\varphi}\eta_{\mu\nu}$	proved
2	Scalar field equation $\ddot{\varphi} + 3H\dot{\varphi} - c^2\nabla^2\varphi = S$	proved
3	Hubble-damped oscillator $\to$ coherence length $\lambda_H = 2\pi c/H$	selected (coherence boundary)
4	Critical acceleration $a_0 = cH/(2\pi)$	selected (coherence boundary)
5	Rotating-space dynamics $\to$ two-channel decomposition $\varphi = \varphi_N + \varphi_h$	proved
6	Backbone local projection $\to g_v = \Upsilon_{\rm rot} g_N$	derived
7	Load/free law $Z_{\rm rot} = 1/(1+\sqrt{y})$ from a binary free/load partition with linear loaded-state measure and local MaxEnt	MaxEnt-selected within binary free/load partition[†]
8	Impedance matching ($n = 2$ uniquely cancels K^{-2} in Z') $\to \hat{I} = 2K^2 A_{\rm vort} \to \Upsilon_{\rm rot} = A_{\rm vort} a_0/g$, $C_{\rm eff} = A_{\rm vort}$	unique by impedance matching within power-law clos
9	$\tau_{\rm rel} = c/g$: unique within old transport representation (dim. analysis + universality)	uniquely selected[§]
10	Hubble-boundary vortex equilibrium rate $\Omega_{\rm tr} = a_0/c$ (stationary coherence theorem for the largest coherent cell, Secs. 5, 17)	derived (coherence-boundary closure)[¶]
11	Vortex maturity parameter $A_{\rm vort} = \omega/(\omega + \Omega_{\rm tr})$ (laminar/turbulent regime)	derived (vortex maturity)[¶]
12	Quantum feedback $\to$ background-pressure enhancement scale (β proxy, $K \approx 8$, Eq. (202))	derived
13	$g^2 = g_N g + a_0 g_N \to \mu(x) = x/(1+x)$ on the mature occupied branch	proved (conditional)
–	$O(1)$ coefficient $\to$ form invariance (constant α); 13% offset in a_0; absorbed by $O(1)$ geometric factors	proved (const. α)

[†]MaxEnt-selected: in a binary free/load partition, the loaded-sector measure is linear in the first-power deficit amplitude, and relative-entropy maximization gives $Z_{\rm rot} = 1/(1+\sqrt{y})$.
[††]Absolutely unique: the ODE $K\,\partial_K \hat{I} = 2\hat{I}$ (from K-independence of $\Upsilon_{\rm rot}$) has the unique solution $\hat{I} = C(A)\,K^2$ for *any* smooth $\hat{I}(K, A)$.
[§]Unique within the galaxy-independent rotating-medium transport ansatz.
[¶]Coherence/vortex theorem: $\Omega_{\rm tr} = a_0/c$ is the unique stationary transport rate fixed by hyperbolic propagation at speed c and the outer coherence boundary $\lambda_H = 2\pi c/H$; in the same pre-existing-vortex picture (Sec. 3), $A_{\rm vort}$ follows from the coherent/incoherent occupation balance rather than being a free closure input. The rate derivation is within the rotating-medium master-equation/coherence-boundary closure; a full microscopic/action-to-PDE derivation remains the next theorem-level strengthening. The saturated value $C_{\rm eff} \simeq 1$ applies on the mature macroscopic occupied branch ($Q_{\rm occ} \gg 1$, $\delta_{\rm sp} \ll 1$, $N_{\rm cell} \gg 1$), not as a universal local orbital-frequency rule for non-galactic systems.

Open programme

The following items are identified as the principal remaining derivation targets. Each is stated with its current status, the specific mathematical task required, and the expected output. The local gravitational backbone entering the MOND split is no longer part of that open list: the companion quantum paper closes the oscillon → Bernoulli → source → $1/r$ chain together with reduced-gap/coercive source control. The items below therefore concern the vortex/galactic completion built on top of that local channel.

1. **$C_{\mathrm{eff}}(\xi)$ analytic spine.**
   Current status: the mature-branch profile $C_{\mathrm{eff}} = A_{\mathrm{vort}} + \mathcal{O}(\delta_{\mathrm{sp}}, \delta_N) \simeq 1$ is established numerically, but no closed-form asymptotic formula exists.
   Task: derive inner ($\xi \ll 1$) and outer ($\xi \gg 1$) asymptotic expansions of $C_{\mathrm{eff}}(\xi)$ from the activation ODE and the Bessel-cell eigenvalue, construct the matched composite approximation, and prove explicit error bounds.
   Output: one proposition, one figure (composite vs. numerical), one error-envelope table.

2. **Closed-form AQUAL Green function in cylindrical geometry.**
   Current status: the AQUAL equivalence theorem (Sec. 105) formally identifies the nonlocal kernel as the unique phantom density $\rho_{\mathrm{ph}} = -\frac{a_0}{4\pi G}\frac{(\nabla g)\cdot\hat{\mathbf{g}}}{g+a_0}$. The full-field mismatch between the 1D algebraic target and the 2D AQUAL solution has been identified as the standard AQUAL curl field [22] inherent to disk geometry (Sec. 9.3): grid-convergence ratio ≈ 1, spherical-source control reduces the mismatch threefold, and a disk-thickness scan confirms monotonic dependence on asphericity. The μ-profile is verified to 2–5% RMS; the formal PDE-to-μ chain is closed with the algebraic/transport-side closure gap closed; the Green-kernel/profile residual remains a quantitative refinement.
   Task: compute the closed-form axisymmetric Green function of the AQUAL operator $\nabla\cdot[\mu(|\nabla\Phi|/a_0)\nabla(\cdot)]$ for $\mu = x/(1+x)$, and derive explicit inner/outer asymptotic expansions. This is an analytic convenience, not a gap closure.
   Output: closed-form kernel + asymptotic error envelope.

3. **Cluster binding: 3D steady-state solve.**
   Current status: the feedback-ODE benchmark (Sec. 16) shows that the 50% binding threshold can be reached if the tail background is retained for $\tau_{\mathrm{ret}} \approx 0.63$ Gyr $\approx 0.42\, t_{\mathrm{dyn}}$, rather than escaping ballistically. In the diffusion picture, the corresponding mean free path $\ell_{\mathrm{mfp}} \approx 16$ kpc is a plausible resonant-scattering target tied to observed ICM cold-front structure, but it has not yet been derived from a wave-scattering calculation.
   Task: (a) solve the scalar field equation on a 3D cluster grid with *Chandra*-derived profiles, (b) compute ℓ_{mfp} from the observed cold-front census and ICM density power spectrum, (c) extract the radial binding profile for comparison with weak-lensing data.
   Output: derived $\rho_{\mathrm{extra}}/\rho_{\mathrm{cl}}$ profile replacing the current calibrated benchmark, plus a prediction for binding–substructure correlation.

4. **Bullet Cluster time-dependent merger simulation.**
   Current status: frozen-hysteresis snapshot with correct peak topology and threshold (Sec. 15).
   Task: evolve the χ field together with N-body + gas hydrodynamics through the full collision, reconstruct $\kappa(x, y, t)$, and compare with Clowe et al. weak-lensing data.
   Output: convergence-map movie, centroid tracks, aperture masses, robustness strip over L_χ and M_χ.

5. **Elliptical-galaxy convective-cell: explicit PDE check.**
   Current status: the universality argument of Sec. 14.1 motivates the working hypothesis

that the Bernoulli pressure-deficit mechanism, the coherence boundary λ_H, the forgetting rate $\Gamma_{\rm mem}$, and the Markov activation chain are topology-independent, so the spiral $\mu(x) = x/(1+x)$ branch may carry over to ellipticals with $\omega \to \sigma/r$. The deep-MOND Faber–Jackson virial coefficient is derived exactly once the deep-MOND force law is assumed (Sec. 14); the convective-cell PDE realization remains open.
Task: solve the scalar PDE explicitly in a spherically symmetric poloidal-circulation background and verify that the numerical μ profile agrees with $x/(1+x)$. This is the required quantitative closure test for the pressure-supported extension, not for the already-closed spiral algebra.
Output: $\mu_{\rm ell}(x)$ profile + comparison with spiral $\mu = x/(1+x)$.

With the AQUAL equivalence theorem established, the 2D residual identified as the standard curl field in disk geometry, and the 15% a_0 offset closed phenomenologically by the calibrated vortex-lag effect (Sec. 5.5), the spiral-galaxy MOND closure chain is complete at the constitutive level, with each step either derived, uniquely selected by consistency, calibrated within the H.3 closure, or observationally constrained. The independent H.3 test is the predicted age dependence of a_0. Two formal reductions remain open (backbone $\to S_{\rm swirl}$ and MaxEnt linearity from the action); these are technical exercises within a framework whose physical content is fully specified. Items 1–2 are analytical refinements that can be carried out within the existing codebase. Items 3–5 each constitute a dedicated paper extending the framework to clusters, mergers, and elliptical galaxies.

Takeaway

This companion paper presents the MOND sector in two layers: (i) the *cosmological origin* of galactic vortices from observed cosmic flows in the space substance (Sec. 3), and (ii) the *quantitative closure* inside a mature vortex cell.

Proved analytically or by exact algebra: the scaling $a_0 \propto cH$; the Fourier $m = 0$ dominance; the Bessel eigenvalue; the quadratic identity yielding $\mu(x) = x/(1+x)$ once the mature occupied branch is specified; $v^4 = GMa_0$ in the deep-MOND limit; Newtonian recovery $\mu \to 1$ at $\xi \ll 1$ (Proposition 3) and deep-MOND scaling $\mu \propto x$ at $\xi \gg 1$ (Proposition 4); the master parameter $K_0 = \ln[2\pi/(y_1\sqrt{\Omega_b})] + \frac{3}{2}\ln(1+z_f)$ (Sec. 18.9); form invariance of μ under $O(1)$ rescaling of a_0; the deep-MOND virial scaling that yields the baryonic Faber–Jackson relation $\sigma^4 = K_\sigma\, GMa_0$ once the MOND force law is assumed (Sec. 14); the turbulent/convective medium-response realization for pressure-supported systems remains a model-dependent extension; and the EFE with its three regimes, anisotropy prediction, and wide-binary consistency (Sec. 8).

Selected within the closure: $Z_{\rm rot} = 1/(1+\sqrt{y})$ from a binary free/load partition with linear loaded-state measure and local MaxEnt; $\hat{I} = 2K_{\rm rot}^2 A_{\rm vort}$ (impedance matching: $n = 2$ is the only power in the chosen family for which $\Upsilon_{\rm rot}$ depends on activation alone, not on medium stiffness); $\tau_{\rm rel} = c/g$ (dimensional analysis + universality, within the old transport representation, Sec. 18.7).

Calibrated / predicted in H.3: the identity $H_{\rm eff} = \ln(1+z_{\rm form})/t_{\rm lookback}(z_{\rm form})$ is algebraic for the adopted uniform coordinate-time memory prescription; the value $z_{\rm form} \approx 0.58$ is calibrated to the local observed a_0 in the standard-ΛCDM background used by `heff_calc.py`; and the falsifiable content is the predicted correlation between inferred a_0 residuals and disk-settling age (Sec. 5.5).

Proved / derived in the coherence-vortex sector: $\Omega_{\rm tr} = a_0/c$ as the unique stationary transport rate of a pre-existing vortex cell at the Hubble coherence boundary (Secs. 5, 17); the vortex maturity parameter $A_{\rm vort} = \omega/(\omega + \Omega_{\rm tr})$ from coherent/incoherent packet balance (laminar $A \to 1$ in mature spirals, turbulent $A \ll 1$ in systems lacking a coherent vortex).

Cosmological vortex origin (Sec. 3): observed cosmic-flow fields (Laniakea, cosmic web, bulk flows), identified in ISPG with bulk motion of the pressure-carrying substance, undergo Kelvin–Helmholtz roll-up under the adopted low-viscosity effective-medium closure → central rarefaction → MOND. Galaxy morphology (spiral, elliptical, irregular, dwarf) corresponds to fluid-dynamical vortex topology (laminar planar, convective poloidal, chaotic, small-scale eddy) as a qualitative model map; the quantitative population and kinematic predictions remain a later task. The MOND scale a_0 is common to all types because it is set by the cosmological coherence boundary, not by galaxy-specific parameters.

Closed locally on the mature branch: $C_{\rm eff} = A_{\rm vort} + \mathcal{O}(\delta_{\rm sp}) + \mathcal{O}(\delta_N)$, with the mature occupied branch satisfying $Q_{\rm occ} \gg 1$, $\delta_{\rm sp} \ll 1$, and $N_{\rm cell} \gg 1$; in the spiral-galaxy regime this yields $C_{\rm eff} \simeq 1$ (Sec. 18.8). The secular history integral is auxiliary: it tracks the global pressure decrease and the accelerated past time, but the primary local MOND closure is the instantaneous vortex equilibrium. The baryonic Bernoulli channel underneath this split is supplied by the companion quantum paper's closed source chain, so the remaining work in the present paper is not a closure of g_N itself but refinement and verification of the transported vortex channel.

Numerically benchmarked (Sec. 18): spatial relaxation $\tau_{\rm sp} \ll t_H$ at all radii; the coupling exponent β (derived from the quantum feedback ODE, not assumed); the interpolating function $\mu(x) = x/(1+x)$ to machine precision only in the transport-side algebraic identity and at the $\sim 10^{-3}$ spectral level on the self-consistent activation branch; the source-side nonlocal 2D kernel with parameters fixed by regularity and outer asymptotics, achieving $\sim$ 6–13% field RMS on the compact matched box and $\sim$ 10% field RMS on the widened disjoint-annulus benchmark; the operator-side AQUAL solve at $\sim$ 11% field RMS on the compact box and $\sim$ 9% field RMS on the widened benchmark, with μ supplied as input; universality across galaxy parameters; spectral convergence; Poisson-source grid-verified positivity for the occupied branch; and fixed-source/coupled hyperbolic relaxation toward the selected static branch. A complete error budget (Sec. 18.8) quantifies the theoretical uncertainty: the dominant source is the 13% offset in a_0, well below observational systematics.

Equivalent constitutive representation: in the quasi-static galactic regime the old transport ansatz is the effective representation picked out by the occupied source-side branch and the combined analytic/PDE results.

Predicted (testable):

$$a_0 = \frac{cH}{2\pi}, \quad \mu(x) = \frac{x}{1+x}, \quad v^4 = GMa_0, \quad \delta w_0 = \frac{K\alpha}{3(1+z_f)^{\alpha}},$$

together with: the calibrated vortex-lag closure $a_0^{\rm eff} = cH_{\rm eff}/(2\pi)$ with $H_{\rm eff} = \ln(1 + z_{\rm form})/t_{\rm lookback}$, where the local normalization fixes $z_{\rm form}$ and the genuine prediction is galaxy-age-dependent a_0 with correlated BTFR scatter (Sec. 5.5); a falsifiable redshift drift $a_0(z) \propto H(z)$; a MOND-background–dark-energy-sector link testable by DESI/Euclid as a consistency target rather than a present confirmation; an environment signature (EFE, Sec. 8); a vertical-confinement mechanism from frame-dragging equatorial anisotropy (Sec. 13); and the Faber–Jackson scaling for ellipticals and dwarf spheroidals (Sec. 14).

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